5

Non-Central and Central Limit Theorems

The material of this chapter is motivated in the following question. Suppose $\{X_n\}_{n\in\mathbb{Z}}$ is a Gaussian series with long-range dependence and let $G : \mathbb{R} \mapsto \mathbb{R}$ be a deterministic function such that $\mathbb{E}G(X_n)^2 < \infty$. What is the limit of the partial sum process

$$\sum_{n=1}^{[Nt]} G(X_n), \quad t \geq 0, \tag{5.0.1}$$

as $N \to \infty$, after suitable centering and normalization? The question is important when dealing with long-range dependent series. As will be seen in this chapter, the limit is not necessarily a Gaussian (or stable) process. If the limit is not Gaussian nor stable, the corresponding limit theorem is known as a *non-central limit theorem* and the limit process is an Hermite process.

We shall also consider a number of generalizations of the setting (5.0.1), and address several related questions such as generation of non-Gaussian series.

We consider nonlinear functions of Gaussian random variables in Section 5.1 and define their Hermite rank in Section 5.2. Convergence of normalized partial sums to the Hermite processes is proved in Section 5.3. The result is stated in Theorem 5.3.1 and is called a *non-central limit theorem* because the Hermite processes are typically non-Gaussian. Section 5.4 deals with short-range dependence, where the limit of the normalized partial sums is Brownian motion. This result is stated in Theorem 5.4.1 and the proof given here is based on a method of moments following the original approach of Breuer and Major [184]. Recently, Nualart and Peccati [771] have shown that it is enough only to prove convergence of moments up to order 4, as indicated in Section 5.5.

In Section 5.6, we focus on non-central and central limit theorems for nonlinear functions of linear time series with long-range or short-range dependence. Multivariate limit theorems are considered in Section 5.7. In Section 5.8, we use nonlinear transformations of Gaussian time series to generate non-Gaussian time series and study the effect of the transformation on the dependence structure.

5.1 Nonlinear Functions of Gaussian Random Variables

In the study of partial sums (5.0.1), Hermite polynomials, defined in Definition 4.1.1, play a fundamental role. As we will see, they form a basis for the space of finite-variance, nonlinear functions of Gaussian random variables, also called Gaussian subordinated variables.

One immediate consequence of the relation (4.1.2) for Hermite polynomials is the following classical result. It is a special case of Proposition 4.3.22 (see also Example 4.3.23). We provide a direct proof based on the properties of Hermite polynomials.

Proposition 5.1.1 *Let $(X, Y)'$ be a Gaussian vector with $\mathbb{E}X = \mathbb{E}Y = 0$ and $\mathbb{E}X^2 = \mathbb{E}Y^2 = 1$. Then, for all $n, m \geq 0$,*

$$\mathbb{E}H_n(X)H_m(Y) = \begin{cases} n!(\mathbb{E}XY)^n, & \text{if } n = m, \\ 0, & \text{if } n \neq m. \end{cases} \tag{5.1.1}$$

Proof Recall that a Gaussian random variables X with $\mathbb{E}X = 0$ has moment generating function $\mathbb{E}e^{tX} = \exp\{\frac{1}{2}\mathbb{E}(tX)^2\}$, and let $F(x,t) = e^{tx-t^2/2}$ as in (4.1.2). Then, for any $s, t \in \mathbb{R}$,

$$\mathbb{E}F(X,s)F(Y,t) = \mathbb{E}\Big(\exp\Big\{sX - \frac{s^2}{2}\Big\}\exp\Big\{tY - \frac{t^2}{2}\Big\}\Big) = \exp\Big\{-\frac{s^2}{2} - \frac{t^2}{2}\Big\}\mathbb{E}\exp\{sX + tY\}$$

$$= \exp\Big\{-\frac{s^2}{2} - \frac{t^2}{2}\Big\}\exp\Big\{\frac{1}{2}\mathbb{E}(sX+tY)^2\Big\} = \exp\{st\mathbb{E}XY\}$$

by using $\mathbb{E}X = \mathbb{E}Y = 0$ and $\mathbb{E}X^2 = \mathbb{E}Y^2 = 1$. Differentiating the left-hand side n times with respect to s and m times with respect to t, setting $s = t = 0$ and using (4.1.3) yields $\mathbb{E}H_n(X)H_m(Y)$. The same operation on the right-hand side yields $n!(\mathbb{E}XY)^n$ if $n = m$, and 0 otherwise. Indeed, this follows from

$$\exp\Big\{st\mathbb{E}XY\Big\} = \sum_{k=0}^{\infty} \frac{s^k t^k (\mathbb{E}XY)^k}{k!}$$

and observing that

$$\frac{\partial^{n+m}}{\partial s^n \partial t^m} s^k t^k \Big|_{s=0,t=0} = \frac{\partial^n}{\partial s^n} s^k \Big|_{s=0} \frac{\partial^m}{\partial t^m} t^k \Big|_{t=0} = \begin{cases} (n!)^2, & \text{if } k = n = m, \\ 0, & \text{otherwise.} \end{cases} \qquad \square$$

Setting $X = Y$ in (5.1.1) yields the following corollary of Proposition 5.1.1. Let

$$\phi(dx) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}dx \tag{5.1.2}$$

be the probability measure on $\mathbb{R}$ associated with a standard normal variable X and $L^2(\phi)$ be the space of measurable, square-integrable functions with respect to $\phi(dx)$. Then, $G \in L^2(\mathbb{R}, \phi)$ if and only if $\mathbb{E}G(X)^2 < \infty$. The space $L^2(\mathbb{R}, \phi)$ is naturally equipped with the inner product

$$(G_1, G_2)_{L^2(\mathbb{R},\phi)} = \int_{\mathbb{R}} G_1(x)G_2(x)\phi(dx) = \int_{\mathbb{R}} G_1(x)G_2(x)\frac{e^{-x^2/2}}{\sqrt{2\pi}}dx = \mathbb{E}G_1(X)G_2(X). \tag{5.1.3}$$

Corollary 5.1.2 *For any $n, m \geq 0$, and standard normal random variable X,*

$$\mathbb{E}H_n(X)H_m(X) = (H_n, H_m)_{L^2(\mathbb{R},\phi)} = \begin{cases} n!, & \text{if } n = m, \\ 0, & \text{if } n \neq m. \end{cases} \tag{5.1.4}$$

In other words, $\{H_n(x)\}_{n\geq 0}$ *is a collection of orthogonal functions in the space* $L^2(\mathbb{R}, \phi)$.

The collection $\{H_n(x)\}_{n\geq 0}$ not only consists of orthogonal functions but it is also a basis for the space $L^2(\mathbb{R}, \phi)$. This is stated in the next result.

Proposition 5.1.3 *The collection* $\{H_n(x)\}_{n\geq 0}$ *is a basis for the space* $L^2(\mathbb{R}, \phi)$.

Proof Let X be a standard normal random variable. It is enough to show that, for $G \in L^2(\mathbb{R}, \phi)$ and all $n \geq 0$,

$$(G, H_n)_{L^2(\mathbb{R},\phi)} = \mathbb{E}G(X)H_n(X) = 0, \tag{5.1.5}$$

implies that $G(x) = 0$ a.e. $\phi(dx)$. If (5.1.5) holds, then

$$\mathbb{E}G(X)F(X, z) = 0,$$

where $F(x, z)$ is defined in (4.1.2) and we can take $z \in \mathbb{C}$. But this means that, for any $z \in \mathbb{C}$,

$$\mathbb{E}G(X)F(X, z)e^{\frac{z^2}{2}} = \mathbb{E}G(X)e^{zX} = \frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}} e^{zx}G(x)e^{-x^2/2}dx = 0.$$

In particular, the $L^2(\mathbb{R})$–Fourier transform of the function $G(x)e^{-x^2/2}$ is zero. Hence, taking the inverse Fourier transform, $G(x)e^{-x^2/2} = 0$ a.e. dx. This yields $G(x) = 0$ a.e. $\phi(dx)$. □

A consequence of Proposition 5.1.3 is that every function $G \in L^2(\mathbb{R}, \phi)$ has a series expansion in Hermite polynomials:

$$G(x) = \sum_{n=0}^{\infty} g_n H_n(x), \tag{5.1.6}$$

where the convergence takes place in $L^2(\mathbb{R}, \phi)$. Note that, since H_n are orthogonal but not orthonormal, we have

$$g_n = \frac{1}{n!}(G, H_n)_{L^2(\mathbb{R},\phi)} = \frac{1}{n!}\int_{\mathbb{R}} G(x)H_n(x)\frac{e^{-x^2/2}dx}{\sqrt{2\pi}} = \frac{1}{n!}\mathbb{E}G(X)H_n(X), \tag{5.1.7}$$

where X is a standard normal variable. The relation (5.1.6) can be written as

$$G(X) = \sum_{n=0}^{\infty} g_n H_n(X), \tag{5.1.8}$$

where the convergence takes place in $L^2(\Omega)$. Indeed, note that

$$\mathbb{E}\Big(G(X) - \sum_{n=0}^{N} g_n H_n(X)\Big)^2 = \Big\| G - \sum_{n=1}^{N} g_n H_n \Big\|^2_{L^2(\mathbb{R},\phi)} \to 0,$$

by using (5.1.6). The following is an immediate corollary.

https://doi.org/10.1017/CBO9781139600347.007 Published online by Cambridge University Press

Proposition 5.1.4 *Let $(X,Y)'$ be a Gaussian vector with $\mathbb{E}X = \mathbb{E}Y = 0$ and $\mathbb{E}X^2 = \mathbb{E}Y^2 = 1$. Suppose $G_1, G_2 \in L^2(\mathbb{R},\phi)$ and let $g_{1,n}$ and $g_{2,n}$, $n \geq 0$, be the coefficients in the Hermite expansions of G_1 and G_2, respectively, as in (5.1.6). Then,*

$$\mathbb{E}G_1(X)G_2(Y) = \sum_{n=0}^{\infty} g_{1,n}g_{2,n}n!\,(\mathbb{E}XY)^n \tag{5.1.9}$$

and

$$\mathrm{Cov}(G_1(X), G_2(Y)) = \sum_{n=1}^{\infty} g_{1,n}g_{2,n}n!\,(\mathbb{E}XY)^n. \tag{5.1.10}$$

Proof As in (5.1.8), $G_1(X) = \sum_{n=0}^{\infty} g_{1,n}H_n(X)$ and $G(Y) = \sum_{m=0}^{\infty} g_{2,m}H_m(Y)$, where the convergence takes place in $L^2(\Omega)$. Then, by Proposition 5.1.1,

$$\mathbb{E}G_1(X)G_2(Y) = \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} g_{1,n}g_{2,m}\mathbb{E}H_n(X)H_m(Y) = \sum_{n=0}^{\infty} g_{1,n}g_{2,n}n!\,(\mathbb{E}XY)^n,$$

which is (5.1.9). Moreover, since $\mathbb{E}G_1(X) = \mathbb{E}G_1(X)H_0(X) = g_{1,0}$, one has $\mathbb{E}G_1(X)\mathbb{E}G_2(Y) = g_{1,0}g_{2,0}$, implying (5.1.10). □

The rate of convergence of the series (5.1.6), through the asymptotic behavior of the coefficients g_n, is considered in Exercise 5.1.

5.2 Hermite Rank

Taking $G_1 = G_2$ in (5.1.10) and using the index k instead of n yield

$$\mathbb{E}G(X)G(Y) = \sum_{k=0}^{\infty} g_k^2 k!\,(\mathbb{E}XY)^k \quad \text{and} \quad \mathrm{Cov}(G(X), G(Y)) = \sum_{k=1}^{\infty} g_k^2 k!\,(\mathbb{E}XY)^k. \tag{5.2.1}$$

In particular, for a Gaussian stationary series $\{X_n\}_{n\in\mathbb{Z}}$ with $\mathbb{E}X_n = 0$, $\mathbb{E}X_n^2 = 1$ and autocovariance function γ_X, the autocovariance of a stationary series $\{G(X_n)\}_{n\in\mathbb{Z}}$ can be expressed as

$$\mathrm{Cov}(G(X_n), G(X_0)) = \sum_{k=1}^{\infty} g_k^2 k!\,(\gamma_X(n))^k. \tag{5.2.2}$$

If the Gaussian series $\{X_n\}_{n\in\mathbb{Z}}$ is long-range dependent in the sense (2.1.5) in condition II, we have $\gamma_X(n) = L_2(n)n^{2d-1}$ for a slowly varying function L_2 and each entry $(\gamma_X(n))^k$ in (5.2.2) becomes $(L_2(n)n^{2d-1})^k$. The dominating series term in (5.2.2) is then determined by the smallest k for which the coefficient g_k is nonzero. This index k has a special name.

Definition 5.2.1 Let ϕ be the standard normal density, $G \in L^2(\mathbb{R},\phi)$ and g_k, $k \geq 0$, be the coefficients in its Hermite expansion (5.1.6). The *Hermite rank* r of G is defined as the smallest index $k \geq 1$ for which $g_k \neq 0$; that is,

$$r = \min\{k \geq 1 : g_k \neq 0\}. \tag{5.2.3}$$

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Remark 5.2.2 One sometimes defines the Hermite rank as

$$r = \min\{k \geq 0 : g_k \neq 0\}. \tag{5.2.4}$$

This definition and (5.2.3) are equivalent, except in the case when the function G is such that $\mathbb{E}G(X) \neq 0$ where $X \sim \mathcal{N}(0,1)$ (e.g., G is a constant $c \neq 0$). Indeed, since $g_0 = \mathbb{E}G(X)H_0(X) = \mathbb{E}G(X) \neq 0$, the Hermite rank equals 0 according to the definition (5.2.4). According to the definition (5.2.3), the Hermite rank of G would be greater than or equal to 1, including the possibility of infinity for a function G equal to a constant $c \neq 0$, since $g_k = (k!)^{-1}\mathbb{E}G(X)H_k(X) = c(k!)^{-1}\mathbb{E}H_k(X) = 0$ for all $k \geq 1$.

One can also define the Hermite rank of G as

$$r = \min\{k \geq 0 : \mathbb{E}[(G(X) - \mathbb{E}G(X))H_k(X)] \neq 0\}. \tag{5.2.5}$$

This definition is equivalent to (5.2.3) since $\mathbb{E}[(G(X) - \mathbb{E}G(X))H_k(X)]$ equals $\mathbb{E}G(X) - \mathbb{E}G(X) = 0$ if $k = 0$ and equals $\mathbb{E}G(X)H_k(X)$ if $k \geq 1$.

Example 5.2.3 Suppose $G(x) = x^2$ and $X \sim \mathcal{N}(0,1)$. Then $\mathbb{E}G(X)H_0(X) = \mathbb{E}X^2 \neq 0$, $\mathbb{E}G(X)H_1(X) = \mathbb{E}X^3 = 0$ and $\mathbb{E}G(X)H_2(X) = \mathbb{E}X^2(X^2-1) = \mathbb{E}X^4 - \mathbb{E}X^2 \neq 0$. Therefore, the Hermite rank of G equals 2 according to (5.2.3) and (5.2.5), and equals to 0 according to (5.2.4). Note that all three definitions imply that $G(x) - \mathbb{E}G(X) = x^2 - 1$ has Hermite rank 2. It is therefore best to consider functions with their mean subtracted, which is what one does in the context of (non)central limit theorems.

Hence the functions $x^{2n} - \mathbb{E}X^{2n}$, $n \geq 1$, have Hermite rank 2 and the functions x^n, $n \geq 1$, n odd, have Hermite rank 1, according to the three definitions.

If $\{X_n\}$ is long-range dependent (LRD), is the sequence $\{G(X_n)\}$ LRD? Not necessarily. The following proposition provides a sufficient condition.

Proposition 5.2.4 *Let $\{X_n\}_{n\in\mathbb{Z}}$ be a stationary Gaussian series with $\mathbb{E}X_n = 0$ and $\mathbb{E}X_n^2 = 1$, which is LRD in the sense (2.1.5) in condition II. Let also $G \in L^2(\mathbb{R}, \phi)$ be a function with Hermite rank r. Then, as $n \to \infty$,*

$$\mathrm{Cov}(G(X_n), G(X_0)) \sim g_r^2 r!(L_2(n))^r n^{(2d-1)r}. \tag{5.2.6}$$

In particular, if

$$\frac{1}{2}\Big(1 - \frac{1}{r}\Big) = \frac{r-1}{2r} < d < \frac{1}{2}, \tag{5.2.7}$$

then the stationary series $\{G(X_n)\}_{n\in\mathbb{Z}}$ is LRD in the sense (2.1.5) in condition II with LRD parameter d_G given by

$$d_G = rd - \frac{r-1}{2} = r\Big(d - \frac{1}{2}\Big) + \frac{1}{2} \in \Big(0, \frac{1}{2}\Big) \tag{5.2.8}$$

and slowly varying function L_G satisfying

$$L_G(u) \sim g_r^2 r!(L_2(u))^r. \tag{5.2.9}$$

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Proof Since G has Hermite rank r, the expansion (5.2.2) starts at $k = r$ and hence

$$\mathrm{Cov}(G(X_n), G(X_0)) = \sum_{k=r}^{\infty} g_k^2 k!\,(\gamma_X(n))^k.$$

But as $n \to \infty$, the leading term in this sum is the term with $k = r$. Therefore,

$$\mathrm{Cov}(G(X_n), G(X_0)) \sim g_r^2 r!(\gamma_X(n))^r.$$

Since, by assumption, $\gamma_X(n) = L_2(n)n^{2d-1}$, we get (5.2.6) or equivalently

$$\mathrm{Cov}(G(X_n), G(X_0)) = L_G(n)n^{2d_G-1},$$

where L_G and d_G satisfy (5.2.9) and (5.2.8), respectively. The condition (5.2.7) ensures that $d_G \in (0, 1/2)$; that is, $\{G(X_n)\}_{n\in\mathbb{Z}}$ is LRD in the sense (2.1.5) in condition II. □

The next result is an immediate consequence of Propositions 5.2.4 and 2.2.5.

Corollary 5.2.5 *Set*

$$H_G = d_G + \frac{1}{2} \tag{5.2.10}$$

and

$$(L_H(u))^2 = \frac{L_G(u)}{d_G(2d_G+1)}. \tag{5.2.11}$$

Under the assumptions and notation of Proposition 5.2.4 and when (5.2.7) holds, one has, as $N \to \infty$,

$$\mathrm{Var}\Big(\sum_{n=1}^{N} G(X_n)\Big) \sim \frac{2g_r^2 r!(L_2(N))^r N^{r(2d-1)+2}}{(r(2d-1)+1)(r(2d-1)+2)} \sim \frac{2L_G(N)N^{2d_G+1}}{2d_G(2d_G+1)} = (L_H(N))^2 N^{2H_G}. \tag{5.2.12}$$

The relation (5.2.12) suggests, in particular, the normalization

$$L_H(N)N^{H_G} = L_H(N)N^{d_G+1/2} = L_H(N)N^{r(d-1/2)+1}$$

for the convergence of partial sums (5.0.1). Table 5.1 displays the interval (5.2.7) and the value and range of d_G in (5.2.8) for the first few values of r. It also displays $H_G = d_G + 1/2$ whose range is always $(1/2, 1)$, and the range of $H_0 = d + 1/2$. The parameter $H_0 = d + 1/2$,

Table 5.1 *Range of d which ensures that $\{G(X_n)\}$ is LRD. Corresponding values of H_0, d_G and H_G.*

r	d	$H_0 = d + \frac{1}{2}$	d_G	$H_G = d_G + \frac{1}{2}$	d_G	H_G
1	$(0, \frac{1}{2})$	$(\frac{1}{2}, 1)$	d	$d + \frac{1}{2}$	$(0, \frac{1}{2})$	$(\frac{1}{2}, 1)$
2	$(\frac{1}{4}, \frac{1}{2})$	$(\frac{3}{4}, 1)$	$2d - \frac{1}{2}$	$2d$	$(0, \frac{1}{2})$	$(\frac{1}{2}, 1)$
3	$(\frac{1}{3}, \frac{1}{2})$	$(\frac{5}{6}, 1)$	$3d - 1$	$3d - \frac{1}{2}$	$(0, \frac{1}{2})$	$(\frac{1}{2}, 1)$
4	$(\frac{3}{8}, \frac{1}{2})$	$(\frac{7}{8}, 1)$	$4d - \frac{3}{2}$	$4d - 1$	$(0, \frac{1}{2})$	$(\frac{1}{2}, 1)$

related to the exponent in (2.1.8), is sometimes used instead of d. The parameter H_G will be denoted H in the next section.

5.3 Non-Central Limit Theorem

We now turn to the study of the limits of partial sums (5.0.1). This section concerns the situation when the limit of (5.0.1) is non-Gaussian in general, and Section 5.4 concerns the case when the limit is Gaussian.

In this section, the limit will involve the process

$$\beta_{k,H} Z_H^{(k)}(t) = \beta_{k,H}^0 \int_{\mathbb{R}^k}'' \frac{e^{it(x_1+\cdots+x_k)}-1}{i(x_1+\cdots+x_k)} \prod_{j=1}^{k} |x_j|^{-\frac{1}{2}+\frac{1-H}{k}} \widehat{B}(dx_1)\ldots\widehat{B}(dx_k), \qquad (5.3.1)$$

where $Z_H^{(k)} = \{Z_H^{(k)}(t)\}_{t\ge 0}$, $1/2 < H < 1$, is a standard Hermite process, defined in (4.2.42),

$$\beta_{k,H}^0 = \Big(2\sin\Big(\Big(\frac{1}{2}-\frac{1-H}{k}\Big)\pi\Big)\Gamma\Big(\frac{2(1-H)}{k}\Big)\Big)^{-k/2} \qquad (5.3.2)$$

and

$$\beta_{k,H} = \frac{\beta_{k,H}^0}{B_{k,H}} = \Big(\frac{k!}{H(2H-1)}\Big)^{1/2}, \qquad (5.3.3)$$

where $B_{k,H}$ is given in (4.2.46). Since $\mathbb{E}Z_H^{(k)}(t)^2 = t^{2H}$, the process (5.3.1) has variance $\mathbb{E}(\beta_{k,H}Z_H^{(k)}(t))^2 = (\beta_{k,H})^2 t^{2H}$. As usual, $\widehat{B}(dx)$ is an Hermitian Gaussian measure with control measure dx. Note also that we specify the Hermite process by its spectral representation as in (4.2.11).

The following is the main result of this section.

Theorem 5.3.1 *(Non-central limit theorem.) Let $\{X_n\}_{n\in\mathbb{Z}}$ be a Gaussian stationary series which is LRD in the sense (2.1.5) in condition II with $d \in (0, 1/2)$. Suppose that $\mathbb{E}X_n = 0$, $\mathbb{E}X_n^2 = 1$. Let G be a function with Hermite rank $k \ge 1$ in the sense of Definition 5.2.1. If*

$$d \in \Big(\frac{1}{2}\Big(1-\frac{1}{k}\Big), \frac{1}{2}\Big), \qquad (5.3.4)$$

then

$$\frac{1}{(L_2(N))^{k/2} N^{k(d-1/2)+1}} \sum_{n=1}^{[Nt]} (G(X_n) - \mathbb{E}G(X_n)) \xrightarrow{fdd} g_k\, \beta_{k,H}\, Z_H^{(k)}(t), \quad t \ge 0, \qquad (5.3.5)$$

where $\xrightarrow{fdd}$ denotes the convergence of finite-dimensional distributions and g_k is the first nonzero coefficient in the Hermite expansion of G in Definition 5.2.1. The process $Z_H^{(k)} = \{Z_H^{(k)}(t)\}_{t\ge 0}$ is the standard Hermite process (4.2.42) of order k with the self-similarity parameter

$$H = k\Big(d-\frac{1}{2}\Big)+1 \in \Big(\frac{1}{2}, 1\Big). \qquad (5.3.6)$$

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The constant $\beta^0_{k,H}$ *given in (5.3.2) equals*

$$\beta^0_{k,H} = \Big(2\sin(d\pi)\Gamma(1-2d)\Big)^{-k/2} \tag{5.3.7}$$

and the normalization $\beta_{k,H}$ *given in (5.3.3) equals*

$$\beta_{k,H} = \Big(\frac{k!}{(k(d-1/2)+1)(k(2d-1)+1)}\Big)^{1/2}. \tag{5.3.8}$$

Remark 5.3.2 By Corollary 5.2.5,

$$\mathrm{Var}\Big(\sum_{n=1}^{N} G(X_n)\Big) \sim \frac{2g_k^2 k!(L_2(N))^k N^{k(2d-1)+2}}{(k(2d-1)+1)(k(2d-1)+2)},$$

as $N \to \infty$. The normalization $(L_2(N))^{k/2}N^{k(d-1/2)+1}$ and the constant $g_k\,\beta_{k,H}$ in (5.3.5) are consistent with this asymptotic result.

Proof We use a number of results found following the proof. By the reduction theorem (Theorem 5.3.3 below), it is enough to prove the result for $G(x) - \mathbb{E}G(X_n) = H_k(x)$, where $H_k(x)$ is the Hermite polynomial of order k.

Step 1 (setup): Let

$$S_{N,k}(t) = \frac{1}{A(N)}\sum_{n=1}^{[Nt]} H_k(X_n)$$

be the corresponding partial sum process, where $A(N) = (L_2(N))^{k/2}N^{k(d-1/2)+1}$ is the normalization.

Now write the series X_n in its spectral representation as

$$X_n = \int_{\mathbb{R}} e^{in\lambda}\widehat{B}_F(d\lambda),$$

where $\widehat{B}_F$ is an Hermitian Gaussian random measure on $\mathbb{R}$ with a symmetric control measure $F(d\lambda)$. The measure $F(d\lambda)$ is the spectral measure of the series $\{X_n\}$ which is concentrated on $(-\pi,\pi]$ and satisfies

$$\gamma_X(n) = \int_{\mathbb{R}} e^{in\lambda}F(d\lambda) \tag{5.3.9}$$

(see Remark 1.3.4). By using Proposition 4.1.2, (b), we have

$$H_k(X_n) = \int''_{\mathbb{R}^k} e^{in(\lambda_1+\dots+\lambda_k)}\widehat{B}_F(d\lambda_1)\dots\widehat{B}_F(d\lambda_k).$$

Then,

$$\begin{aligned} S_{N,k}(t) &= \frac{1}{A(N)}\int''_{\mathbb{R}^k}\sum_{n=1}^{[Nt]} e^{in(\lambda_1+\dots+\lambda_k)}\widehat{B}_F(d\lambda_1)\dots\widehat{B}_F(d\lambda_k) \\ &= \frac{1}{A(N)}\int''_{\mathbb{R}^k} e^{i(\lambda_1+\dots+\lambda_k)}\frac{e^{i[Nt](\lambda_1+\dots+\lambda_k)}-1}{e^{i(\lambda_1+\dots+\lambda_k)}-1}\widehat{B}_F(d\lambda_1)\dots\widehat{B}_F(d\lambda_k). \end{aligned}$$

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By making the change of variables $\lambda_1 = x_1/N, \dots, \lambda_k = x_k/N$, and using the change of variables formula (B.2.15), the process $S_{N,k}(t)$ has the same distribution as (we use the same notation)

$$S_{N,k}(t) = \int_{\mathbb{R}^k}'' K_{N,t}(x_1,\dots,x_k)\widehat{B}_{F_N}(dx_1)\dots\widehat{B}_{F_N}(dx_k),$$

where

$$K_{N,t}(x_1,\dots,x_k) = \frac{e^{i\frac{[Nt]}{N}(x_1+\dots+x_k)}-1}{N(e^{i\frac{1}{N}(x_1+\dots+x_k)}-1)} \tag{5.3.10}$$

and $\widehat{B}_{F_N}$ is an Hermitian Gaussian random measure on $\mathbb{R}$ with a control measure satisfying

$$F_N(A) = \frac{N^{1-2d}}{L_2(N)}F(N^{-1}A), \quad A \in \mathcal{B}(\mathbb{R}). \tag{5.3.11}$$

For the convergence of finite-dimensional distributions, we consider a linear combination

$$\sum_{j=1}^{J}\theta_j S_{N,k}(t_j) = \int_{\mathbb{R}^k}'' k_N(x_1,\dots,x_k)\widehat{B}_{F_N}(dx_1)\dots\widehat{B}_{F_N}(dx_k),$$

where $\theta_j \in \mathbb{R}$, $t_j \geq 0$, $j = 1,\dots,J$,

$$k_N(x_1,\dots,x_k) = \sum_{j=1}^{J}\theta_j K_{N,t_j}(x_1,\dots,x_k).$$

Step 2 (goal): We want to conclude that

$$\sum_{j=1}^{J}\theta_j S_{N,k}(t_j) \xrightarrow{d} \int_{\mathbb{R}^k}'' k_0(x_1,\dots,x_k)\widehat{B}_{F_0}(dx_1)\dots\widehat{B}_{F_0}(dx_k), \tag{5.3.12}$$

where

$$k_0(x_1,\dots,x_k) = \sum_{j=1}^{J}\theta_j K_{t_j}(x_1,\dots,x_k), \quad K_t(x_1,\dots,x_k) = \frac{e^{it(x_1+\dots+x_k)}-1}{i(x_1+\dots+x_k)} \tag{5.3.13}$$

and $\widehat{B}_{F_0}$ is an Hermitian Gaussian random measure with a control measure

$$\mathbb{E}|\widehat{B}_{F_0}(dx)|^2 = \frac{|x|^{-2d}dx}{2\sin(d\pi)\Gamma(1-2d)} =: F_0(dx).$$

Since the Hermite process is normalized, namely, $\mathbb{E}(Z_H^{(k)}(1))^2 = 1$, this would yield the desired convergence (5.3.5).

Step 3 (method): To show (5.3.12), we shall apply Proposition 5.3.6 below with F_N and F_0 as defined above, and

$$\widehat{K}_N(x_1,\dots,x_k) = k_N(x_1,\dots,x_k), \quad \widehat{K}_0(x_1,\dots,x_k) = k_0(x_1,\dots,x_k).$$

We now verify the assumptions of that proposition.

(i) The sequence F_N converges vaguely[1] to F_0 by Proposition 5.3.4.

(ii) The measure F_N is without atoms using Theorem 9.6 in Zygmund [1030], Chapter III, Section 9.

(iii) The sequence $k_N(x_1, \dots, x_k)$ converges to $k_0(x_1, \dots, x_k)$ uniformly in any rectangle $[-A, A]^k$ since, for fixed t, $K_{N,t}(x_1, \dots, x_k)$ converges to $K_t(x_1, \dots, x_k)$ in (5.3.13) uniformly in any rectangle $[-A, A]^k$. For example, if $t = 1$, the latter can be seen from

$$\begin{aligned}
|K_{N,1}(x_1, \dots, x_k) - K_1(x_1, \dots, x_k)| &= \Big|\frac{e^{i(x_1+\dots+x_k)} - 1}{N(e^{i\frac{1}{N}(x_1+\dots+x_k)} - 1)} - \frac{e^{i(x_1+\dots+x_k)} - 1}{i(x_1 + \dots + x_k)}\Big| \\
&= |e^{i(x_1+\dots+x_k)} - 1|\Big|\frac{N(e^{i\frac{1}{N}(x_1+\dots+x_k)} - 1) - i(x_1 + \dots + x_k)}{N(e^{i\frac{1}{N}(x_1+\dots+x_k)} - 1)(x_1 + \dots + x_k)}\Big| \\
&\le C_1|e^{i(x_1+\dots+x_k)} - 1|\frac{|\frac{1}{N}(x_1 + \dots + x_k)^2|}{|N\frac{1}{N}(x_1 + \dots + x_k)||x_1 + \dots + x_k|} \le \frac{C_2}{N},
\end{aligned}$$

for all $(x_1, \dots, x_k) \in [-A, A]^k$, since $(x_1 + \dots + x_k)/N$ is arbitrarily small for all $(x_1, \dots, x_k) \in [-A, A]^k$.

(iv) It remains to check that the condition (5.3.26) below holds. Let

$$\begin{aligned}
\mu_N(dx_1, \dots, dx_k) &= |k_N(x_1, \dots, x_k)|^2 F_N(dx_1) \dots F_N(dx_k), \\
\mu_0(dx_1, \dots, dx_k) &= |k_0(x_1, \dots, x_k)|^2 F_0(dx_1) \dots F_0(dx_k)
\end{aligned}$$

be measures on $\mathcal{B}(\mathbb{R}^k)$. The measures μ_N are finite and concentrated on $(-N\pi, N\pi]^k$. Since F_N converge vaguely to F_0 and k_N converge to k_0 uniformly on compact intervals, the measures μ_N converge vaguely to μ_0. In view of (A.3.5) in Appendix A.3, to show (5.3.26), it is enough to prove that μ_N converges not only vaguely but also weakly (with the limit being μ_0). To do so, we shall apply Lemma A.3.1 in Appendix A.3 with $\ell = k$ and $C_N = N$. We need to show that

$$\phi_N(s_1, \dots, s_k) = \int_{\mathbb{R}^k} e^{i(\frac{[Ns_1]}{N}x_1+\dots+\frac{[Ns_k]}{N}x_k)}|k_N(x_1, \dots, x_k)|^2 F_N(dx_1) \dots F_N(dx_k)$$

tends to a continuous limit as $N \to \infty$. But by (5.3.10),

$$\begin{aligned}
\phi_N(s_1, \dots, s_k) = &\sum_{j_1,j_2=1}^{J} \theta_{j_1}\theta_{j_2} \int_{\mathbb{R}^k} e^{i(\frac{[Ns_1]}{N}x_1+\dots+\frac{[Ns_k]}{N}x_k)} \\
&\times K_{N,t_{j_1}}(x_1, \dots, x_k)\overline{K_{N,t_{j_2}}(x_1, \dots, x_k)} F_N(dx_1) \dots F_N(dx_k) \\
= &\sum_{j_1,j_2=1}^{J} \theta_{j_1}\theta_{j_2} \frac{1}{N^2} \sum_{n_1=0}^{[Nt_{j_1}]-1} \sum_{n_2=0}^{[Nt_{j_2}]-1} \int_{\mathbb{R}^k} e^{i(\frac{[Ns_1]}{N}x_1+\dots+\frac{[Ns_k]}{N}x_k)} \\
&\times\ e^{i\frac{n_1}{N}(x_1+\dots+x_k)}e^{-i\frac{n_2}{N}(x_1+\dots+x_k)} F_N(dx_1) \dots F_N(dx_k).
\end{aligned}$$

By (5.3.11),

[1] The notions of vague and weak convergence are discussed in Appendix A.3.

$$\phi_N(s_1,\dots,s_k) = \sum_{j_1,j_2=1}^{J} \theta_{j_1}\theta_{j_2}\frac{1}{N^2}\sum_{n_1=0}^{[Nt_{j_1}]-1}\sum_{n_2=0}^{[Nt_{j_2}]-1}\int_{\mathbb{R}^k} e^{i([Ns_1]+n_1-n_2)\lambda_1}\dots e^{i([Ns_k]+n_1-n_2)\lambda_k}$$
$$\times \frac{1}{(L_2(N))^k N^{k(2d-1)}} F(d\lambda_1)\dots F(d\lambda_k)$$
$$= \sum_{j_1,j_2=1}^{J} \theta_{j_1}\theta_{j_2}\frac{1}{N^2}\sum_{n_1=0}^{[Nt_{j_1}]-1}\sum_{n_2=0}^{[Nt_{j_2}]-1}\frac{\gamma_X([Ns_1]+n_1-n_2)}{L_2(N)N^{2d-1}}\dots\frac{\gamma_X([Ns_k]+n_1-n_2)}{L_2(N)N^{2d-1}},$$

where $\gamma_X(\cdot)$ is the autocovariance (5.3.9) given in (2.1.5). One can show (Exercise 5.4, (*b*)) that, for each $s_1,\dots,s_k\in\mathbb{R}$,

$$\phi_N(s_1,\dots,s_k) \to \sum_{j_1,j_2=1}^{J}\theta_{j_1}\theta_{j_2}\int_0^{t_{j_1}} dx_1\int_0^{t_{j_2}} dx_2\, |s_1+x_1-x_2|^{2d-1}\dots|s_k+x_1-x_2|^{2d-1}. \tag{5.3.14}$$

The limit function is continuous (Exercise 5.4, (*b*)), and Lemma A.3.1 applies. □

The next results were used in the proof of Theorem 5.3.1 above. The first result allowed to replace G in the partial sum process by an Hermite polynomial.

Theorem 5.3.3 (*Reduction theorem.*) *In the setting and under the assumptions of Theorem 5.3.1, let g_k be the first nonzero coefficient in the Hermite expansion of G and $H_k(x)$ be the Hermite polynomial of order k. If the convergence (5.3.5) holds for $G(x) = g_k H_k(x)$, then it also holds for the general function G with Hermite rank k.*

Proof Since G has Hermite rank k, we have

$$G(x) = g_k H_k(x) + \sum_{j=k+1}^{\infty} g_j H_j(x) =: g_k H_k(x) + G^*(x).$$

Note that, by Corollary 5.2.5,

$$\mathrm{Var}\Big(\sum_{n=1}^{[Nt]} G^*(X_n)\Big) \sim C(L_2([Nt]))^{k+1}[Nt]^{(k+1)(2d-1)+2},$$

as $N\to\infty$. This implies that

$$\frac{1}{(L_2(N))^{k/2}N^{k(d-1/2)+1}}\sum_{n=1}^{[Nt]} G^*(X_n) \xrightarrow{p} 0,$$

for each t. The result now follows from Billingsley [154], Theorem 3.1. □

The next result shows that a properly normalized spectral measure of the series $\{X_n\}$ converges vaguely. The notion of vague convergence is discussed in Appendix A.3.

Proposition 5.3.4 *Let $\{X_n\}_{n\in\mathbb{Z}}$ by a stationary series which is LRD in the sense (2.1.5) in condition II with $d\in(0,1/2)$. Let F be the spectral measure of $\{X_n\}_{n\in\mathbb{Z}}$, and define*

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$$F_N(A) = \frac{N^{1-2d}}{L_2(N)} F(N^{-1}A), \quad A \in \mathcal{B}(\mathbb{R}),\ N \geq 1. \tag{5.3.15}$$

Then,

$$F_N \xrightarrow{v} F_0 \tag{5.3.16}$$

in the sense of vague convergence, where

$$\frac{F_0(dx)}{dx} = C|x|^{-2d} \tag{5.3.17}$$

and

$$C^{-1} = 2\sin(d\pi)\,\Gamma(1-2d). \tag{5.3.18}$$

Proof $a)$[2] Consider a measure μ_N defined by

$$\mu_N(dx) = |K_N(x)|^2 F_N(dx),$$

where

$$K_N(x) = \frac{1}{N}\sum_{j=0}^{N-1} e^{i\frac{j}{N}x} = \begin{cases} \frac{e^{ix}-1}{N(e^{i\frac{x}{N}}-1)}, & \text{if } \frac{x}{2\pi N} \notin \mathbb{Z}, \\ 1, & \text{if } \frac{x}{2\pi N} \in \mathbb{Z}. \end{cases}$$

We now want to apply Lemma A.3.1 in Appendix A.3 to show that the measure μ_N converges weakly. We will use this fact to conclude the convergence of F_N in (5.3.16).

Take $\ell = 1$ and $C_N = N$ in Lemma A.3.1. Note that the function ϕ_N in (A.3.6) can be expressed as

$$\begin{aligned}
\phi_N(t) &= \int_{\mathbb{R}} e^{i\frac{[Nt]}{N}x}\Big|\frac{1}{N}\sum_{j=0}^{N-1} e^{i\frac{j}{N}x}\Big|^2 \frac{N^{1-2d}}{L_2(N)} F\Big(d\frac{x}{N}\Big) = \int_{\mathbb{R}} e^{i[Nt]\lambda}\Big|\frac{1}{N}\sum_{j=0}^{N-1} e^{ij\lambda}\Big|^2 \frac{N^{1-2d}}{L_2(N)} F(d\lambda) \\
&= \frac{1}{N^2N^{2d-1}L_2(N)} \sum_{j_1,j_2=0}^{N-1} \int_{\mathbb{R}} e^{i([Nt]+j_1-j_2)\lambda} F(d\lambda) = \frac{1}{N^2N^{2d-1}L_2(N)} \sum_{j_1,j_2=0}^{N-1} \gamma_X([Nt]+j_1-j_2) \\
&= \frac{1}{N^2N^{2d-1}L_2(N)} \sum_{j=-(N-1)}^{N-1} (N-|j|)\gamma_X([Nt]+j) = \frac{1}{N}\sum_{j=-(N-1)}^{N-1}\Big(1-\frac{|j|}{N}\Big)\frac{\gamma_X([Nt]+j)}{N^{2d-1}L_2(N)},
\end{aligned}$$

where $\gamma_X(n) = \int_{\mathbb{R}} e^{in\lambda}F(d\lambda)$ is the autocovariance function. One can show (Exercise 5.4, (a)) that, for every t, as $N \to \infty$,

$$\phi_N(t) \to \int_{-1}^{1} (1-|x|)|t+x|^{2d-1}dx =: \phi(t). \tag{5.3.19}$$

We have $\phi(t) = \phi(-t)$, $t \in \mathbb{R}$. One can show (Exercise 5.4, (a)) that, for $t > 0$,

$$\phi(t) = \frac{1}{2d(2d+1)}\Big\{(t+1)^{2d+1} + |t-1|^{2d+1} - 2t^{2d+1}\Big\}. \tag{5.3.20}$$

Since the function $\phi(t)$ is continuous at $t = 0$, Lemma A.3.1 shows that

$$\mu_N \xrightarrow{w} \mu_0,$$

[2] The proof in this first part is similar to the end of the proof of Theorem 5.3.1.

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where μ_0 is a finite measure having $\phi(t)$ for its Fourier transform:

$$\phi(t) = \int_{\mathbb{R}} e^{itx} \mu_0(dx). \tag{5.3.21}$$

b) We show next that F_N also converges locally weakly. For $B \in \mathcal{B}(\mathbb{R})$, note that

$$F_N(B) = \int_B |K_N(x)|^{-2} \mu_N(dx).$$

The functions $|K_N(x)|^2$ converge uniformly on compact intervals to $|K_0(x)|^2$, where

$$K_0(x) = \begin{cases} \frac{e^{ix}-1}{ix}, & \text{if } x \neq 0, \\ 1, & \text{if } x = 0. \end{cases}$$

Since $|K_0(x)|^2$ is continuous and non-vanishing in $[-\pi/2, \pi/2]$, the weak convergence of μ_N to μ_0 implies that, for every $B \in \mathcal{B}([-\pi/2, \pi/2])$ such that $\mu_0(\partial B) = 0$,

$$\lim_{N\to\infty} F_N(B) = \int_B |K_0(x)|^{-2} \mu_0(dx) =: F_0(B). \tag{5.3.22}$$

This defines a finite measure F_0 on $\mathcal{B}([-\pi/2, \pi/2])$.

c) We want to conclude next that, for $B_1, B_2 \in \mathcal{B}([-\pi/2, \pi/2])$ satisfying $B_1 = tB_2 = \{tx : x \in B_2\}$ with some $t > 0$,

$$F_0(B_1) = F_0(tB_2) = t^{1-2d} F_0(B_2). \tag{5.3.23}$$

Observe that

$$\begin{aligned} F_N(tB) &= \frac{N^{1-2d}}{L_2(N)} F(N^{-1}tB) = \frac{N^{1-2d}}{\left[\frac{N}{t}\right]^{1-2d}} \frac{L_2\left(\left[\frac{N}{t}\right]\right)}{L_2(N)} \frac{\left[\frac{N}{t}\right]^{1-2d}}{L_2\left(\left[\frac{N}{t}\right]\right)} F\left(\left[\frac{N}{t}\right]^{-1} \frac{\left[\frac{N}{t}\right]}{\frac{N}{t}} B\right) \\ &= \frac{N^{1-2d}}{\left[\frac{N}{t}\right]^{1-2d}} \frac{L_2\left(\left[\frac{N}{t}\right]\right)}{L_2(N)} F_{\left[\frac{N}{t}\right]}\left(\frac{\left[\frac{N}{t}\right]}{\frac{N}{t}} B\right). \end{aligned}$$

For any $t > 0$ and $B \in \mathcal{B}([-\pi/2, \pi/2])$ such that $F_0(\partial B) = 0$ and $F_0(\partial(tB)) = 0$, this implies by (5.3.22) that

$$F_0(tB) = \lim_{N\to\infty} F_N(tB) = t^{1-2d} F_0(B). \tag{5.3.24}$$

This proves (5.3.23) when $F_0(\partial B_1) = F_0(\partial B_2) = 0$. We now want to remove these restrictions. In particular, (5.3.23) holds with $B_2 = (a, b)$ for $-\pi/2 \leq a, at < 0 < b, bt \leq \pi/2$ as long as $F_0(\{a\}) = F_0(\{at\}) = F_0(\{b\}) = F_0(\{bt\}) = 0$. Since the set $\{x \in [-\pi/2, \pi/2] : F_0(\{x\}) > 0\}$ is at most countable, the relation (5.3.23) holds for intervals $B_2 = (a, b)$ for $-\pi/2 \leq a, at < 0 < b, bt \leq \pi/2$, except perhaps for a countable number of a, b and t. For any fixed t, a and b satisfying $-\pi/2 \leq a, at < 0 < b, bt \leq \pi/2$, we can select $a_n \downarrow a$, $b_n \uparrow b$ and $t_n \uparrow t$ so that the relation (5.3.23) holds with $B_2 = (a_n, b_n)$ and $B_1 = (a_n t_n, b_n t_n)$. Since F_0 is a measure, and $(a_n, b_n) \uparrow (a, b)$ and $(a_n t_n, b_n t_n) \uparrow (at, bt)$, passing in the limit yields (5.3.23) with $B_2 = (a, b)$ for all

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$-\pi/2 \le a, at < 0 < b, bt \le \pi/2$. Since intervals generate Borel sets, this shows (5.3.23) without the restrictions $F_0(\partial B_1) = F_0(\partial B_2) = 0$.

d) F_0 was defined on $\mathcal{B}([-\pi/2, \pi/2])$. We now want to define it for every bounded set $B \in \mathcal{B}(\mathbb{R})$. We do it as follows. If $B \in \mathcal{B}(\mathbb{R})$ satisfies $B \subset [-K\pi/2, K\pi/2]$, set

$$F_0(B) = K^{1-2d} F_0\Big(\frac{1}{K}B\Big).$$

This definition does not depend on K because of (5.3.23). It coincides with the definition when $B \in \mathcal{B}([-\pi/2, \pi/2])$. The set measure F_0 can be extended to a locally finite measure F_0 by setting, for example,

$$F_0(B) = \lim_{K\to\infty} F_0(B \cap [-K, K]).$$

The relation holds for all bounded $B \in \mathcal{B}(\mathbb{R})$ and, taking $t = 1$ in (5.3.24), this implies that F_N tends locally weakly to F_0.

e) The relation (5.3.23) holds for all $B_1, B_2 \in \mathcal{B}(\mathbb{R})$ and hence, for all $t > 0$ and $B \in \mathcal{B}(\mathbb{R})$,

$$F_0(tB) = t^{1-2d} F_0(B).$$

In particular, by taking $B = [0, 1]$,

$$F_0([0, t]) = t^{1-2d} F_0([0, 1]).$$

Since $1 - 2d > 0$, this shows that the measure F_0 is absolutely continuous with respect to the Lebesgue measure on $\mathcal{B}([0, \infty))$, and that

$$\frac{F_0(dx)}{dx} = Cx^{-2d}, \quad x \ge 0,$$

for some constant C. A similar argument yields an analogous relation for $x < 0$. The constant C is the same since F_N and hence F_0 are symmetric. This yields (5.3.17).

f) Finally, for the constant C in (5.3.18), we have from (5.3.21) and (5.3.22) that

$$\begin{aligned}\phi(t) &= \int_{\mathbb{R}} e^{itx}\mu_0(dx) = \int_{\mathbb{R}} e^{itx}|K_0(x)|^2 F_0(dx) = C\int_{\mathbb{R}} e^{itx}|K_0(x)|^2|x|^{-2d}dx\\ &= 2C\int_{\mathbb{R}} e^{itx}\frac{1-\cos x}{x^2}|x|^{-2d}dx = 4C\int_0^\infty \cos(tx)(1-\cos x)x^{-2d-2}dx =: 4C\psi_d(t).\end{aligned}$$

If $0 < -2d-1 < 1$ or $-1 < d < -1/2$, $\psi_d(t)$ can be evaluated for $t > 0$ by using Formula 3.762.3 in Gradshteyn and Ryzhik [421]:

$$\begin{aligned}\psi_d(t) &= \int_0^\infty \cos(tx)x^{-2d-2}dx - \int_0^\infty \cos(tx)\cos(x)x^{-2d-2}dx\\ &= \cos\Big(\frac{(2d+1)\pi}{2}\Big)\Gamma(-2d-1)t^{2d+1}\\ &\quad - \cos\Big(\frac{(2d+1)\pi}{2}\Big)\Gamma(-2d-1)\frac{1}{2}\Big\{(t+1)^{2d+1} + |t-1|^{2d+1}\Big\}\\ &= \frac{1}{2}\sin(d\pi)\Gamma(-2d-1)\Big\{(t+1)^{2d+1} + |t-1|^{2d+1} - 2t^{2d+1}\Big\} =: \tilde{\psi}_d(t).\end{aligned} \tag{5.3.25}$$

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On the other hand, for fixed $t > 0$, the functions $\psi_d(t)$ and $\tilde{\psi}_d(t)$ are analytic for complex-valued d satisfying $-2 < \Re(-2d-1) < 1$ or $-1 < \Re(d) < 1/2$, and $\Re(-2d-1) \neq 0, -1$ or $\Re(d) \neq -1/2, 0$. Since they coincide for $-1 < d < -1/2$, they must coincide for $0 < d < 1/2$. By comparing (5.3.25) and (5.3.20), and using the relation (2.2.13), we deduce that

$$\frac{1}{2d(2d+1)}\Big\{(t+1)^{2d+1} + |t-1|^{2d+1} - 2t^{2d+1}\Big\}$$
$$= 2C\sin(d\pi)\Gamma(-2d-1)\Big\{(t+1)^{2d+1} + |t-1|^{2d+1} - 2t^{2d+1}\Big\}$$

or

$$C = \frac{1}{2\sin(d\pi)(-2d)(-2d-1)\Gamma(-2d-1)} = \frac{1}{2\sin(d\pi)\Gamma(1-2d)}. \qquad \square$$

The next proposition was used to show the convergence of multiple integrals. Its proof is based on the following classical lemma (Theorem 3.2, Billingsley [154]).

Lemma 5.3.5 *Suppose that:*

(1) For each M, $X_{N,M} \xrightarrow{d} X_M$ as $N \to \infty$,

(2) $X_M \xrightarrow{d} X$ as $M \to \infty$,

(3) $\limsup_M \limsup_N \mathbb{P}(|X_{N,M} - Y_N| > \epsilon) = 0$, for all $\epsilon > 0$.

Then, $Y_N \xrightarrow{d} X$ as $N \to \infty$.

Finally, the next proposition was used in the proof of Theorem 5.3.1.

Proposition 5.3.6 *Let F_N, $N \geq 1$, F_0 be symmetric, locally finite measures without atoms on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ so that $F_N \xrightarrow{v} F_0$ in the sense of vague convergence. Let W_{F_N} and W_{F_0} be Hermitian Gaussian random measures on $\mathbb{R}$ with control measures F_N and F_0, respectively. Let also $\widehat{K}_N(x_1, \ldots, x_k)$ be a sequence of Hermitian, measurable functions on $\mathbb{R}^k$ tending to a continuous function $\widehat{K}_0(x_1, \ldots, x_k)$ uniformly in any rectangle $[-A, A]^k$, $A > 0$. Moreover, suppose that $\widehat{K}_0$ and $\widehat{K}_N$ satisfy the relation*

$$\lim_{A\to\infty} \sup_{N\geq 0} \int_{\mathbb{R}^k \setminus [-A,A]^k} |\widehat{K}_N(x_1, \ldots, x_k)|^2 F_N(dx_1) \ldots F_N(dx_k) = 0. \tag{5.3.26}$$

Then,

$$\int''_{\mathbb{R}^k} \widehat{K}_N(x_1, \ldots, x_k) W_{F_N}(dx_1) \ldots W_{F_N}(dx_k) \xrightarrow{d} \int''_{\mathbb{R}^k} \widehat{K}_0(x_1, \ldots, x_k) W_{F_0}(dx_1) \ldots W_{F_0}(dx_k), \tag{5.3.27}$$

where the last integral exists.

Proof Let $\widehat{H}^k$ consist of functions g which are Hermitian symmetric in the sense of (B.2.12) in Appendix B.2 and have a representation (B.2.13) with bounded sets A_j. Let $\overline{H}^k_{F_0}$ consist of those functions in $\widehat{H}^k$ for which $F_0(\partial A_j) = 0$, $\pm j = 1, \ldots, n$, where A_j are the sets in their representation (B.2.13). Since $F_N \xrightarrow{v} F_0$ by assumption and $F_0(\partial A_j) = 0$, $\pm j = 1, \ldots, n$, we have (Appendix A.3) that $F_N(A_j) \to F_0(A_j)$, $\pm j = 1, \ldots, n$. This implies that a Gaussian vector $(W_{F_N}(A_j), \pm j = 1, \ldots, n)$

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converges in distribution to a Gaussian vector $(W_{F_0}(A_j), \pm j = 1, \dots, n)$. For $g \in \overline{H}^k_{F_0}$, $\int''_{\mathbb{R}^k} g(x_1, \dots, x_k) W_F(dx_1) \dots W_F(dx_k)$ is a polynomial of $W_F(A_j)$s (where $F = F_0$ or F_N). Since the $W_{F_N}(A_j)$s converge in distribution to $W_F(A_j)$s , it follows that

$$\int''_{\mathbb{R}^k} g(x_1, \dots, x_k) W_{F_N}(dx_1) \dots W_{F_N}(dx_k) \xrightarrow{d} \int''_{\mathbb{R}^k} g(x_1, \dots, x_k) W_{F_0}(dx_1) \dots W_{F_0}(dx_k). \tag{5.3.28}$$

To show (5.3.27), it is enough to prove that, for any $\epsilon > 0$, there is $g \in \overline{H}^k_{F_0}$ such that

$$\int_{\mathbb{R}^k} |\widehat{K}_N(x_1, \dots, x_k) - g(x_1, \dots, x_k)|^2 F_N(dx_1) \dots F_N(dx_k) < \epsilon, \tag{5.3.29}$$

for $N = 0$ and $N > N(\epsilon)$. Indeed, denote a multiple integral with respect to W_{F_N} by $\widehat{I}_{F_N}(\cdot)$. Note that the integral $\widehat{I}_{F_0}(\widehat{K}_0)$ is well-defined by using (5.3.26) for $N = 0$ and the fact that $\widehat{K}_0$ is continuous (hence square integrable on bounded sets). By (5.3.29) for $N = 0$, there is a sequence $g_M \in \overline{H}^k_{F_0}$ such that $\widehat{I}_{F_0}(g_M) \to \widehat{I}_{F_0}(\widehat{K}_0)$ in $L^2(\Omega)$ and hence in distribution. In fact, by (5.3.29), the sequence $g_M \in \overline{H}^k_{F_0}$ can be chosen such that

$$\limsup_M \limsup_N \mathbb{E}|\widehat{I}_{F_N}(g_M) - \widehat{I}_{F_N}(\widehat{K}_N)|^2 = 0.$$

By (5.3.28), for each fixed M, $\widehat{I}_{F_N}(g_M) \to \widehat{I}_{F_0}(g_M)$ in distribution. Hence, the convergence $\widehat{I}_{F_N}(\widehat{K}_N) \to \widehat{I}_{F_0}(\widehat{K}_0)$ in (5.3.27) follows from Lemma 5.3.5 with

$$X_{N,M} = \widehat{I}_{F_N}(g_M), \quad X_M = \widehat{I}_{F_0}(g_M), \quad Y_N = \widehat{I}_{F_N}(\widehat{K}_N) \quad \text{and} \quad X = \widehat{I}_{F_0}(\widehat{K}_0).$$

By using (5.3.26), note that (5.3.29) is equivalent to the same relation where the integration over $\mathbb{R}^k$ is replaced by $[-A, A]^k$ (for large enough A). Then for $N \geq 1$, for example, write

$$\int_{[-A,A]^k} |\widehat{K}_N(x_1, \dots, x_k) - g(x_1, \dots, x_k)|^2 F_N(dx_1) \dots F_N(dx_k) \leq 2\Big(F_N([-A, A])\Big)^k$$

$$\times \sup_{(x_1,\dots,x_k)\in[-A,A]^k} \Big(|\widehat{K}_N(x_1, \dots, x_k) - \widehat{K}_0(x_1, \dots, x_k)|^2 + |\widehat{K}_0(x_1, \dots, x_k) - g(x_1, \dots, x_k)|^2\Big). \tag{5.3.30}$$

Since $F_N([-A, A]) \to F_0([-A, A])$ (F_0 has no atoms and $F_0(\partial[-A, A]) = F_0(\{-A, A\}) = 0$) and F_0 is locally finite, we have $\sup_{N\geq 1} F_N([-A, A]) < \infty$. Since $\widehat{K}_N \to \widehat{K}_0$ and $\widehat{K}_0$ is continuous on $[-A, A]^k$, the supremum of the difference $|\widehat{K}_N(x_1, \dots, x_k) - \widehat{K}_0(x_1, \dots, x_k)|$ is arbitrarily small for large N. On the other hand, since $\widehat{K}_0$ is continuous on $[-A, A]^k$, we can choose $g \in \overline{H}^k_{F_0}$ such that the supremum of the difference $|\widehat{K}_0(x_1, \dots, x_k) - g(x_1, \dots, x_k)|$ is arbitrarily small. (To construct such g, one can take a grid of $[-A, A]^k$ and define g to be equal to a value of $\widehat{K}_0$ on the grid rectangle. The function g approximates $\widehat{K}_0$ uniformly as the grid gets finer since $\widehat{K}_0$ is continuous.) This shows that the bound (5.3.30) can be made arbitrarily small for large N and suitable g. The argument for $N = 0$ is analogous. □

5.4 Central Limit Theorem

Non-central limit theorem (Theorem 5.3.1) applies for a Gaussian stationary series exhibiting long-range dependence when $(k-1)/2k < d < 1/2$, where k is the Hermite rank of a transformation of the series. What happens in the case $0 < d < (k-1)/2k$, possible when $k \geq 2$? The following result will show that in this case, the limit of partial sum processes is the usual Brownian motion (see Corollary 5.4.5 below). The result is stated under more general conditions.

Theorem 5.4.1 *Let $\{X_n\}_{n\in\mathbb{Z}}$ be a Gaussian stationary series with autocovariance function γ_X. Suppose that $\mathbb{E}X_n = 0$, $\mathbb{E}X_n^2 = 1$. Let G be a function with Hermite rank $k \geq 1$ in the sense of Definition 5.2.1, and γ_G be the autocovariance function of $\{G(X_n)\}_{n\in\mathbb{Z}}$. If*

$$\sum_{n=1}^{\infty} |\gamma_X(n)|^k < \infty \tag{5.4.1}$$

or, equivalently,

$$\sum_{n=1}^{\infty} |\gamma_G(n)| < \infty \quad \textit{and} \quad \gamma_X(n) \to 0, \ \textit{as } n \to \infty, \tag{5.4.2}$$

then

$$\frac{1}{N^{1/2}} \sum_{n=1}^{[Nt]} (G(X_n) - \mathbb{E}G(X_n)) \xrightarrow{fdd} \sigma B(t), \quad t \geq 0, \tag{5.4.3}$$

where $\{B(t)\}_{t\geq 0}$ is a standard Brownian motion and

$$\sigma^2 = \sum_{m=k}^{\infty} g_m^2 m! \sum_{n=-\infty}^{\infty} (\gamma_X(n))^m, \tag{5.4.4}$$

where g_m are the coefficients in the Hermite expansion (5.1.6) of G.

Proof To see the equivalence of (5.4.1) and (5.4.2), note from (5.2.2) that

$$\gamma_G(n) = \sum_{m=k}^{\infty} g_m^2 m! (\gamma_X(n))^m = (\gamma_X(n))^k \sum_{m=k}^{\infty} g_m^2 m! (\gamma_X(n))^{m-k}.$$

Moreover, $|\gamma_X(n)| \leq 1$ since X_n has variance 1. This shows that

$$\sum_{n=1}^{\infty} |\gamma_G(n)| \leq \sum_{n=1}^{\infty} |\gamma_X(n)|^k \sum_{m=k}^{\infty} g_m^2 m!$$

and hence (5.4.1) implies (5.4.2). Conversely, since $\gamma_X(n) \to 0$ in (5.4.2), we have $|\gamma_G(n)| \geq |\gamma_X(n)|^k g_k^2 k!/2$ for large enough n and hence (5.4.2) implies (5.4.1).

Denote the partial sum process in (5.4.3) by

$$S_{N,t}(G) = \frac{1}{N^{1/2}} \sum_{n=1}^{[Nt]} (G(X_n) - \mathbb{E}G(X_n)),$$

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where

$$G(X_n) - \mathbb{E}G(X_n) = \sum_{m=k}^{\infty} g_m H_m(X_n).$$

By using Proposition 5.1.1, we have

$$\mathbb{E}(S_{N,t}(G))^2 = \sum_{m=k}^{\infty} g_m^2 \mathbb{E}(S_{N,t}(H_m))^2$$

and

$$\mathbb{E}(S_{N,t}(H_m))^2 = \frac{1}{N} \sum_{n_1,n_2=1}^{[Nt]} \mathbb{E}H_m(X_{n_1})H_m(X_{n_2}) = \frac{m!}{N} \sum_{n_1,n_2=1}^{[Nt]} (\gamma_X(n_1 - n_2))^m$$
$$= \frac{m!}{N} \sum_{n=-[Nt]+1}^{[Nt]-1} ([Nt] - |n|)(\gamma_X(n))^m \le tm! \sum_{n=-[Nt]+1}^{[Nt]-1} |\gamma_X(n)|^m \le tm! \sum_{n=-\infty}^{\infty} |\gamma_X(n)|^k, \tag{5.4.5}$$

for $m \ge k$. As $N \to \infty$,

$$\mathbb{E}(S_{N,t}(H_m))^2 = m!t \sum_{n=-[Nt]+1}^{[Nt]-1} \Big(\frac{[Nt]}{Nt} - \frac{|n|}{Nt}\Big)(\gamma_X(n))^m \to m!t \sum_{n=-\infty}^{\infty} (\gamma_X(n))^m$$

and

$$\mathbb{E}(S_{N,t}(G))^2 \to t \sum_{m=k}^{\infty} g_m^2 m! \sum_{n=-\infty}^{\infty} (\gamma_X(n))^m.$$

This shows that the limiting variance exists, is finite and equals to $\sigma^2 t$ with σ^2 given by (5.4.4). Note also that, by using (5.4.5),

$$\mathbb{E}\Big(S_{N,t}(G) - S_{N,t}\Big(\sum_{m=k}^{M} g_m H_m\Big)\Big)^2 = \mathbb{E}\Big(S_{N,t}\Big(\sum_{m=M+1}^{\infty} g_m H_m\Big)\Big)^2$$
$$= \sum_{m=M+1}^{\infty} g_m^2 \mathbb{E}(S_{N,t}(H_m))^2 \le t \sum_{n=-\infty}^{\infty} |\gamma_X(n)|^k \sum_{m=M+1}^{\infty} g_m^2 m!$$

is arbitrarily small for large enough M and all N. It is therefore enough to prove (5.4.3) for $G(x) - \mathbb{E}G(X_n) = \sum_{m=k}^{M} g_m H_m(x)$ for fixed $M < \infty$; that is, when G is a polynomial. The variance σ^2 in (5.4.4) is replaced by $\sigma^2 = \sum_{m=k}^{M} g_m^2 m! \sum_{n=-\infty}^{\infty} (\gamma_X(n))^m$.

For

$$G(x) - \mathbb{E}G(X_n) = \sum_{m=k}^{M} g_m H_m(x), \tag{5.4.6}$$

set

$$S_N(G) = \sum_{j=1}^{J} \theta_j S_{N,t_j}(G),$$

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where $\theta_j \in \mathbb{R}$ and $t_1, \dots, t_J \geq 0$. By the method of moments (e.g., Gut [442], Theorem 8.6 in Chapter 5 and Section 4.10), it is enough to show that, for all integers $p \geq 1$,

$$\mathbb{E}\Big(S_N(G)\Big)^p \to \sigma^p \mathbb{E}\Big(\sum_{j=1}^{J} \theta_j B(t_j)\Big)^p. \tag{5.4.7}$$

This is equivalent to: for $p \geq 1$,

$$\chi_p\Big(S_N(G)\Big) \to \sigma^p \chi_p\Big(\sum_{j=1}^{J} \theta_j B(t_j)\Big), \tag{5.4.8}$$

where χ_p denotes the pth cumulant (see Definition 4.3.7). Since the limit is Gaussian and has zero mean,

$$\chi_p\Big(\sum_{j=1}^{J} \theta_j B(t_j)\Big) = \begin{cases} \mathbb{E}\Big(\sum_{j=1}^{J} \theta_j B(t_j)\Big)^2, & \text{if } p = 2, \\ 0, & \text{if } p = 1, 3, 4, \dots \end{cases} \tag{5.4.9}$$

Note that, by using (5.4.6) and the multilinearity of joint cumulants,[3]

$$\begin{aligned}
\chi_p(S_N(G)) &= \chi_p\Big(\sum_{j=1}^{J} \sum_{m=k}^{M} \theta_j g_m \frac{1}{N^{1/2}} \sum_{n=1}^{[Nt_j]} H_m(X_n)\Big) \\
&= \chi\Big(\sum_{j_1=1}^{J} \sum_{m_1=k}^{M} \theta_{j_1} g_{m_1} \frac{1}{N^{1/2}} \sum_{n_1=1}^{[Nt_{j_1}]} H_{m_1}(X_{n_1}), \dots, \sum_{j_p=1}^{J} \sum_{m_p=k}^{M} \theta_{j_p} g_{m_p} \frac{1}{N^{1/2}} \sum_{n_p=1}^{[Nt_{j_p}]} H_{m_p}(X_{n_p})\Big) \\
&= \sum_{j_1,\dots,j_p=1}^{J} \sum_{m_1,\dots,m_p=k}^{M} \Big(\prod_{\ell=1}^{p} \theta_{j_\ell} g_{m_\ell}\Big) \frac{1}{N^{p/2}} \sum_{n_1=1}^{[Nt_{j_1}]} \cdots \sum_{n_p=1}^{[Nt_{j_p}]} \chi\Big(H_{m_1}(X_{n_1}), \dots, H_{m_p}(X_{n_p})\Big).
\end{aligned}$$

By using (4.3.26), we obtain that

$$\begin{aligned}
\chi_p(S_N(G)) = & \sum_{j_1,\dots,j_p=1}^{J} \sum_{m_1,\dots,m_p=k}^{M} \Big(\prod_{\ell=1}^{p} \theta_{j_\ell} g_{m_\ell}\Big) \frac{1}{N^{p/2}} \sum_{n_1=1}^{[Nt_{j_1}]} \cdots \sum_{n_p=1}^{[Nt_{j_p}]} m_1! \dots m_p! \\
& \times \sum_{A \in \mathcal{A}_c(m_1,\dots,m_p)} g(A) \prod_{w \in E(A)} \gamma_X(n_{d_1(w)} - n_{d_2(w)}) \\
= & \sum_{j_1,\dots,j_p=1}^{J} \sum_{m_1,\dots,m_p=k}^{M} \Big(\prod_{\ell=1}^{p} \theta_{j_\ell} g_{m_\ell} m_\ell!\Big) \\
& \times \sum_{A \in \mathcal{A}_c(m_1,\dots,m_p)} g(A) \frac{1}{N^{p/2}} T_N(t_{j_1}, \dots, t_{j_p}, m_1, \dots, m_p, A),
\end{aligned}$$

[3] $\chi(\sum_{j_1} a_{1,j_1} X_{1,j_1}, \dots, \sum_{j_p} a_{p,j_p} X_{p,j_p}) = \sum_{j_1,\dots,j_p} a_{1,j_1} \dots a_{p,j_p} \chi(X_{1,j_1}, \dots, X_{p,j_p})$. Also $\chi_p(X) = \chi(X, \dots, X)$.

where $E(A)$ is the set of lines (edges) of a multigraph A, $d_1(w) < d_2(w)$ are the two points connected by the line w, and

$$T_N(t_{j_1},\ldots,t_{j_p},m_1,\ldots,m_p,A) = \sum_{n_1=1}^{[Nt_{j_1}]}\cdots\sum_{n_p=1}^{[Nt_{j_p}]}\prod_{w\in E(A)}\gamma_X(n_{d_1(w)}-n_{d_2(w)}). \quad (5.4.10)$$

Since for $p \geq 3$, any connected multigraph $A \in \mathcal{A}_c(m_1,\ldots,m_p)$ is nonstandard (see Definition 5.4.2 below), Lemma 5.4.3 implies that, for $p \geq 3$,

$$T_N(t_{j_1},\ldots,t_{j_p},m_1,\ldots,m_p,A) = o(N^{p/2}),$$

as $N \to \infty$. Thus, for $p \geq 3$, $\chi_p(S_N(G)) \to 0$, as $N \to \infty$. In view of (5.4.9), this establishes (5.4.8) for $p \geq 3$.

For $p = 1$, the convergence (5.4.8) is trivial since $\mathbb{E}S_N(G) = 0$ and hence both sides are zero. For $p = 2$, we need to show that

$$\begin{aligned}\mathbb{E}\Big(S_N(G)\Big)^2 &\to \sigma^2\mathbb{E}\Big(\sum_{j=1}^{J}\theta_j B(t_j)\Big)^2\\ &= \sigma^2\sum_{j_1,j_2=1}^{J}\theta_{j_1}\theta_{j_2}(\mathbb{E}B(t_{j_1})B(t_{j_2})) = \sigma^2\sum_{j_1,j_2=1}^{J}\theta_{j_1}\theta_{j_2}(t_{j_1}\wedge t_{j_2}),\end{aligned} \quad (5.4.11)$$

where $x \wedge y = \min\{x, y\}$. Note that, by using Proposition 5.1.1,

$$\begin{aligned}\mathbb{E}\Big(S_N(G)\Big)^2 &= \mathbb{E}\Big(\sum_{j=1}^{J}\sum_{m=k}^{M}\theta_j g_m\frac{1}{N^{1/2}}\sum_{n=1}^{[Nt_j]}H_m(X_n)\Big)^2\\ &= \sum_{j_1,j_2=1}^{J}\sum_{m_1,m_2=k}^{M}\theta_{j_1}\theta_{j_2}g_{m_1}g_{m_2}\frac{1}{N}\sum_{n_1=1}^{[Nt_{j_1}]}\sum_{n_2=1}^{[Nt_{j_2}]}\Big(\mathbb{E}H_{m_1}(X_{n_1})H_{m_2}(X_{n_2})\Big)\\ &= \sum_{j_1,j_2=1}^{J}\sum_{m=k}^{M}\theta_{j_1}\theta_{j_2}g_m^2 m!\frac{1}{N}\sum_{n_1=1}^{[Nt_{j_1}]}\sum_{n_2=1}^{[Nt_{j_2}]}(\gamma_X(n_1-n_2))^m\\ &= \sum_{j_1,j_2=1}^{J}\sum_{m=k}^{M}\theta_{j_1}\theta_{j_2}g_m^2 m!\Big(S_{N,1}+S_{N,2}\Big),\end{aligned}$$

where, with $a = t_{j_1}\wedge t_{j_2}$ and $b = t_{j_1}\vee t_{j_2} = \max\{t_{j_1}, t_{j_2}\}$,

$$S_{N,1} = \frac{1}{N}\sum_{n_1=1}^{[Na]}\sum_{n_2=1}^{[Na]}(\gamma_X(n_1-n_2))^m, \quad S_{N,2} = \frac{1}{N}\sum_{n_1=1}^{[Na]}\sum_{n_2=[Na]+1}^{[Nb]}(\gamma_X(n_1-n_2))^m.$$

We have

$$S_{N,1} = a\sum_{n=-[Na]}^{[Na]}\Big(\frac{[Na]}{Na}-\frac{|n|}{Na}\Big)(\gamma_X(n))^m \to a\sum_{n=-\infty}^{\infty}(\gamma_X(n))^m = (t_{j_1}\wedge t_{j_2})\sum_{n=-\infty}^{\infty}(\gamma_X(n))^m.$$

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By Lemma 5.4.4 below, $S_{N,2} \to 0$. It follows that

$$\mathbb{E}\Big(S_N(G)\Big)^2 \to \sum_{j_1,j_2=1}^{J} \sum_{m=k}^{M} \theta_{j_1}\theta_{j_2} g_m^2 m!(t_{j_1} \wedge t_{j_2}) \sum_{n=-\infty}^{\infty} (\gamma_X(n))^m = \sigma^2 \sum_{j_1,j_2=1}^{J} \theta_{j_1}\theta_{j_2}(t_{j_1} \wedge t_{j_2})$$

and hence that (5.4.11) holds. □

The following result was used in the proof of Theorem 5.4.1 above. It will be stated here under more general conditions. It involves another notion related to multigraphs considered in Section 4.3.2.

Definition 5.4.2 A multigraph will be called *standard* if its points can be paired in such a way that no line joins points in different pairs. Otherwise, a multigraph is called *nonstandard*.

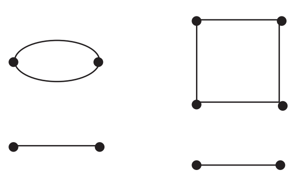
An example of a standard multigraph is given in Figure 5.1, (a). It consists of two pairs of points with no line joining the two pairs. All multigraphs with an odd number of points are nonstandard, as are connected multigraphs when the number of points is greater than two. A non-connected multigraph, however, is not necessarily standard. See Figure 5.1, (b), where the four points connected by lines making a square cannot be separated into two pairs with no line joining the two pairs.

Lemma 5.4.3 *Suppose the assumptions of Theorem 5.4.1. Fix a multigraph $A \in \mathcal{A}(m_1,\dots,m_p)$, $m_1,\dots,m_p \geq k$, and fix $j_1,\dots,j_p \in \{1,\dots,J\}$. Consider the sum $T_N(t_{j_1},\dots,t_{j_p},m_1,\dots,m_p,A)$ in (5.4.10). If A is a nonstandard multigraph, then as $N \to \infty$,*

$$T_N(t_{j_1},\dots,t_{j_p},m_1,\dots,m_p,A) = o(N^{p/2}). \tag{5.4.12}$$

Proof Denote $T_N(t_{j_1},\dots,t_{j_p},m_1,\dots,m_p,A)$ by $T_N(A)$. For simplicity, assume that $t_1 = \dots = t_j = 1$ in (5.4.10). The proof is analogous in the general case. Also, without loss of generality, suppose that $m_1 \leq \cdots \leq m_p$. Note that

$$|T_N(A)| \leq \sum_{n_1,\dots,n_p=1}^{N} \prod_{i=1}^{p} \prod_{\substack{w\in E(A):\\ d_1(w)=i}} |\gamma_X(n_i - n_{d_2(w)})|.$$



(a) (b)

Figure 5.1 (a) A standard multigraph. (b) A non-connected and nonstandard multigraph.

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For $i = 1, \dots, p$, let $k_A(i) = \#\{w \in E(A) : d_1(w) = i\}$ be the number of lines incident to point $d_1(w) = i$ and which join points $d_2(w) > i$. By using the inequality $(a_1 \dots a_k)^{1/k} \le (a_1 + \cdots + a_k)/k$, $k \ge 1$, $a_i \ge 0$, between the geometric and arithmetic means, we have

$$\prod_{\substack{w\in E(A):\\ d_1(w)=i}} |\gamma_X(n_i - n_{d_2(w)})| = \Big(\prod_{\substack{w\in E(A):\\ d_1(w)=i}} |\gamma_X(n_i - n_{d_2(w)})|^{k_A(i)} \Big)^{1/k_A(i)}$$
$$\le \frac{1}{k_A(i)} \sum_{\substack{w\in E(A):\\ d_1(w)=i}} |\gamma_X(n_i - n_{d_2(w)})|^{k_A(i)}.$$

Observe that

$$\sum_{n_i=1}^{N} |\gamma_X(n_i - n_{d_2(w)})|^{k_A(i)} \le \sum_{|n|\le N} |\gamma_X(n)|^{k_A(i)},$$

since $1 \le n_i, n_{d_2(w)} \le N$. Hence,

$$|T_N(A)| \le \sum_{n_1,\dots,n_p=1}^{N} \prod_{i=1}^{p} \frac{1}{k_A(i)} \sum_{\substack{w\in E(A):\\ d_1(w)=i}} |\gamma_X(n_i - n_{d_2(w)})|^{k_A(i)}$$
$$= \prod_{i=1}^{p} \sum_{n_i=1}^{N} \frac{1}{k_A(i)} \sum_{\substack{w\in E(A):\\ d_1(w)=i}} |\gamma_X(n_i - n_{d_2(w)})|^{k_A(i)}$$
$$\le \prod_{i=1}^{p} \frac{1}{k_A(i)} \sum_{\substack{w\in E(A):\\ d_1(w)=i}} \sum_{|n|\le N} |\gamma_X(n)|^{k_A(i)} = \prod_{i=1}^{p} \sum_{|n|\le N} |\gamma_X(n)|^{k_A(i)}. \tag{5.4.13}$$

Let now $h_A(i) = k_A(i)/m_i$. If $k_A(i) = 0$, then $h_A(i) = 0$ and $\sum_{|n|\le N} |\gamma_X(n)|^{k_A(i)} = 2N + 1 \le CN = CN^{1-h_A(i)}$. If $k_A(i) = m_i$, then $h_A(i) = 1$ and $\sum_{|n|\le N} |\gamma_X(n)|^{k_A(i)} = \sum_{|n|\le N} |\gamma_X(n)|^{m_i} \le C = CN^{1-h_A(i)}$, since $|\gamma_X(n)| \le 1$ and $\sum_n |\gamma_X(n)|^{m_i} < \infty$ for $m_i \ge k$. Thus, if $k_A(i) = 0$ or $k_A(i) = m_i$, we have

$$\sum_{|n|\le N} |\gamma_X(n)|^{k_A(i)} \le CN^{1-h_A(i)}. \tag{5.4.14}$$

On the other hand, if $0 < k_A(i) < m_i$, then we want to show that

$$\sum_{|n|\le N} |\gamma_X(n)|^{k_A(i)} = o(N^{1-h_A(i)}). \tag{5.4.15}$$

Indeed, for any fixed $\epsilon > 0$, there is $N(\epsilon)$ such that $\sum_{|n|>N(\epsilon)} |\gamma_X(n)|^{m_i} < \epsilon$. Then, by Hölder's inequality,

$$\sum_{|n|\le N} |\gamma_X(n)|^{k_A(i)} \le \sum_{|n|\le N\wedge N(\epsilon)} |\gamma_X(n)|^{k_A(i)} + \sum_{N(\epsilon)<|n|\le N} |\gamma_X(n)|^{k_A(i)}$$

$$\le C(\epsilon) + \Big(\sum_{N(\epsilon)<|n|\le N} |\gamma_X(n)|^{m_i}\Big)^{\frac{k_A(i)}{m_i}} \Big(\sum_{N(\epsilon)<|n|\le N} 1\Big)^{1-\frac{k_A(i)}{m_i}} \le C(\epsilon) + C\epsilon^{h_A(i)} N^{1-h_A(i)}.$$

The relation (5.4.15) follows since $\epsilon > 0$ is arbitrarily small and $1 - h_A(i) > 0$.

For a nonstandard multigraph A, there is at least one i such that $0 < k_A(i) < m_i$. Combining (5.4.14) and (5.4.15), it follows from (5.4.13) that, for a nonstandard multigraph,

$$|T_N(A)| \le o\Big(N^{p-\sum_{i=1}^p h_A(i)}\Big).$$

It is now enough to show that

$$\sum_{i=1}^p h_A(i) \ge \frac{p}{2}. \tag{5.4.16}$$

Let $p_1(w) = m_{d_1(w)}$ and $p_2(w) = m_{d_2(w)}$. They are the degrees of the points $d_1(w)$ and $d_2(w)$ joined by the line w. Since it is assumed in (5.4.10) that $d_1(w) \le d_2(w)$ and since we assumed here that $m_1 \le \cdots \le m_p$, we have $p_1(w) \le p_2(w)$. Then,

$$2\sum_{i=1}^p h_A(i) = 2\sum_{i=1}^p \frac{k_A(i)}{m_i} = 2\sum_{i=1}^p \sum_{\substack{w\in E(A):\\ d_1(w)=i}} \frac{1}{m_i} = 2\sum_{i=1}^p \sum_{\substack{w\in E(A):\\ d_1(w)=i}} \frac{1}{m_{d_1(w)}}$$

$$= 2\sum_{i=1}^p \sum_{\substack{w\in E(A):\\ d_1(w)=i}} \frac{1}{p_1(w)} = 2\sum_{w\in E(A)} \frac{1}{p_1(w)} \ge \sum_{w\in E(A)} \Big(\frac{1}{p_1(w)} + \frac{1}{p_2(w)}\Big).$$

For fixed i, the number of $w \in E(A)$ for which $d_1(w) = i$ or $d_2(w) = i$ (equivalently, $p_1(w) = m_i$ or $p_2(w) = m_i$) is exactly m_i. This implies that

$$\sum_{w\in E(A)} \Big(\frac{1}{p_1(w)} + \frac{1}{p_2(w)}\Big) = p$$

and hence the bound (5.4.16) follows. □

The following auxiliary lemma was also used above. Its proof is left as Exercise 5.5.

Lemma 5.4.4 *Let* $0 < a < b < \infty$ *and* $\sum_{n=1}^\infty |c(n)| < \infty$ *. Then, as* $N \to \infty$*,*

$$\frac{1}{N}\sum_{n_1=1}^{[Na]} \sum_{n_2=[Na]+1}^{[Nb]} c(n_2 - n_1) \to 0. \tag{5.4.17}$$

The following result addresses the question raised in the beginning of this section.

Corollary 5.4.5 *Let $\{X_n\}_{n\in\mathbb{Z}}$ be a Gaussian stationary series which is LRD in the sense (2.1.5) in condition II with $d \in (0, 1/2)$. Suppose that $\mathbb{E}X_n = 0$, $\mathbb{E}X_n^2 = 1$. Let G be a function with Hermite rank $k \geq 1$ in the sense of Definition 5.2.1. If*

$$d \in \left(0, \frac{1}{2}\left(1 - \frac{1}{k}\right)\right), \tag{5.4.18}$$

then the convergence (5.4.3) to Brownian motion holds.

Proof The result follows from Theorem 5.4.1 by observing that the conditions (5.4.2) hold. Indeed, the second condition in (5.4.2) holds by condition II of LRD in (2.1.5). For the first condition in (5.4.2), Proposition 5.2.4 implies that

$$\gamma_G(n) \sim L(n)n^{k(2d-1)},$$

where L is a slowly varying function at infinity. Then, $\sum_{n=1}^{\infty} |\gamma_G(n)| < \infty$ if $k(2d-1)+1 < 0$ or $d < (1 - 1/k)/2$, which follows from the assumption (5.4.18). □

5.5 The Fourth Moment Condition

Nualart and Peccati [771] have obtained the following remarkable result: when trying to prove a central limit theorem for a multiple Wiener–Itô integral to converge to a Gaussian distribution, it is not necessary to prove convergence of all the moments to those of the Gaussian but only to prove convergence of moments (or equivalently, cumulants) up to order four. This then guarantees convergence of all the higher moments (or cumulants). There are also additional equivalent conditions involving convergence of contractions[4] $\|f_N \otimes_r f_N\|$, which are often easier to use in practice than moments.

Theorem 5.5.1 *Let $\xi_N = I_k(f_N)$ with symmetric $f_N \in L^2(E^k, m^k)$, $k \geq 2$. Suppose that*

$$\lim_{N\to\infty} k!\|f_N\|^2_{L^2(E^k,m^k)} = \lim_{N\to\infty} \mathbb{E}\xi_N^2 = 1. \tag{5.5.1}$$

Then the following conditions are equivalent:

1. $\lim_{N\to\infty} \chi_4(I_k(f_N)) = 0$.
2. *For every $r = 1, \ldots, k-1$,*

$$\lim_{N\to\infty} \|f_N \otimes_r f_N\|_{L^2(E^{2(k-r)},m^{2(k-r)})} = 0.$$

3. *As $N \to \infty$, the sequence $\{I_k(f_N),\ N \geq 1\}$ converges in distribution towards $\mathcal{N}(0, 1)$.*

For a proof, see Peccati and Taqqu [797], p. 182. The condition (5.5.1) standardizes the random variables. An application to a special case of Theorem 5.4.1 can be found in Nourdin [760]. For a proof of Theorem 5.4.1 using Theorem 5.5.1, see Nourdin and Peccati [763].

[4] Contractions are defined in Appendix B.2. See the relation (B.2.17).

5.6 Limit Theorems in the Linear Case

In Sections 5.3 and 5.4, we presented limit theorems for partial sums of nonlinear functions of Gaussian stationary series with long-range dependence. We consider here extensions of the results to nonlinear functions of linear time series with long-range dependence. Linear time series with long-range dependence will be defined as in condition I of Section 2.1 with i.i.d. innovations $\{\mathbb{Z}_n\}$ and supposing $\mu = 0$. We thus consider:

Condition L The time series $\{X_n\}_{n\in\mathbb{Z}}$ has a linear representation

$$X_n = \sum_{k=0}^{\infty} \psi_k \epsilon_{n-k}$$

with i.i.d. innovations ϵ_n, $n \in \mathbb{Z}$, satisfying $\mathbb{E}\epsilon_n = 0$, $\mathbb{E}\epsilon_n^2 < \infty$ and a sequence $\{\psi_k\}_{k\geq 0}$ satisfying

$$\psi_k = L_1(k)k^{d-1}, \tag{5.6.1}$$

where $d \in (0, 1/2)$ and L_1 is a slowly varying function at infinity.

We are interested in the limits of partial sums (5.0.1) when $\{X_n\}$ satisfies condition L. Limit theorems based on two different approaches are given in Sections 5.6.1 and 5.6.2 below. In both cases, the focus is on noncentral limit theorems.

5.6.1 Direct Approach for Entire Functions

In this approach, G in (5.0.1) is supposed to be an entire function; that is, analytic on the whole complex plane, expressed as

$$G(z) = \sum_{k=0}^{\infty} c_k z^k, \quad z \in \mathbb{C}. \tag{5.6.2}$$

We shall use the notion of the so-called *exponent rank* of G. It will play a role similar to that of the Hermite rank in the context of Gaussian series.

Definition 5.6.1 We shall say that G is of *exponent rank* $m \geq 1$ (with respect to X_1) if m is the smallest integer such that

$$\mathbb{E}G^{(m)}(X_1) \neq 0, \tag{5.6.3}$$

where $G^{(m)}$ denotes the mth derivative of G. The corresponding *rank coefficient* is $\mathbb{E}G^{(m)}(X_1)$.

Remark 5.6.2 Note that, when X_1 is a standard Gaussian variable and G satisfies suitable smoothness and integrability assumptions, the exponent rank and the Hermite rank coincide. Indeed, by differentiating G in (5.1.6) m times, we get

$$G^{(m)}(x) = \sum_{n=m}^{\infty} g_n H_n^{(m)}(x)$$

or, by using the property $H_n^{(1)}(x) = nH_{n-1}(x)$ of Hermite polynomials,

$$G^{(m)}(x) = \sum_{n=m}^{\infty} g_n n(n-1)\dots(n-m+1)H_{n-m}(x) = g_m m! + \sum_{k=1}^{\infty} g_{m+k}\frac{(m+k)!}{k!}H_k(x).$$

This yields $\mathbb{E}G^{(m)}(X_1) = g_m m!$. Hence both the Hermite rank and the exponent rank equal m, if m is the smallest integer such that $g_n = 0$ for all $n < m$. The preceding argument clearly holds for polynomial functions G. One can make fewer restrictions on G; see, for example, Proposition 5.8.4 below.

The next result, due to Surgailis [935], is a non-central limit theorem for partial sums of functions of linear time series. The following sketch of proof outlines, in particular, how the exponent rank arises.

Theorem 5.6.3 *Let $\{X_n\}_{n\in\mathbb{Z}}$ be a linear time series satisfying condition L above with LRD parameter d. Let G be an entire function defined by (5.6.2) such that its exponent rank (with respect to X_1) is m and*

$$\sum_{j,k=0}^{\infty} |c_j||c_k|(j!k!)^2 2^{2(j+k)}\bar{\mu}_{j+k} < \infty,$$

where $\bar{\mu}_k = \mathbb{E}|\epsilon_1|^k$, $k \geq 0$. If

$$d \in \Big(\frac{1}{2}\Big(1-\frac{1}{m}\Big), \frac{1}{2}\Big),$$

then, as $N \to \infty$,

$$\frac{1}{(L_1(N))^m N^{m(d-1/2)+1}} \sum_{n=1}^{[Nt]} (G(X_n) - \mathbb{E}G(X_n)), \quad t \geq 0,$$

has the same limit in finite-dimensional distributions as

$$\mathbb{E}G^{(m)}(X_1)\frac{Y_{[Nt],m}}{(L_1(N))^m N^{m(d-1/2)+1}}, \quad t \geq 0,$$

where

$$Y_{[Nt],m} = \sum_{n=1}^{[Nt]} \sum_{0\leq j_1<j_2<\dots<j_m<\infty} \psi_{j_1}\epsilon_{n-j_1}\dots\psi_{j_m}\epsilon_{n-j_m}. \tag{5.6.4}$$

Moreover, that limit is

$$\frac{(\mathbb{E}G^{(m)}(X_1))}{m!A_{m,H}} Z_H^{(m)}(t),$$

where $Z_H^{(m)} = \{Z_H^{(m)}(t)\}_{t\geq 0}$ is the standard Hermite process (4.2.42) of order m with the self-similarity parameter

$$H = m\Big(d-\frac{1}{2}\Big)+1 \in \Big(\frac{1}{2}, 1\Big),$$

and $A_{m,H}$ is given by (4.2.7).

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Sketch of Proof Note from (5.6.2) that $G(X_n) = \sum_{k=0}^{\infty} c_k (X_n)^k$. The idea of the proof consists of decomposing

$$(X_n)^k = \sum_{p_1, p_2, \dots, p_k=0}^{\infty} \psi_{p_1}\psi_{p_2}\dots\psi_{p_k}\epsilon_{n-p_1}\epsilon_{n-p_2}\dots\epsilon_{n-p_k} \tag{5.6.5}$$

in terms of the cardinality $|\{p_1, \dots, p_k\}|$ of the set $\{p_1, \dots, p_k\}$. When $|\{p_1, \dots, p_k\}| = k$, the term $\psi_{p_1}\psi_{p_2}\dots\psi_{p_k}\epsilon_{n-p_1}\epsilon_{n-p_2}\dots\epsilon_{n-p_k}$ is not modified. When $|\{p_1, \dots, p_k\}| < k$ and for instance equal to $k-1$ with $p_1 = p_2$, it is split in two parts:

$$\begin{aligned} \psi_{p_1}\psi_{p_2}\dots\psi_{p_k}(\epsilon_{n-p_1}\epsilon_{n-p_2}\dots\epsilon_{n-p_k}) &= \psi_{p_1}^2\psi_{p_3}\dots\psi_{p_k}(\epsilon_{n-p_1}^2\epsilon_{n-p_3}\dots\epsilon_{n-p_k}) \\ = \psi_{p_1}^2\psi_{p_3}\dots\psi_{p_k}\,(\mu_2\,\epsilon_{n-p_3}\dots\epsilon_{n-p_k}) &+ \psi_{p_1}^2\psi_{p_3}\dots\psi_{p_k}\,(\eta_{p_1}(2)\,\epsilon_{n-p_3}\dots\epsilon_{n-p_k}), \end{aligned} \tag{5.6.6}$$

where $\mu_\ell = \mathbb{E}\epsilon_1^\ell$ and $\eta_p(\ell) = \epsilon_p^\ell - \mu_\ell$. One then shows that the second term with $\eta_{p_1}(2)$ is negligible in the limit theorem consideration since it exhibits short-range dependence. More generally, $(X_n)^k$ can be written as

$$(X_n)^k = \sum_{\ell=0}^{k} \binom{k}{\ell} \sum_{(p)_\ell} \psi_{p_1}\epsilon_{n-p_1}\dots\psi_{p_\ell}\epsilon_{n-p_\ell} \sum_{(V)(k-\ell)} \sum_{(q)_r:(q)_r\cap(p)_\ell=\emptyset} \psi_{q_1}^{v_1}\epsilon_{n-q_1}^{v_1}\dots\psi_{q_r}^{v_r}\epsilon_{n-q_r}^{v_r}, \tag{5.6.7}$$

where the summation over $(p)_\ell$ corresponds to the summation over the sets $\{p_1, \dots, p_\ell\}$ of cardinality $|\{p_1, \dots, p_\ell\}| = \ell$, that is, over $p_1, \dots, p_\ell$ which take different values. The sum $\sum_{(V)(k-\ell)}$ is taken over the rest, namely, over all partitions of the set $\{1, \dots, k-\ell\}$ of cardinality $v_1, \dots, v_r$ such that $v_i \geq 2$, for all i. By convention, this sum equals 1 for $(V)(0)$. The sum over $(q)_r : (q)_r \cap (p)_\ell = \emptyset$ is the sum over $\{q_1, \dots, q_r\}$ of cardinality r such that $\{q_1, \dots, q_r\} \cap \{p_1, \dots, p_\ell\} = \emptyset$. Thus in (5.6.8), the qs involve different indices than the ps.

As is done in (5.6.6), we now introduce centered random variables in (5.6.7) and thus write $\epsilon_{n-q_1}^{v_1} = \eta_{q_1}(v_1) + \mu_{v_1} \dots, \epsilon_{n-q_r}^{v_r} = \eta_{q_r}(v_r) + \mu_{v_r}$. Plugging this in (5.6.7) and expanding the product, we drop the terms involving any centered random variables $\eta_{q_1}(v_1), \dots, \eta_{q_r}(v_r)$ (e.g., the second term in (5.6.6)). These terms can be shown to be negligible since they exhibit short-range dependence. This results in

$$(X_n)_1^k = \sum_{\ell=0}^{k} \binom{k}{\ell} \sum_{(p)_\ell} \psi_{p_1}\epsilon_{n-p_1}\dots\psi_{p_\ell}\epsilon_{n-p_\ell} \sum_{(V)(k-\ell)} \sum_{(q)_r:(q)_r\cap(p)_\ell=\emptyset} \psi_{q_1}^{v_1}\mu_{v_1}\dots\psi_{q_r}^{v_r}\mu_{v_r}. \tag{5.6.8}$$

So the difference between $(X_n)^k$ in (5.6.7) and $(X_n)_1^k$ in (5.6.8) is that when there is an ϵ_{n-p}^ℓ with $\ell > 1$ in (5.6.5), it is replaced by $\mu_\ell = \mathbb{E}\epsilon_1^\ell$ in (5.6.8). Then we replace it further by

$$(Z_n)_1^k = \sum_{\ell=0}^{k} \binom{k}{\ell} \sum_{(p)_\ell} \psi_{p_1}\epsilon_{n-p_1}\dots\psi_{p_\ell}\epsilon_{n-p_\ell} \sum_{(V)(k-\ell)} \sum_{(q)_r} \psi_{q_1}^{v_1}\mu_{v_1}\dots\psi_{q_r}^{v_r}\mu_{v_r}. \tag{5.6.9}$$

$(Z_n)_1^k$ further modifies (5.6.8) by removing the condition $(q)_r \cap (p)_\ell = \emptyset$. In contrast to $(X_n)^k$, the notation $(Z_n)_1^k$ is a shorthand for the right-hand side of (5.6.9) and does not mean Z_n to the power k. The difference between (5.6.8) and (5.6.9) is that in (5.6.9), we also include qs that have same values as the ps.

Observe that the summands of (5.6.8) and $(Z_n)_1^k$ with $\ell > 0$ have zero mean. Then,

$$\mathbb{E}(X_n)^k = \mathbb{E}(Z_n)_1^k = \sum_{(V)(k)} \sum_{(q)_r} \psi_{q_1}^{v_1} \mu_{v_1} \dots \psi_{q_r}^{v_r} \mu_{v_r},$$

for $k \geq 0$ and hence for $0 \leq s \leq k$,

$$\mathbb{E}(X_n)^{k-s} = \mathbb{E}(Z_n)_1^{k-s} = \sum_{(V)(k-s)} \sum_{(q)_r} \psi_{q_1}^{v_1} \mu_{v_1} \dots \psi_{q_r}^{v_r} \mu_{v_r}. \tag{5.6.10}$$

Let us now define formally $G_1(Z_n) = \sum_{s \geq 0} c_s (Z_n)_1^s$, where $(Z_n)_1^k$ is given in (5.6.9), and prove that

$$G_1(Z_n) = \sum_{s \geq 0} c_s (Z_n)_1^s = \sum_{s \geq 0} \frac{e_s}{s!} \sum_{(p)_s} \psi_{p_1} \epsilon_{n-p_1} \dots \psi_{p_s} \epsilon_{n-p_s}, \tag{5.6.11}$$

where

$$e_s = \mathbb{E} G^{(s)}(X_n).$$

Observe that by (5.6.10),

$$\frac{e_s}{s!} = \frac{\mathbb{E} G^{(s)}(X_n)}{s!} = \sum_{k \geq s} c_k \binom{k}{s} \mathbb{E}(X_n)^{k-s} = \sum_{k \geq s} c_k \binom{k}{s} \mathbb{E}(Z_n)_1^{k-s}.$$

Using this and again (5.6.10), note that the right-hand side of (5.6.11) can be expressed as

$$\begin{aligned}
&\sum_{s \geq 0} \frac{e_s}{s!} \sum_{(p)_s} \psi_{p_1} \epsilon_{n-p_1} \dots \psi_{p_s} \epsilon_{n-p_s} \\
&= \sum_{s \geq 0} \Big(\sum_{k \geq s} c_k \binom{k}{s} \sum_{(V)(k-s)} \sum_{(q)_r} \psi_{q_1}^{v_1} \mu_{v_1} \dots \psi_{q_r}^{v_r} \mu_{v_r} \Big) \sum_{(p)_s} \psi_{p_1} \epsilon_{n-p_1} \dots \psi_{p_s} \epsilon_{n-p_s} \\
&= \sum_{k \geq 0} c_k \Big(\sum_{s=0}^{k} \binom{k}{s} \sum_{(p)_s} \psi_{p_1} \epsilon_{n-p_1} \dots \psi_{p_s} \epsilon_{n-p_s} \sum_{(V)(k-s)} \sum_{(q)_r} \psi_{q_1}^{v_1} \mu_{v_1} \dots \psi_{q_r}^{v_r} \mu_{v_r} \Big) = \sum_{k \geq 0} c_k (Z_n)_1^k,
\end{aligned}$$

by (5.6.9), hence proving (5.6.11).

The next step in the proof consists in showing that $\sum_{n=1}^{[Nt]} G(X_n)$ can be replaced by $\sum_{n=1}^{[Nt]} G_1(Z_n)$ and that the leading term in $\sum_{n=1}^{[Nt]} G_1(Z_n)$ is the term corresponding to $s = m$, that is by (5.6.11),

$$e_m Y_{[Nt],m} = \frac{e_m}{m!} \sum_{n=1}^{[Nt]} \sum_{(p)_m} \psi_{p_1} \epsilon_{n-p_1} \dots \psi_{p_m} \epsilon_{n-p_m}.$$

More precisely, it is proved in Lemmas 2 and 3 of Surgailis [935] that

$$\mathbb{E} \Big(\sum_{n=1}^{N} (G(X_n) - G_1(Z_n)) \Big)^2 \leq CN,$$

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where C is a positive constant and that

$$\mathbb{E}\Big(\sum_{n=1}^{N}\sum_{\ell\geq m+1}\frac{e_\ell}{\ell!}\sum_{(p)_\ell}\psi_{p_1}\epsilon_{p_1}\dots\psi_{p_\ell}\epsilon_{p_\ell}\Big)^2=\mathbb{E}\Big(\sum_{\ell\geq m+1}e_\ell Y_{N,\ell}\Big)^2=o\Big((L_1(N))^{2m}N^{m(2d-1)+2}\Big),$$

as N tends to infinity.

To conclude, it remains to study the asymptotic behavior of the main term $e_m Y_{[Nt],m}$ and show that, as $N\to\infty$,

$$e_m\frac{Y_{[Nt],m}}{(L_1(N))^m N^{m(d-1/2)+1}}\xrightarrow{fdd}e_m Z_H^{(m)}(t).$$

The basic idea is to express $Y_{[Nt],m}$ as a discrete multiple stochastic integral as follows. Note that

$$\begin{aligned}\frac{Y_{[Nt],m}}{(L_1(N))^m N^{m(d-1/2)+1}}&=\frac{1}{(L_1(N))^m N^{m(d-1/2)+1}}\sum_{n=1}^{[Nt]}\sum_{i_m<\dots<i_2<i_1\leq n}\prod_{s=1}^{m}\psi_{n-i_s}\epsilon_{i_s}\\&=\sum_{i_1,i_2,\dots,i_m}h_{t,N}(i_1,i_2,\dots,i_m)\epsilon_{i_1}\epsilon_{i_2}\dots\epsilon_{i_m}=:J_m(h_{t,N}),\end{aligned}\tag{5.6.12}$$

where

$$h_{t,N}(i_1,\dots,i_m)=\begin{cases}N^{-m/2}\sum_{n=1}^{[Nt]}\Big(\prod_{s=1}^{m}\frac{\psi_{n-i_s}}{L(N)N^{d-1}}1_{\{i_s\leq n\}}\Big)\frac{1}{N}, & \text{if } i_m<\dots<i_2<i_1,\\ 0, & \text{otherwise.}\end{cases}\tag{5.6.13}$$

In view of (5.6.13), there is effectively no summation on diagonals in (5.6.12). The sums $J_m(h)$ as in (5.6.12) are known as discrete multiple integrals, and have similar second-order properties as multiple integrals $I_m(g)$ (with respect to Gaussian random measures). By Proposition 5.6.4, if

$$\tilde{h}_N(v_1,\dots,v_m)=N^{m/2}h_N([v_1N],\dots,[v_mN])$$

converges in $L^2(\mathbb{R}^m)$ to some function h, then $J_m(h_N)\xrightarrow{d}I_m(h)$. In the case (5.6.13) considered here, it can be shown[5] (and certainly seen informally) that

$$\begin{aligned}N^{m/2}h_{t,N}([Nv_1],\dots,[Nv_m])&\to h_t(v_1,\dots,v_m)\\&=\begin{cases}\int_0^t\prod_{s=1}^{m}(u-v_s)_+^{d-1}du, & \text{if } v_m<\dots<v_1,\\ 0, & \text{otherwise}\end{cases}\end{aligned}$$

in $L^2(\mathbb{R}^m)$, and hence that

$$e_m\frac{Y_{[Nt],m}}{(L_1(N))^m N^{1+m(d-1/2)}}\xrightarrow{fdd}\frac{e_m}{m!}\int_{\mathbb{R}^m}'\Big\{\int_0^t\prod_{j=1}^{m}(u-v_j)_+^{d-1}du\Big\}B(dv_1)\dots B(dv_m),$$

which yields the desired result. □

We used the following fact in the proof of Theorem 5.6.3. Let h be a function defined in $\mathbb{Z}^m$ such that $\sum'_{(i_1,\dots,i_m)\in\mathbb{Z}^m}h(i_1,\dots,i_m)^2<\infty$, where $'$ indicates the exclusion of the

[5] See (4.7.21) in Giraitis, Koul, and Surgailis [406].

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diagonals $i_p = i_q$, $p \neq q$, and let $\{\epsilon_i, i \in \mathbb{Z}\}$ be the standardized i.i.d. random variables. The following result relates the *discrete multiple stochastic integral*

$$J_m(h) = \sum_{(i_1,\ldots,i_m)\in\mathbb{Z}^m}{}' h(i_1, \ldots, i_m)\epsilon_{i_1} \ldots \epsilon_{i_m} \tag{5.6.14}$$

to a continuous one.

Proposition 5.6.4 *Let $h_{j,N}$ be square-summable functions defined on $\mathbb{Z}^{m_j}$, $j = 1, \ldots, k$. Let*

$$\tilde{h}_{j,N}(x_1, \ldots, x_{m_j}) = N^{m_j/2} h_{j,N}\left([Nx_1], \ldots, [Nx_{m_j}]\right), \quad j = 1, \ldots, k.$$

Suppose that there exist $h_j \in L^2(\mathbb{R}^{m_j})$, $j = 1, \ldots, k$, such that

$$\|\tilde{h}_{j,N} - h_j\|_{L^2(\mathbb{R}^{m_j})} \to 0$$

as $N \to \infty$. Then, as $N \to \infty$,

$$\Big(J_{m_1}(h_{1,N}), \ldots, J_{m_k}(h_{k,N})\Big) \xrightarrow{d} \Big(I_{m_1}(h_1), \ldots, I_{m_k}(h_k)\Big),$$

where each $I_{m_j}(h_j) = \int'_{\mathbb{R}^{m_j}} h_j(x_1, \ldots, x_{m_j}) B(dx_1) \ldots B(dx_{m_j})$, $j = 1, \ldots, k$, denotes the m_j-tuple Wiener–Itô integral with respect to the same standard Brownian motion.

For a proof, see Surgailis [935], Lemma 7 or Giraitis et al. [406], Proposition 14.3.2. The proof is similar to that of Proposition 5.3.6 where h and h_ϵ are approximated by “simple” functions, and then the convergence of $N^{-1/2} \sum_{k=1}^{[Nu]} \epsilon_k$ to Brownian motion is used.

5.6.2 Approach Based on Martingale Differences

We shall study the asymptotic behavior of $\sum_{n=1}^{[Nt]} G(X_n)$ using a martingale difference approach. We need to introduce first some notation. Let F be the distribution of the linear time series $X_n = \sum_{k=0}^{\infty} \psi_k \epsilon_{n-k}$ and F_j the distribution of the truncated series

$$X_{n,j} = \sum_{k=0}^{j-1} \psi_k \epsilon_{n-k},$$

defined for $j \geq 0$, with the convention $X_{n,0} = 0$. Let

$$G_j(x) = \int_{\mathbb{R}} G(x+y) dF_j(y), \quad G_\infty(x) = \int_{\mathbb{R}} G(x+y) dF(y), \tag{5.6.15}$$

and

$$G_j^{(r)}(x) = \frac{d^r}{dx^r} \int_{\mathbb{R}} G(x+y) dF_j(y), \quad G_\infty^{(r)}(x) = \frac{d^r}{dx^r} \int_{\mathbb{R}} G(x+y) dF(y). \tag{5.6.16}$$

If the rth derivative $G_j^{(r)}$ of G_j exists, define

$$G_{j,\lambda}^{(r)}(x) = \sup_{|y|\leq\lambda} |G_j^{(r)}(x+y)|, \quad \lambda \geq 0.$$

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We shall say that G satisfies *Condition* $C(r, j, \lambda)$ if

1. $G_j^{(r)}(x)$ exists for all x and $G_j^{(r)}$ is continuous.
2. For all $x \in \mathbb{R}$,

$$\sup_{I\subset\{0,1,\dots\}} \mathbb{E}\Big\{G_{j,\lambda}^{(r)}(x + \sum_{i\in I} \psi_i\epsilon_i)\Big\}^4 < \infty,$$

   where the supremum is taken over all subsets I of $\{0, 1, \dots\}$.

Let us comment on Condition $C(r, j, \lambda)$. It is satisfied if the rth derivative of G is bounded and continuous, in which case the condition does not depend on j. Moreover, if G is any polynomial, then $C(r, j, \lambda)$ holds provided that ϵ_i has finite moments of sufficiently high order.

The novelty here is that $C(r, j, \lambda)$ can hold without G being smooth. An important example is the indicator function.

Example 5.6.5 (*Condition* $C(r, 1, \lambda)$ *and the indicator function*) If $G(x) = 1_{\{x\le u\}}$, for some fixed u, let us prove that G satisfies $C(r, 1, \lambda)$ for all positive λ as soon as the probability density function g of ϵ_1 has a continuous and integrable rth derivative.

Since $X_{n,1} = \psi_0\epsilon_n$ has distribution F_1, we have

$$G_1(x) = \int_{\mathbb{R}} G(x+y)dF_1(y) = \int_{\mathbb{R}} G(x+\psi_0 y_1)g(y_1)dy_1 = \psi_0^{-1}\int_{\mathbb{R}} G(z)g\left(\frac{z-x}{\psi_0}\right)dz.$$

Note that

$$\frac{d^r}{dx^r}\left\{G(z)g\left(\frac{z-x}{\psi_0}\right)\right\} = \frac{(-1)^r}{\psi_0^r}G(z)g^{(r)}\left(\frac{z-x}{\psi_0}\right).$$

Since, by assumption, $\int_{\mathbb{R}} |g^{(r)}(y)|dy < \infty$, we get

$$G_1^{(r)}(x) = \frac{(-1)^r}{\psi_0^{r+1}}\int_{\mathbb{R}} G(z)g^{(r)}\left(\frac{z-x}{\psi_0}\right)dz = \frac{(-1)^r}{\psi_0^r}\int_{\mathbb{R}} G(x+\psi_0 y_1)g^{(r)}(y_1)dy_1.$$

Moreover, $G_1^{(r)}$ is a continuous function since $g^{(r)}$ is assumed to be a continuous function. This gives (5.6.2) of $C(r, 1, \lambda)$. Let us now check (5.6.2) of $C(r, 1, \lambda)$. For all subset I of $\{1, 2, \dots\}$, we have

$$\begin{aligned}
&\mathbb{E}\Big\{\sup_{|y|\le\lambda}\Big|G_1^{(r)}\Big(x + \sum_{i\in I}\psi_i\epsilon_i + y\Big)\Big|\Big\}^4 \\
&= \frac{1}{\psi_0^{4r}}\mathbb{E}\Big\{\sup_{|y|\le\lambda}\Big|\int_{\mathbb{R}} G\Big(x + \sum_{i\in I}\psi_i\epsilon_i + y + \psi_0 y_1\Big)g^{(r)}(y_1)dy_1\Big|\Big\}^4,
\end{aligned}$$

which is bounded by $\psi_0^{-4r}(\int_{\mathbb{R}} |g^{(r)}(y_1)|dy_1)^4 < \infty$. Thus ensures that $G(x) = 1_{\{x\le u\}}$ satisfies Condition $C(r, 1, \lambda)$.

Observe that the indicator function $h(x) = 1_{\{x\le u\}}$ is allowed in the Gaussian case (Section 5.3) but not in the situation considered in Section 5.6.1. As we have just seen, it is allowed in the methodology discussed in this section.

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The idea here is to use a decomposition into martingale differences (also known as a mixingale decomposition), as explained in the sketch of the proof below, and to prove that the leading term is once again $Y_{[Nt],m}$, defined in (5.6.4), where here m is the power rank of G defined as follows.

Definition 5.6.6 We shall say that G is of *power rank* $m \geq 1$ (with respect to X_1) if it is the smallest integer such that

$$G_{\infty}^{(m)}(0) = \frac{d^m}{dx^m} \mathbb{E}G(x + X_1)\Big|_{x=0} \neq 0. \tag{5.6.17}$$

The corresponding *rank coefficient* is $G_{\infty}^{(m)}(0)$.

Remark 5.6.7 If G satisfies suitable smoothness and integrability assumptions, the differentiation in (5.6.17) can be taken inside the expectation leading to

$$G_{\infty}^{(m)}(0) = \mathbb{E}G^{(m)}(X_1),$$

that is, the exponent and power ranks of the function G are the same. This is clearly the case for polynomial functions G.

Here is the precise result due to Ho and Hsing [477].

Theorem 5.6.8 *Let $\{X_n\}_{n\in\mathbb{Z}}$ be a linear time series satisfying condition L above with LRD parameter d. Suppose that the power rank (with respect to X_1) of a function G is m. Suppose also that, for some j and λ, the condition $C(r, j, \lambda)$ holds for $r = 0, 1, \dots, m+2$ and $\mathbb{E}G(X_1)^2 < \infty$. If*

$$d \in \left(\frac{1}{2}\Big(1 - \frac{1}{m}\Big), \frac{1}{2}\right)$$

and $\mathbb{E}|\epsilon_1|^{2m\vee 8} < \infty$, then, as $N \to \infty$,

$$\frac{1}{(L_1(N))^m N^{m(d-1/2)+1}} \sum_{n=1}^{[Nt]} (G(X_n) - \mathbb{E}G(X_n)), \quad t \geq 0,$$

has the same limit in finite-dimensional distributions as

$$G_{\infty}^{(m)}(0) \frac{Y_{[Nt],m}}{(L_1(N))^m N^{m(d-1/2)+1}}, \quad t \geq 0,$$

where $Y_{[Nt],m}$ is defined in (5.6.4). Moreover, that limit is

$$\frac{G_{\infty}^{(m)}(0)}{m! A_{m,H}} Z_H^{(m)}(t),$$

where $Z_H^{(m)} = \{Z_H^{(m)}(t)\}_{t\geq 0}$ is the standard Hermite process (4.2.42) of order m with the self-similarity parameter

$$H = m\Big(d - \frac{1}{2}\Big) + 1 \in \Big(\frac{1}{2}, 1\Big),$$

and $A_{m,H}$ is given by (4.2.7).

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Sketch of Proof The second parts of Theorems 5.6.3 and 5.6.8 are similar. This shows that the key is to reduce the original $\sum_{n=1}^{[Nt]} G(X_n)$ to $Y_{[Nt],m}$. It is $Y_{[Nt],m}$ which converges to the limit after suitable normalization.

The idea of the proof is to condition on the σ–fields

$$\mathcal{F}_k = \sigma\{\epsilon_i : i \leq k\},$$

using the telescoping expression

$$G(X_n) - \mathbb{E}G(X_n) = \sum_{j\geq 0}\Big(\mathbb{E}(G(X_n)|\mathcal{F}_{n-j}) - \mathbb{E}(G(X_n)|\mathcal{F}_{n-j-1})\Big), \tag{5.6.18}$$

since the extreme summands are such that $\mathbb{E}(G(X_n)|\mathcal{F}_n) = G(X_n)$ and $\lim_{m\to-\infty}\mathbb{E}(G(X_n)|\mathcal{F}_m) = \mathbb{E}(G(X_n)|\mathcal{F}_{-\infty}) = \mathbb{E}G(X_n)$ by the martingale convergence theorem (e.g., Gut [442], Corollary 12.1 in Chapter 10). Now write

$$\mathbb{E}(G(X_n)|\mathcal{F}_{n-j}) = \mathbb{E}(G(X_{n,j} + \tilde{X}_{n,j})|\mathcal{F}_{n-j}),$$

with

$$X_{n,j} = \sum_{k=0}^{j-1}\psi_k\epsilon_{n-k} \quad \text{and} \quad \tilde{X}_{n,j} = \sum_{k\geq j}\psi_k\epsilon_{n-k}.$$

Since $\tilde{X}_{n,j}$ is $\mathcal{F}_{n-j}$–measurable and $X_{n,j}$ is independent of $\mathcal{F}_{n-j}$,

$$\mathbb{E}(G(X_n)|\mathcal{F}_{n-j}) = \int_{\mathbb{R}} G(x + \tilde{X}_{n,j})dF_j(x) = G_j(\tilde{X}_{n,j}), \tag{5.6.19}$$

where F_j is the distribution of $X_{n,j}$ and $G_j(y) = \mathbb{E}G(X_{n,j}+y)$. Using (5.6.18) and (5.6.19), we get that

$$\begin{aligned}
\sum_{n=1}^{[Nt]}(G(X_n) - \mathbb{E}G(X_n)) &= \sum_{n=1}^{[Nt]}\sum_{j_1\geq 0}(G_{j_1}(\tilde{X}_{n,j_1}) - G_{j_1+1}(\tilde{X}_{n,j_1+1}))\\
&= \sum_{n=1}^{[Nt]}\sum_{j_1\geq 0}(\tilde{X}_{n,j_1} - \tilde{X}_{n,j_1+1})G^{(1)}_{j_1+1}(\tilde{X}_{n,j_1+1})\\
&\quad + \Big(G_{j_1}(\tilde{X}_{n,j_1}) - G_{j_1+1}(\tilde{X}_{n,j_1+1})\\
&\quad - (\tilde{X}_{n,j_1} - \tilde{X}_{n,j_1+1})G^{(1)}_{j_1+1}(\tilde{X}_{n,j_1+1})\Big)\\
&= \sum_{n=1}^{[Nt]}\sum_{j_1\geq 0}\psi_{j_1}\epsilon_{n-j_1}G^{(1)}_{j_1+1}(\tilde{X}_{n,j_1+1})\\
&\quad + \Big(G_{j_1}(\tilde{X}_{n,j_1}) - G_{j_1+1}(\tilde{X}_{n,j_1+1}) - \psi_{j_1}\epsilon_{n-j_1}G^{(1)}_{j_1+1}(\tilde{X}_{n,j_1+1})\Big)
\end{aligned} \tag{5.6.20}$$

$$\approx \sum_{n=1}^{[Nt]}\sum_{j_1\geq 0}\psi_{j_1}\epsilon_{n-j_1}G^{(1)}_{j_1+1}(\tilde{X}_{n,j_1+1}), \tag{5.6.21}$$

after proving that the terms in the parentheses can be neglected.

We have introduced the ϵs in (5.6.21). We need now to introduce $G_\infty^{(1)}(0), \dots, G_\infty^{(m)}(0)$. To do so, we express the summands in the remaining term (5.6.21) as

$$\begin{aligned}\psi_{j_1}\epsilon_{n-j_1}G_{j_1+1}^{(1)}(\tilde{X}_{n,j_1+1}) &= \psi_{j_1}\epsilon_{n-j_1}G_\infty^{(1)}(0) + \psi_{j_1}\epsilon_{n-j_1}(G_{j_1+1}^{(1)}(\tilde{X}_{n,j_1+1}) - G_\infty^{(1)}(0))\\ &= \psi_{j_1}\epsilon_{n-j_1}G_\infty^{(1)}(0) + \psi_{j_1}\epsilon_{n-j_1}\sum_{j_2\ge j_1+1}(G_{j_2}^{(1)}(\tilde{X}_{n,j_2}) - G_{j_2+1}^{(1)}(\tilde{X}_{n,j_2+1})). \end{aligned} \tag{5.6.22}$$

Focusing on the term in brackets, we write as before (see (5.6.20)),

$$\begin{aligned} &G_{j_2}^{(1)}(\tilde{X}_{n,j_2}) - G_{j_2+1}^{(1)}(\tilde{X}_{n,j_2+1}) = \psi_{j_2}\epsilon_{n-j_2}G_{j_2+1}^{(2)}(\tilde{X}_{n,j_2+1})\\ &\quad + \left(G_{j_2}^{(1)}(\tilde{X}_{n,j_2}) - G_{j_2+1}^{(1)}(\tilde{X}_{n,j_2+1}) - \psi_{j_2}\epsilon_{n-j_2}G_{j_2+1}^{(2)}(\tilde{X}_{n,j_2+1})\right)\\ &\approx \psi_{j_2}\epsilon_{n-j_2}G_{j_2+1}^{(2)}(\tilde{X}_{n,j_2+1}) = \psi_{j_2}\epsilon_{n-j_2}G_\infty^{(2)}(0) + \psi_{j_2}\epsilon_{n-j_2}(G_{j_2+1}^{(2)}(\tilde{X}_{n,j_2+1}) - G_\infty^{(2)}(0)), \end{aligned} \tag{5.6.23}$$

where in that last equality, we proceeded as in (5.6.22). Relations (5.6.22) and (5.6.23) yield

$$\begin{aligned}&\psi_{j_1}\epsilon_{n-j_1}G_{j_1+1}^{(1)}(\tilde{X}_{n,j_1+1}) \approx \psi_{j_1}\epsilon_{n-j_1}G_\infty^{(1)}(0)\\ &\quad + \psi_{j_1}\epsilon_{n-j_1}\sum_{j_2\ge j_1+1}\left(\psi_{j_2}\epsilon_{n-j_2}G_\infty^{(2)}(0) + \psi_{j_2}\epsilon_{n-j_2}(G_{j_2+1}^{(2)}(\tilde{X}_{n,j_2+1}) - G_\infty^{(2)}(0))\right).\end{aligned}$$

Thus, we get

$$\begin{aligned}\sum_{n=1}^{[Nt]}(G(X_n) - \mathbb{E}G(X_n)) &= Y_{[Nt],1}G_\infty^{(1)}(0) + Y_{[Nt],2}G_\infty^{(2)}(0)\\ &\quad + \sum_{n=1}^{[Nt]}\sum_{j_2>j_1\ge 0}\psi_{j_1}\epsilon_{n-j_1}\psi_{j_2}\epsilon_{n-j_2}(G_{j_2}^{(2)}(\tilde{X}_{n,j_2}) - G_\infty^{(2)}(0)),\end{aligned} \tag{5.6.24}$$

where

$$Y_{N,r} = \sum_{n=1}^{N}\sum_{0\le j_1<j_2<\dots<j_r}\prod_{s=1}^{r}\psi_{j_s}\epsilon_{n-j_s}.$$

Iterating, we get

$$\sum_{n=1}^{[Nt]}(G(X_n) - \mathbb{E}G(X_n)) = Y_{[Nt],1}G_\infty^{(1)}(0) + Y_{[Nt],2}G_\infty^{(2)}(0) + \dots + Y_{[Nt],m}G_\infty^{(m)}(0) + R_N, \tag{5.6.25}$$

where R_N is shown to be a negligible remainder term in Ho and Hsing [477]. Hence, the first term of the expansion (5.6.25) is given by $Y_{[Nt],m}G_\infty^{(m)}(0)$, where m is the power rank. This is the same $Y_{[Nt],m}$ as in (5.6.4). One concludes by applying the last part of Theorem 5.6.3. □

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5.7 Multivariate Limit Theorems

We are interested here in the limit behavior of the finite-dimensional distributions of the following vector as $N \to \infty$:

$$V_N(t) = \Big(\frac{1}{A_r(N)} \sum_{n=1}^{[Nt]} (G_r(X_n) - \mathbb{E}G_r(X_n))\Big)_{r=1,\dots,R}, \quad t \geq 0, \tag{5.7.1}$$

where G_r, $r = 1, \dots, R$, are deterministic functions and $A_r(N)$s are appropriate normalizations. We first focus on the case when $\{X_n\}_{n\in\mathbb{Z}}$ is a Gaussian stationary series. We have to distinguish between the situations where the resulting limit law for (5.7.1) is:

(a) a multivariate Gaussian process with dependent Brownian motion marginals, or
(b) a multivariate process with dependent Hermite processes as marginals, or
(c) a combination.

The cases (a), (b) and (c) will be referred to as, respectively, the SRD, LRD and mixed cases. In Section 5.7.4, we state the corresponding results for multivariate multilinear processes,

$$Y_N(t) = \Big(\frac{1}{A_r(N)} \sum_{n=1}^{[Nt]} \sum_{1\leq i_1<\dots<i_{k_r}<\infty} \psi_{r,i_1} \dots \psi_{r,i_{k_r}} \epsilon_{n-i_1} \dots \epsilon_{n-i_{k_r}}\Big)_{r=1,\dots,R}. \tag{5.7.2}$$

But first, we focus on (5.7.1).

5.7.1 The SRD Case

The first result concerns the SRD case, generalizing Theorem 5.4.1.

Theorem 5.7.1 *(SRD case.) Let $\{X_n\}_{n\in\mathbb{Z}}$ be a Gaussian stationary series with autocovariance function γ_X. Suppose that $\mathbb{E}X_n = 0$, $\mathbb{E}X_n^2 = 1$. Let G_r, $r = 1, \dots, R$, be deterministic functions with respective Hermite ranks $k_r \geq 1$, $r = 1, \dots, R$, in the sense of Definition 5.2.1. If*

$$\sum_{n=1}^{\infty} |\gamma_X(n)|^{k_r} < \infty, \quad r = 1, \dots, R,^{6} \tag{5.7.3}$$

then

$$V_N(t) \stackrel{fdd}{\longrightarrow} V(t), \quad t \geq 0, \tag{5.7.4}$$

where $V_N(t)$ is given in (5.7.1) with $A_r(N) = N^{1/2}$, $r = 1, \dots, R$. The limit process $V(t) = (\sigma_1 B_1(t), \dots, \sigma_R B_R(t))'$ is multivariate Brownian motion, where

$$\sigma_r^2 = \sum_{m=k_r}^{\infty} g_{r,m}^2 m! \sum_{n=-\infty}^{\infty} (\gamma_X(n))^m, \quad r = 1, \dots, R, \tag{5.7.5}$$

with $g_{r,m}$ being the coefficients in the Hermite expansion (5.1.6) of G_r, $\{B_r(t)\}_{t\in\mathbb{R}}$, $r = 1, \dots, R$, are standard Brownian motions with the cross-covariance: for $t_1, t_2 \geq 0$,

[6] See (5.4.1) and (5.4.2) for an equivalent condition.

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$$\mathbb{E}B_{r_1}(t_1)B_{r_2}(t_2) = (t_1 \wedge t_2)\frac{\sigma_{r_1,r_2}}{\sigma_{r_1}\sigma_{r_2}}, \tag{5.7.6}$$

where

$$\sigma_{r_1,r_2} = \sum_{m=k_{r_1}\vee k_{r_2}}^{\infty} g_{r_1,m}g_{r_2,m}m! \sum_{n=-\infty}^{\infty} \gamma_X(n)^m \tag{5.7.7}$$

($a \wedge b = \min\{a, b\}$ and $a \vee b = \max\{a, b\}$).

Proof The proof follows that of Theorem 5.4.1 in the univariate case. As in that proof (see the argument leading to (5.4.6)), it is enough to consider the functions

$$G_r(x) - \mathbb{E}G_r(X_n) = \sum_{m=k_r}^{M} g_{r,m}H_m(x), \tag{5.7.8}$$

where M is finite and fixed. Set

$$S_N = \sum_{r=1}^{R}\sum_{j=1}^{J_r}\theta_{r,j}V_{r,N}(t_{r,j}),$$

where $V_{r,N}(t)$ are the components of $V_N(t) = (V_{1,N}(t), \dots, V_{R,N}(t))'$, $\theta_{r,j} \in \mathbb{R}$ and $t_{r,j} \geq 0$. By the method of moments (e.g., Gut [442], Theorem 8.6 in Chapter 5 and Section 4.10), it is enough to show that, for all integers $p \geq 1$,

$$\mathbb{E}S_N^p \to \mathbb{E}\Big(\sum_{r=1}^{R}\sum_{j=1}^{J_r}\theta_{r,j}V_r(t_{r,j})\Big)^p, \tag{5.7.9}$$

where $V_r(t)$ are the components of the limit vector $V(t) = (V_1(t), \dots, V_R(t))'$. This is equivalent to: for $p \geq 1$,

$$\chi_p(S_N) \to \chi_p\Big(\sum_{r=1}^{R}\sum_{j=1}^{J_r}\theta_{r,j}V_r(t_{r,j})\Big), \tag{5.7.10}$$

where χ_p denotes the pth cumulant. Since the limit is Gaussian and has zero mean,

$$\chi_p\Big(\sum_{r=1}^{R}\sum_{j=1}^{J_r}\theta_{r,j}V_r(t_{r,j})\Big) = \begin{cases} \mathbb{E}\Big(\sum_{r=1}^{R}\sum_{j=1}^{J_r}\theta_{r,j}V_r(t_{r,j})\Big)^2, & \text{if } p = 2, \\ 0, & \text{if } p = 1, 3, 4, \dots \end{cases} \tag{5.7.11}$$

As in the proof of Theorem 5.4.1 (see the argument following (5.4.9)), by using the multilinearity of joint cumulants and the relation (4.3.26),

$$\begin{aligned} \chi_p(S_N) &= \chi_p\Big(\sum_{r=1}^{R}\sum_{j=1}^{J_r}\sum_{m=k_r}^{M}\theta_{r,j}g_{r,m}\frac{1}{N^{1/2}}\sum_{n=1}^{[Nt_{r,j}]}H_m(X_n)\Big) \\ &= \sum_{r_1,\dots,r_p=1}^{R}\sum_{j_1=1}^{J_{r_1}}\cdots\sum_{j_p=1}^{J_{r_p}}\sum_{m_1=k_{r_1}}^{M}\cdots\sum_{m_p=k_{r_p}}^{M}\Big(\prod_{\ell=1}^{p}\theta_{r_\ell,j_\ell}g_{r_\ell,m_\ell}\Big) \end{aligned}$$

$$\times \frac{1}{N^{p/2}} \sum_{n_1=1}^{[Nt_{r_1,j_1}]} \cdots \sum_{n_p=1}^{[Nt_{r_p,j_p}]} \chi\Big(H_{m_1}(X_{n_1}), \dots, H_{m_p}(X_{n_p})\Big)$$

$$= \sum_{r_1,\dots,r_p=1}^{R} \sum_{j_1=1}^{J_{r_1}} \cdots \sum_{j_p=1}^{J_{r_p}} \sum_{m_1=k_{r_1}}^{M} \cdots \sum_{m_p=k_{r_p}}^{M} \Big(\prod_{\ell=1}^{p} \theta_{r_\ell,j_\ell} g_{r_\ell,m_\ell}\Big)$$

$$\times \sum_{A\in\mathcal{A}_c(m_1,\dots,m_p)} g(A) \frac{1}{N^{p/2}} T_N(t_{r_1,j_1}, \dots, t_{r_p,j_p}, m_1, \dots, m_p, A),$$

where T_Ns are defined in (5.4.10). As in the proof of Theorem 5.4.1, by using Lemma 5.4.3, we conclude that, for $p \geq 3$, $\chi_p(S_N) \to 0$, as $N \to \infty$.

For $p = 1$, the convergence (5.7.10) is trivial since $\mathbb{E}S_N = 0$ and hence both sides are zero. For $p = 2$, we need to show that

$$\mathbb{E}S_N^2 \to \mathbb{E}\Big(\sum_{r=1}^{R}\sum_{j=1}^{J_r} \theta_{r,j} V_r(t_{r,j})\Big)^2 = \sum_{r_1,r_2=1}^{R} \sum_{j_1=1}^{J_{r_1}} \sum_{j_2=1}^{J_{r_2}} \theta_{r_1,j_1}\theta_{r_2,j_2} \mathbb{E}V_{r_1}(t_{r_1,j_1}) V_{r_2}(t_{r_2,j_2})$$

$$= \sum_{r_1,r_2=1}^{R} \sum_{j_1=1}^{J_{r_1}} \sum_{j_2=1}^{J_{r_2}} \theta_{r_1,j_1}\theta_{r_2,j_2} (t_{r_1,j_1} \wedge t_{r_2,j_2}) \sum_{m=k_{r_1}\vee k_{r_2}}^{\infty} g_{r_1,m} g_{r_2,m} m! \sum_{n=-\infty}^{\infty} \gamma_X(n)^m. \tag{5.7.12}$$

Observe that

$$\mathbb{E}S_N^2 = \mathbb{E}\Big(\sum_{r=1}^{R}\sum_{j=1}^{J_r}\sum_{m=k_r}^{M} \theta_{r,j} g_{r,m} \frac{1}{N^{1/2}} \sum_{n=1}^{[Nt_{r,j}]} H_m(X_n)\Big)^2$$

$$= \sum_{r_1,r_2=1}^{R} \sum_{j_1=1}^{J_{r_1}} \sum_{j_2=1}^{J_{r_2}} \sum_{m_1=k_{r_1}}^{M} \sum_{m_2=k_{r_2}}^{M} \theta_{r_1,j_1}\theta_{r_2,j_2} g_{r_1,m_1} g_{r_2,m_2} \frac{1}{N} \sum_{n_1=1}^{[Nt_{r_1,j_1}]} \sum_{n_2=1}^{[Nt_{r_2,j_2}]} \mathbb{E}H_{m_1}(X_{n_1}) H_{m_2}(X_{n_2})$$

$$= \sum_{r_1,r_2=1}^{R} \sum_{j_1=1}^{J_{r_1}} \sum_{j_2=1}^{J_{r_2}} \sum_{m=k_{r_1}\vee k_{r_2}}^{M} \theta_{r_1,j_1}\theta_{r_2,j_2} g_{r_1,m} g_{r_2,m} m! \frac{1}{N} \sum_{n_1=1}^{[Nt_{r_1,j_1}]} \sum_{n_2=1}^{[Nt_{r_2,j_2}]} (\gamma_X(n_1 - n_2))^m.$$

As in the proof of Theorem 5.4.1, by using Lemma 5.4.4,

$$\mathbb{E}S_N^2 \to \sum_{r_1,r_2=1}^{R} \sum_{j_1=1}^{J_{r_1}} \sum_{j_2=1}^{J_{r_2}} \theta_{r_1,j_1}\theta_{r_2,j_2} (t_{r_1,j_1} \wedge t_{r_2,j_2}) \sum_{m=k_{r_1}\vee k_{r_2}}^{\infty} g_{r_1,m} g_{r_2,m} m! \sum_{n=-\infty}^{\infty} \gamma_X(n)^m$$

which is the desired limit (5.7.12). This limit can be expressed as

$$\sum_{r_1,r_2=1}^{R} \sum_{j_1=1}^{J_{r_1}} \sum_{j_2=1}^{J_{r_2}} \theta_{r_1,j_1}\theta_{r_2,j_2} (t_{r_1,j_1} \wedge t_{r_2,j_2}) \sigma_{r_1,r_2} = \mathbb{E}\Big(\sum_{r=1}^{R}\sum_{j=1}^{J_r} \theta_{r,j}\sigma_r B_r(t_{r,j})\Big)^2.$$

This concludes the proof. □

The next result is analogous to Corollary 5.4.5.

Corollary 5.7.2 *(SRD case.) Let $\{X_n\}_{n\in\mathbb{Z}}$ be a Gaussian stationary series which is LRD in the sense (2.1.5) in condition II with $d \in (0, 1/2)$. Suppose that $\mathbb{E}X_n = 0$, $\mathbb{E}X_n^2 = 1$. Let G_r, $r = 1, \dots, R$, be deterministic functions with respective Hermite ranks $k_r \geq 1$, $r = 1, \dots, R$, in the sense of Definition 5.2.1. If*

$$d \in \Big(0, \frac{1}{2}\Big(1 - \frac{1}{k_r}\Big)\Big), \quad r = 1, \dots, R, \tag{5.7.13}$$

then the convergence (5.7.4) holds.

Example 5.7.3 (*Bivariate limit result in SRD case*) Under the setting of Corollary 5.7.2, suppose $d \in (0, 1/4)$, let $R = 2$ and

$$G_1(x) = aH_2(x) + bH_3(x) = bx^3 + ax^2 - 3bx - a, \quad G_2(x) = cH_3(x) = cx^3 - 3cx.$$

Here with $r = 1$, we have $k_1 = 2$ and $g_{1,2} = a$, $g_{1,3} = b$, and with $r = 2$, we have $k_2 = 3$ and $g_{2,3} = c$. In particular, for $m = k_{r_1} \vee k_{r_2} = 2 \vee 3 = 3$, we have $g_{r_1,m}g_{r_2,m} = g_{1,3}g_{2,3} = bc$. Then, in (5.7.5),

$$\sigma_1^2 = 2a^2 \sum_{n=-\infty}^{\infty} \gamma_X(n)^2 + 6b^2 \sum_{n=-\infty}^{\infty} \gamma_X(n)^3, \quad \sigma_2^2 = 6c^2 \sum_{n=-\infty}^{\infty} \gamma_X(n)^3,$$

and

$$\Big(\frac{1}{N^{1/2}} \sum_{n=1}^{[Nt]} G_1(X_n), \frac{1}{N^{1/2}} \sum_{n=1}^{[Nt]} G_2(X_n)\Big)' \xrightarrow{fdd} \Big(\sigma_1 B_1(t), \sigma_2 B_2(t)\Big)', \quad t \geq 0,$$

where the Brownian motions B_1 and B_2 have the covariance structure: for $t_1, t_2 \geq 0$,

$$\mathrm{Cov}\,(B_1(t_1), B_2(t_2)) = 6bc \frac{t_1 \wedge t_2}{\sigma_1\sigma_2} \sum_{n=-\infty}^{\infty} \gamma_X(n)^3.$$

B_1 and B_2 are independent when $b = 0$.

5.7.2 The LRD Case

The next result concerns the LRD case, and extends Theorem 5.3.1. In Appendix B.2 (see (B.2.14)), $\widehat{I}_k(f)$ is used to denote a multiple integral with respect to an Hermitian Gaussian random measure on $\mathbb{R}$. To indicate its dependence on the corresponding symmetric control measure m, we shall denote the multiple integral by $\widehat{I}_{m,k}(f)$ throughout this section, and reserve the notation $\widehat{I}_k(f)$ for the case when the control measure m is the Lebesgue measure.

We also let

$$f_{H,k,t}(x_1, \dots, x_k) = B_{k,H} \frac{e^{it(x_1+\cdots+x_k)} - 1}{i(x_1 + \cdots + x_k)} \prod_{j=1}^{k} |x_j|^{\frac{1-H}{k} - \frac{1}{2}} \tag{5.7.14}$$

be the kernel function appearing in the representation (4.2.42) of a standard Hermite process of order k, where $B_{k,H}$ is the constant in (4.2.46). Note that the limit process in Theorem 5.3.1 is $g_k\beta_{k,H}\widehat{I}_k(f_{H,k,t})$. It can be seen from the proof of Theorem 5.3.1 that the limit process can also be represented as $g_k \widehat{I}_{F_0,k}(K_t)$, where K_t is given in (5.3.13) and the measure F_0 is defined by (5.3.17)–(5.3.18) (and depends on d).

Theorem 5.7.4 *(LRD case.) Let $\{X_n\}_{n\in\mathbb{Z}}$ be a Gaussian stationary series which is LRD in the sense (2.1.5) in condition II with $d \in (0, 1/2)$. Suppose that $\mathbb{E}X_n = 0$, $\mathbb{E}X_n^2 = 1$. Let G_r, $r = 1, \ldots, R$, be deterministic functions with respective Hermite ranks $k_r \geq 1$, $r = 1, \ldots, R$, in the sense of Definition 5.2.1. If*

$$d \in \Big(\frac{1}{2}\Big(1 - \frac{1}{k_r}\Big), \frac{1}{2}\Big), \quad r = 1, \ldots, R, \tag{5.7.15}$$

then

$$V_N(t) \stackrel{fdd}{\longrightarrow} V^{(d)}(t), \quad t \geq 0, \tag{5.7.16}$$

where $V_N(t)$ is given in (5.7.1) with

$$A_r(N) = (L_2(N))^{k_r/2} N^{k_r(d-1/2)+1}, \quad r = 1, \ldots, R.$$

The limit process can be represented as

$$V^{(d)}(t) \stackrel{d}{=} \Big(g_{r,k_r}\beta_{k_r,H_r}\widehat{I}_{k_r}(f_{H_r,k_r,t})\Big)_{r=1,\ldots,R} \stackrel{d}{=} \Big(g_{r,k_r}\widehat{I}_{F_0,k_r}(K_t)\Big)_{r=1,\ldots,R} \tag{5.7.17}$$

in the sense of finite-dimensional distributions, where

$$H_r = k_r\Big(d - \frac{1}{2}\Big) + 1 \in \Big(\frac{1}{2}, 1\Big), \quad r = 1, \ldots, R, \tag{5.7.18}$$

g_{r,k_r} is the first nonzero coefficient in the Hermite expansion of G_r in Definition 5.2.1, $\beta_{k,H}$ is the constant given in (5.3.3), $f_{H,k,t}$ is the kernel function defined in (5.7.14), K_t is given in (5.3.13) and the measure F_0 is defined by (5.3.17)–(5.3.18) (and depends on d).

Remark 5.7.5 The individual component processes $\widehat{I}_{k_r}(f_{H_r,k_r,t})$ above are standard Hermite process of orders k_r with the self-similarity parameter H_r. They are multiple integrals with respect to the same Hermitian Gaussian random measure. It can be shown that they are dependent (Bai and Taqqu [74], Proposition 1).

Proof The proof follows closely that of Theorem 5.3.1. By the reduction theorem (Theorem 5.3.3), it is enough to prove the result for $G_r(x) - \mathbb{E}G_r(X_n) = H_{k_r}(x)$, where $H_{k_r}(x)$ is the Hermite polynomial of order k_r. As in Step 1 of that proof, one can write

$$V_{r,N}(t) = \widehat{I}_{F_N,k_r}(K_{N,t}),$$

where $K_{N,t}$ is the function given in (5.3.10) (with the number of variables k_r, suggested by the order of the multiple integral k_r) and the measure F_N satisfies (5.3.11). As in Step 2 of that proof, we would like to conclude now that

$$\sum_{r=1}^{R}\sum_{j=1}^{J_r}\theta_{r,j}V_{r,N}(t_{r,j}) = \sum_{r=1}^{R}\sum_{j=1}^{J_r}\theta_{r,j}\widehat{I}_{F_N,k_r}(K_{N,t_{r,j}})$$

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$$\to \sum_{r=1}^{R}\sum_{j=1}^{J_r}\theta_{r,j}\widehat{I}_{F_0,k_r}(K_{t_{r,j}}) = \sum_{r=1}^{R}\sum_{j=1}^{J_r}\theta_{r,j}V_r(t_{r,j}),$$

where $\theta_{r,j} \in \mathbb{R}$, $t_{r,j} \geq 0$. This can be shown as in Step 3 of that proof by using Proposition 5.7.6 below, which extends Proposition 5.3.6 to the multivariate case. □

The next auxiliary result, which was used in the preceding proof, can be established in the same way as Proposition 5.3.6.

Proposition 5.7.6 *Let F_N, $N \geq 1$, F_0 be symmetric, locally finite measures without atoms on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ so that $F_N \xrightarrow{v} F_0$ in the sense of vague convergence. Let $\widehat{K}_{N,k_r}(x_1, \ldots, x_{k_r})$, $r = 1, \ldots, R$, be a sequence of Hermitian, measurable functions on $\mathbb{R}^{k_r}$ tending to a continuous function $\widehat{K}_{0,k_r}(x_1, \ldots, x_{k_r})$ uniformly in any rectangle $[-A, A]^{k_r}$, $A > 0$. Moreover, suppose that $\widehat{K}_{0,k_r}$ and $\widehat{K}_{N,k_r}$ satisfy the relation*

$$\lim_{A\to\infty}\sup_{N\geq 0}\int_{\mathbb{R}^{k_r}\setminus[-A,A]^{k_r}} |\widehat{K}_{N,k_r}(x_1, \ldots, x_{k_r})|^2 F_N(dx_1)\ldots F_N(dx_{k_r}) = 0, \quad r = 1, \ldots, R. \tag{5.7.19}$$

Then,

$$\Big(\widehat{I}_{F_N,k_r}(\widehat{K}_{N,k_r})\Big)_{r=1,\ldots,R} \xrightarrow{d} \Big(\widehat{I}_{F_0,k_r}(\widehat{K}_{0,k_r})\Big)_{r=1,\ldots,R}, \tag{5.7.20}$$

where the last integrals exist.

Example 5.7.7 (*Bivariate limit result in LRD case*) Under the setting of Theorem 5.7.4, suppose $d \in (1/4, 1/2)$ and $L_2(u) \sim 1$ in (2.1.5) of condition II. Let also $R = 2$, $G_1(x) = H_1(x) = x$, $G_2(x) = H_2(x) = x^2 - 1$. In this case, the convergence (5.7.17) can be expressed as

$$\Big(\frac{1}{N^{d+1/2}}\sum_{n=1}^{[Nt]}X_n, \frac{1}{N^{2d}}\sum_{n=1}^{[Nt]}(X_n^2 - 1)\Big)' \xrightarrow{fdd} \Big(\frac{1}{\sqrt{d(2d+1)}}Z_{H_1}^{(1)}(t), \frac{1}{\sqrt{d(4d-1)}}Z_{H_2}^{(2)}(t)\Big)', t \geq 0,$$

where the standard fractional Brownian motion $\{Z_{H_1}^{(1)}(t)\}_{t\geq 0}$ and the standard Rosenblatt process $\{Z_{H_2}^{(2)}(t)\}_{t\geq 0}$ share the same random measure in the multiple integral representations. The components $\{Z_{H_1}^{(1)}(t)\}_{t\geq 0}$ and $\{Z_{H_2}^{(2)}(t)\}_{t\geq 0}$ are uncorrelated but dependent (as stated in Remark 5.7.5). The constant factors in the limit are consistent with (5.2.12) in Corollary 5.2.5.

5.7.3 The Mixed Case

The final result concerns the mixed case. We need to modify slightly the setting used in Theorems 5.7.1 and 5.7.4. We let

$$S_N(t) = \Big(\frac{1}{A_{S,r_1}(N)}\sum_{n=1}^{[Nt]}(G_{S,r_1}(X_n) - \mathbb{E}G_{S,r_1}(X_n))\Big)_{r_1=1,\ldots,R_S}, \tag{5.7.21}$$

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$$L_N(t) = \Big(\frac{1}{A_{L,r_2}(N)} \sum_{n=1}^{[Nt]} (G_{L,r_2}(X_n) - \mathbb{E}G_{L,r_2}(X_n))\Big)_{r_2=1,\dots,R_L}, \qquad (5.7.22)$$

where the letters S and L refer to the expected SRD and LRD behaviors of the two vector processes, respectively. We will suppose that the Hermite ranks $k_{S,r_1} \geq 1$ and $k_{L,r_2} \geq 1$ are such that $k_{S,r_1} > k_{L,r_2}$, indicating that G_{S,r_1} involves higher-order polynomials than G_{L,r_2}, which allows the $G_{S,r_1}(X_n)$s to be "less dependent" than the $G_{L,r_2}(X_n)$s.

Theorem 5.7.8 *(Mixed case.) Let $\{X_n\}_{n\in\mathbb{Z}}$ be a Gaussian stationary series which is LRD in the sense (2.1.5) in condition II with $d \in (0, 1/2)$. Suppose that $\mathbb{E}X_n = 0$, $\mathbb{E}X_n^2 = 1$. Let G_{S,r_1}, $r_1 = 1,\dots,R_S$, be functions with respective Hermite ranks $k_{S,r_1} \geq 1$, $r_1 = 1,\dots,R_S$, and G_{L,r_2}, $r_2 = 1,\dots,R_L$, be functions with respective Hermite ranks $k_{L,r_2} \geq 1$, $r_2 = 1,\dots,R_L$ in the sense of Definition 5.2.1. Suppose*

$$\frac{1}{2}\Big(1 - \frac{1}{k_{L,r_2}}\Big) < d < \frac{1}{2}\Big(1 - \frac{1}{k_{S,r_1}}\Big), \quad r_1 = 1,\dots,R_S,\ r_2 = 1,\dots,R_L, \qquad (5.7.23)$$

which implies $k_{S,r_1} > k_{L,r_2}$. Then,

$$(S_N(t), L_N(t))' \xrightarrow{fdd} (V(t), V^{(d)}(t))', \quad t \geq 0, \qquad (5.7.24)$$

where $V(t)$ denotes the limit of $S_N(t)$ according to Theorem 5.7.1 and $V^{(d)}(t)$ denotes the limit of $L_N(t)$ according to Theorem 5.7.4. Moreover, the processes $\{V(t)\}_{t\geq 0}$ and $\{V^{(d)}(t)\}_{t\geq 0}$ are independent.

Proof Using the reduction arguments as in the proofs of Theorems 5.7.1 and 5.7.4, we can replace $G_{S,r} - \mathbb{E}G_{S,r}$ in (5.7.21) with $\sum_{m=k_{S,r}}^{M} g_{S,r,m} H_m$, and we can replace $G_{L,r}$ in (5.7.22) with $g_{L,r,k_{L,r}} H_{k_{L,r}}$, where $k_{S,r} > k_{L,r}$ are the corresponding Hermite ranks and $g_{S,r,m}$, $g_{L,r,m}$ are the corresponding coefficients of the Hermite expansions.

For fixed finite time points t_j, $j = 1,\dots,J$, we need to consider the joint convergence of the vector

$$\begin{aligned}&(S_{j,r_1,N}, L_{j,r_2,N})_{j,r_1,r_2}\\ &:= \Big(\frac{1}{A_{S,r_1}(N)} \sum_{m=k_{S,r_1}}^{M} g_{S,r_1,m} S_{N,t_j}(H_m), \frac{1}{A_{L,r_2}(N)} g_{L,r_2,k_{L,r_2}} S_{N,t_j}(H_{k_{L,r_2}})\Big)_{j,r_1,r_2},\end{aligned} \qquad (5.7.25)$$

where $j = 1,\dots,J$, $r_1 = 1,\dots,R_S$, $r_2 = 1,\dots,R_L$, and $S_{N,t}(G) = \sum_{n=1}^{[Nt]} G(X_n)$.

As in the proof of Theorem 5.3.1, we can express Hermite polynomials as multiple integrals:

$$S_{j,r_1,N} = \sum_{m=k_{S,r_1}}^{M} I_m(f_{m,j,r_1,N}), \quad L_{j,r_2,N} = I_{k_{L,r_2}}(f_{j,r_2,N}),$$

where $f_{m,j,r_1,N}$, $f_{j,r_2,N}$ are some symmetric square-integrable functions.

Express the vector in (5.7.25) as $(S_N, L_N)'$, where $S_N := (S_{j,r_1,N})_{j,r_1}$, $L_N := (L_{j,r_2,N})_{j,r_2}$. By Theorem 5.7.1, S_N converges in distribution to some multivariate normal distribution, and by Theorem 5.7.4, L_N converges to a multivariate distribution expressed through multiple integrals. The joint convergence follows from Corollary 5.7.11 below. □

Example 5.7.9 (*Bivariate limit result in mixed case*) In the setting of Theorem 5.7.8, suppose $d \in (1/3, 1/4)$ and $L_2(u) \sim 1$ in (2.1.5) of condition II. Let also $G_1(x) = H_3(x) = x^3 - 3x$, $G_2(x) = H_2(x) = x^2 - 1$. Then, $\sigma^2 = 6\sum_{n=-\infty}^{\infty} \gamma_X(n)^3$ and

$$\Big(\frac{1}{N^{1/2}}\sum_{n=1}^{[Nt]}(X_n^3 - 3X_n), \frac{1}{N^{2d}}\sum_{n=1}^{[Nt]}(X_n^2 - 1)\Big)' \xrightarrow{fdd} \Big(\sigma B(t), \frac{1}{\sqrt{d(4d-1)}} Z_H^{(2)}(t)\Big)', \quad t \geq 0,$$

where the standard Rosenblatt process $\{Z_H^{(2)}(t)\}_{t\geq 0}$ and the standard Brownian motion $\{B(t)\}_{t\geq 0}$ are independent.

In the proof of Theorem 5.7.8, we used Corollary 5.7.11, which is based on the following theorem. It is a special case of Proposition 1.5 of Nourdin, Nualart, and Peccati [768].

Theorem 5.7.10 *Consider two vectors of multiple integrals (with respect to a Gaussian random measure):*

$$\xi_N = (I_{p_{r_1}}(f_{r_1,N}))_{r_1=1,\dots,R_1}, \quad \eta_N = (I_{q_{r_2}}(g_{r_2,N}))_{r_2=1,\dots,R_2}, \tag{5.7.26}$$

where $f_{r_1,N}$s and $g_{r_2,N}$s are symmetric square-integrable functions. Let ξ and η be two independent random vectors taking values in $\mathbb{R}^{R_1}$ and $\mathbb{R}^{R_2}$, respectively. Suppose that we have the following marginal convergence in distribution: as $N \to \infty$,

$$\xi_N \xrightarrow{d} \xi, \quad \eta_N \xrightarrow{d} \eta.$$

Assume that we have the pairwise contractions[7] *vanishing:*

$$\lim_{N\to\infty} \|f_{r_1,N} \otimes_u g_{r_2,N}\| = 0, \tag{5.7.27}$$

for all $u = 1, \dots, p_{r_1} \wedge q_{r_2}$, $r_1 = 1, \dots, R_1$ and $r_2 = 1, \dots, R_2$, where $\|\cdot\|$ denotes the $L^2(\mathbb{R}^{p_{r_1}+q_{r_2}-2u})$ norm. Then, we have the joint convergence: as $N \to \infty$,

$$(\xi_N, \eta_N)' \xrightarrow{d} (\xi, \eta)',$$

namely, ξ_N and η_N are asymptotically independent.

This result implies the following corollary.

Corollary 5.7.11 *Consider*

$$S_N = (I_{k_{S,r_1}}(f_{S,r_1,N}))_{r_1=1,\dots,R_S}, \quad L_N = (I_{k_{L,r_2}}(f_{L,r_2,N}))_{r_2=1,\dots,R_L},$$

where $k_{S,r_1} > k_{L,r_2}$, $r_1 = 1, \dots, R_S$, $r_2 = 1, \dots, R_L$. Suppose that, as $N \to \infty$, S_N converges in distribution to a multivariate normal law, and L_N converges in distribution to a multivariate law. Then, there are independent random vectors X and Y such that

$$(S_N, L_N)' \xrightarrow{d} (X, Y)'.$$

[7] Contractions are defined in Appendix B.2. See the relation (B.2.17).

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Proof By Theorem 5.7.10, we need only to check (5.7.27). Let $(\cdot,\cdot)$ denote the inner product in a suitable dimension. By Fubini's theorem, one can write

$$
\begin{aligned}
\|f\otimes_u g\|^2 &= \int_{\mathbb{R}^{p-u}} dx_1\dots dx_{p-u}\int_{\mathbb{R}^{q-u}} dx_{p-u+1}\dots dx_{p+q-2u}\\
&\quad\times\Big[\int_{\mathbb{R}^u} dy_1\dots dy_u f(x_1,\dots,x_{p-u},y_1,\dots,y_u)g(x_{p-u+1},\dots,x_{p+q-2u},y_1,\dots,y_u)\Big]^2\\
&= \int_{\mathbb{R}^u} dy_1\dots dy_u\int_{\mathbb{R}^u} dz_1\dots dz_u\int_{\mathbb{R}^{p-u}} dx_1\dots dx_{p-u}\\
&\quad\times f(x_1,\dots,x_{p-u},y_1,\dots,y_u)f(x_1,\dots,x_{p-u},z_1,\dots,z_u)\\
&\quad\times\int_{\mathbb{R}^{q-u}} dx_{p-u+1}\dots dx_{p+q-2u}\\
&\quad\times g(x_{p-u+1},\dots,x_{p+q-2u},y_1,\dots,y_u)g(x_{p-u+1},\dots,x_{p+q-2u},z_1,\dots,z_u)\\
&= (f\otimes_{p-u} f, g\otimes_{q-u} g),
\end{aligned}
$$

where $u = 1,\dots,p\wedge q$, and f, g have p and q variables, respectively. We get by the Cauchy–Schwarz inequality that for $u = 1,\dots,k_{L,r_2}$, $r_1 = 1,\dots,R_S$, $r_2 = 1,\dots,R_L$,

$$
\begin{aligned}
\|f_{S,r_1,N}\otimes_u f_{L,r_2,N}\|^2 &= (f_{S,r_1,N}\otimes_{k_{S,r_1}-u} f_{S,r_1,N}, f_{L,r_2,N}\otimes_{k_{L,r_2}-u} f_{L,r_2,N})\\
&\le \|f_{S,r_1,N}\otimes_{k_{S,r_1}-u} f_{S,r_1,N}\|\|f_{L,r_2,N}\otimes_{k_{L,r_2}-u} f_{L,r_2,N}\|\to 0
\end{aligned}
$$

because $\|f_{S,r_1,N}\otimes_{k_{S,r_1}-u} f_{S,r_1,N}\|\to 0$ for all $u = 1,\dots,k_{S,r_1}-1$ by the Nualart–Peccati central limit theorem [771]. Since $k_{S,r_1} > k_{L,r_2}$, this indeed covers all $u = 1,\dots,k_{L,r_2}$. For the second term, one has by Fubini's theorem and the Cauchy–Schwarz inequality (see Peccati and Taqqu [797], Lemma 6.2.2) that

$$
\|f_{L,r_2,N}\otimes_{k_{L,r_2}-u} f_{L,r_2,N}\| \le \|f_{L,r_2,N}\|^2, \tag{5.7.28}
$$

for all $u = 1,\dots,k_{S,r_1}-1$, and the right-hand side of (5.7.28) is bounded due to the tightness of the distribution of $I_{k_{L,r_2}}(f_{L,r_2,N})$ (Lemma 2.1 of Nourdin and Rosiński [764]). Therefore (5.7.27) holds. □

We presented above central and non-central limit theorems in the multivariate case when the underlying stationary series is Gaussian. In the linear case discussed in Section 5.6, the non-central limit behavior is typically determined by partial sums of multilinear forms (5.6.4). Multivariate results for multilinear forms are presented in the next Section 5.7.4 and are in the same spirit as those described here.

5.7.4 Multivariate Limits of Multilinear Processes

We consider here processes called *multilinear processes* (or *discrete-chaos processes*), which are defined as

$$
X_n = \sum_{1\le i_1<\dots<i_k<\infty} \psi_{i_1}\dots\psi_{i_k}\epsilon_{n-i_1}\dots\epsilon_{n-i_k},\quad n\in\mathbb{Z}, \tag{5.7.29}
$$

where

$$
\sum_{i=1}^{\infty}\psi_i^2 < \infty, \tag{5.7.30}
$$

ϵ_is are i.i.d. with zero mean and unit variance, and $k \geq 1$ is the *order*. X_n, $n \in \mathbb{Z}$, is also said to belong to a *discrete chaos* of order k. Observe that the summation in (5.7.29) excludes the diagonals, in contrast to the situation considered in Theorem 5.6.3. We are thus focusing on what we may call a "discrete multiple integral."

If $\psi_i = L_1(i)i^{d-1}$ with $0 < d < 1/2$ and a slowly varying function L_1, and when $k > 1$, that is, except for linear processes, the partial sum of X_n when suitably normalized no longer converges to fractional Brownian motion. But depending on d and k, it either converges to an *Hermite process* if $\{X_n\}$ is still LRD, or it converges to Brownian motion if $\{X_n\}$ is SRD. See Giraitis et al. [406] for more details.

Consider the multilinear processes

$$X_{r,n} = \sum_{1 \leq i_1 < \cdots < i_{k_r} < \infty} \psi_{r,i_1} \dots \psi_{r,i_{k_r}} \epsilon_{n-i_1} \dots \epsilon_{n-i_{k_r}}, \quad r = 1, \dots, R, \tag{5.7.31}$$

where $k_1, \dots, k_r$ are the orders for $X_{1,n}, \dots, X_{R,n}$ respectively, and $\{\psi_{r,i}\}$ are coefficients. Let

$$Y_{r,N}(t) = \frac{1}{A_r(N)} \sum_{n=1}^{[Nt]} X_{r,n}, \quad t \geq 0, \tag{5.7.32}$$

where $A_r(N)$ is a normalization factor such that $\lim_{N\to\infty} \mathrm{Var}(Y_{r,N}(1)) = 1$, $r = 1, \dots, R$. We want to study the limit of the following vector process as $N \to \infty$:

$$Y_N(t) = (Y_{1,N}(t), \dots, Y_{R,N}(t))'. \tag{5.7.33}$$

Depending on $\{\psi_{r,i}\}$ and k_r, the components of $Y_N(t)$ can be either SRD, or LRD, or a mixture of SRD and LRD.

We start with some facts about multilinear processes $\{X_n\}$ in (5.7.29). Note first that the condition (5.7.30) guarantees that X_n is well-defined in $L^2(\Omega)$, since

$$\mathbb{E}X_n^2 = \sum_{1 \leq i_1 < \cdots < i_k < \infty} \psi_{i_1}^2 \dots \psi_{i_k}^2 < \infty.$$

We use throughout a convention $\psi_i = 0$ for $i \leq 0$. One can compute the autocovariance of $\{X_n\}$ as

$$\gamma_X(n) = \sum_{1 \leq i_1 < \cdots < i_k < \infty} \psi_{n+i_1} \psi_{i_1} \dots \psi_{n+i_k} \psi_{i_k}, \quad n \in \mathbb{Z}. \tag{5.7.34}$$

The following proposition describes the asymptotic behavior of $\gamma_X(n)$ under suitable assumptions.

Proposition 5.7.12 *Suppose $\gamma_X(n)$ is defined in (5.7.34), $\psi_i = L_1(i)i^{d-1}$, $i \geq 1$, with $0 < d < 1/2$ where L_1 is slowly varying at infinity. Then,*

$$\gamma_X(n) = L^*(n) n^{2d_X - 1}, \tag{5.7.35}$$

for a slowly varying function L^* and

$$d_X = k\Big(d - \frac{1}{2}\Big) + \frac{1}{2}. \tag{5.7.36}$$

Proof By (5.7.34), as $n \to \infty$,

$$\gamma_X(n) \sim (k!)^{-1} \Big(\sum_{i=1}^{\infty} \psi_{n+i} \psi_i \Big)^k$$

(the diagonal terms with $i_p = i_q$ are negligible as $n \to \infty$). Proposition 2.2.9 then yields (5.7.35) with (5.7.36) and $L^*(n) \sim (k!)^{-1} B(d, 1-2d)^k L_1(n)^{2k}$. □

Remark 5.7.13 According to Proposition 5.7.12, when $d < \frac{1}{2}(1 - \frac{1}{k})$, we have $\sum |\gamma_X(n)| < \infty$, and when $d > \frac{1}{2}(1-\frac{1}{k})$, we have $\sum |\gamma_X(n)| = \infty$. So for $\psi_i = L_1(i) i^{d-1}$, $0 < d < 1/2$, the quantity $\frac{1}{2}(1 - \frac{1}{k})$ is the boundary between SRD and LRD.

We now define precisely what SRD and LRD mean for a multilinear process $\{X_n\}$, and from then on we use this definition whenever we talk about SRD or LRD.

Definition 5.7.14 *Let $\{X_n\}$ be a multilinear process given in (5.7.29) with coefficients ψ_i, autocovariances $\gamma_X(n)$ and order k. We say that $\{X_n\}$ is:*

(a) SRD if for some $d \in (-\infty, \frac{1}{2}(1 - \frac{1}{k}))$ and some constant $c > 0$,

$$|\psi_i| \le c i^{d-1}, \quad i \ge 1, \quad \sum_{n=-\infty}^{\infty} \gamma_X(n) > 0; \tag{5.7.37}$$

(b) LRD if for some $d \in (\frac{1}{2}(1 - \frac{1}{k}), \frac{1}{2})$ and some L_1 slowly varying at infinity,

$$\psi_i = L_1(i) i^{d-1}, \quad i \ge 1. \tag{5.7.38}$$

Remark 5.7.15 The ds in (5.7.37) and (5.7.38) are different. In the SRD case, $\{\psi_i\}$ is only assumed to decay faster than a power function, which implies

$$\sum_{n=0}^{\infty} |\gamma_X(n)| \le \sum_{n=0}^{\infty} \Big(\sum_{i=1}^{\infty} |\psi_{n+i} \psi_i| \Big)^k \le c^{2k} \sum_{n=0}^{\infty} \Big(\sum_{i=1}^{\infty} (i+n)^{d-1} i^{d-1} \Big)^k. \tag{5.7.39}$$

The last bound can be shown to be finite when $d < \frac{1}{2}(1 - \frac{1}{k})$ (Exercise 5.8). In contrast, in the LRD case, the regular variation assumption on $\{\psi_i\}$ yields a *memory parameter* $d_X = k(d - 1/2) + 1/2$ given by (5.7.36).

Next we consider the cross-covariance between two multilinear processes defined by

$$X_{1,n} = \sum_{1 \le i_1 < \cdots < i_p < \infty} \psi_{i_1} \dots \psi_{i_p} \epsilon_{n-i_1} \dots \epsilon_{n-i_p}, \tag{5.7.40}$$

$$X_{2,n} = \sum_{1 \le i_1 < \cdots < i_q < \infty} \phi_{i_1} \dots \phi_{i_q} \epsilon_{n-i_1} \dots \epsilon_{n-i_q}. \tag{5.7.41}$$

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$\{X_{1,n}\}$ and $\{X_{2,n}\}$ share the same series $\{\epsilon_i\}$ but the sequences $\{\psi_i\}$ and $\{\phi_i\}$ can be different. Then, the cross-covariance is

$$\operatorname{Cov}(X_{1,n}, X_{2,0}) = \begin{cases} 0, & \text{if } p \neq q, \\ \sum_{1\le i_1<\cdots<i_k<\infty} \psi_{i_1}\phi_{n+i_1}\ldots\psi_{i_k}\phi_{n+i_k}, & \text{if } p = q = k, \end{cases} \tag{5.7.42}$$

for any $n \in \mathbb{Z}$.

We now state the multivariate joint convergence results for the vector process $Y_N(t)$ in (5.7.33) (the proofs can be found in Bai and Taqqu [75]). Recall that Y_N is normalized so that the asymptotic variance of every component at $t = 1$ equals 1.

Theorem 5.7.16 *(SRD case.) If all the components in Y_N defined in (5.7.33) are SRD in the sense of (5.7.37), then*

$$Y_N(t) \xrightarrow{fdd} B(t) = (B_1(t), \ldots, B_R(t))', \quad t \ge 0,$$

where $\{B(t)\}$ is a multivariate Gaussian process with $B_1(t), \ldots, B_R(t)$ being standard Brownian motions with

$$\operatorname{Cov}(B_{r_1}(t_1), B_{r_2}(t_2)) = (t_1 \wedge t_2)\frac{\sigma_{r_1,r_2}}{\sigma_{r_1}\sigma_{r_2}} \tag{5.7.43}$$

and

$$\sigma_r^2 = \sum_{n=-\infty}^{\infty} \operatorname{Cov}(X_{r,n}, X_{r,0}), \quad \sigma_{r_1,r_2} = \sum_{n=-\infty}^{\infty} \operatorname{Cov}(X_{r_1,n}, X_{r_2,0}).$$

The normalization $A_r(N)$ in (5.7.32) satisfies $A_r(N) \sim \sigma_r\sqrt{N}$ as $N \to \infty$.

Remark 5.7.17 In view of (5.7.42) and (5.7.43), if all the components of $Y_N(t)$ have a different order, then the limit components $B_r(t)$ are uncorrelated and hence independent. Otherwise, they are in general dependent and their covariance is given by (5.7.43).

Theorem 5.7.18 *(LRD case.) If all the components in Y_N defined in (5.7.33) are LRD in the sense of (5.7.38) with $d = d_1, \ldots, d_R$, respectively, then*

$$Y_N(t) \xrightarrow{fdd} Z^{(d,k)}(t) = (Z_{H_1}^{(k_1)}(t), \ldots, Z_{H_R}^{(k_R)}(t))', \quad t \ge 0,$$

where

$$Z_H^{(k)}(t) = A_{k,H}\int'_{\mathbb{R}^k}\Big(\int_0^t \prod_{j=1}^k (s-u_j)_+^{d-1}ds\Big)W(du_1)\ldots W(du_k), \tag{5.7.44}$$

where $W(du)$ is a Gaussian random measure on $\mathbb{R}$ with control measure du, $H = k(d - 1/2) + 1$ and $A_{k,H}$ is given by (4.2.7). The standard Hermite processes $Z_{H_r}^{(k_r)}(t)$s are sharing the same Gaussian random measure $W(du)$ in their multiple integral representations, and are dependent across r. The normalization $A_r(N)$ in (5.7.32) satisfies: for some $c_r > 0$, as $N \to \infty$,

$$A_r(N) \sim c_r N^{k_r(d_r-1/2)+1} L_1(N)^{k_r/2}.$$

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Finally, we consider the mixed SRD and LRD case.

Theorem 5.7.19 *(Mixed case.) Break Y_N in (5.7.33) into 3 parts: $Y_N = (Y_{S_1,N}, Y_{S_2,N}, Y_{L,N})'$, where within $Y_{S_1,N}$ (R_{S_1}–dimensional) every component is SRD and has order $k_{S_1,r} = 1$, within $Y_{S_2,N}$ (R_{S_2}–dimensional) every component is SRD and has order $k_{S_2,r} \geq 2$, and within $Y_{L,N}$ (R_L–dimensional) every component is LRD. Then,*

$$Y_N(t) = (Y_{S_1,N}(t), Y_{S_2,N}(t), Y_{L,N}(t))' \xrightarrow{fdd} (W^{(1)}(t), B(t), Z^{(d,k)}(t))', \quad t \geq 0, \tag{5.7.45}$$

where $B(t) = (B_1(t), \dots, B_{R_{S_2}}(t))'$ is the multivariate Gaussian process appearing in Theorem 5.7.16, $Z^{(d,k)}(t)$ is the multivariate Hermite process appearing in Theorem 5.7.18,

$$W^{(1)}(t) = (W(t), \dots, W(t))', \tag{5.7.46}$$

where $W(t) = \int_0^t W(du)$ is the Brownian motion integrator for defining $Z^{(d,k)}(t)$ (see (5.7.44)). Moreover, $\{B(t)\}$ is independent of $\{W^{(1)}(t), Z^{(d,k)}(t)\}$.

Remark 5.7.20 To understand heuristically why $\{B(t)\}$ and $\{W^{(1)}(t), Z^{(d,k)}(t)\}$ are independent, note that $Y_{S_2,N}(t)$ belongs to the chaos of order greater than or equal to two, and is thus uncorrelated with $Y_{S_1,N}(t)$ which belongs to the first-order chaos, and also uncorrelated with the random noise $\{\epsilon_i\}$ which also belongs to the first-order chaos, and which after summing becomes asymptotically the Gaussian random measure W defining $Z^{(d,k)}(t)$. It is therefore simpler to establish independence in the context of Theorem 5.7.19 than in the context of Theorem 5.7.8.

5.8 Generation of Non-Gaussian Long- and Short-Range Dependent Series

Nonlinear transformations of Gaussian series provide interesting means to generate non-Gaussian stationary series. By non-Gaussian, we mean a stationary series having a particular non-Gaussian marginal distribution function. Thus, consider a stationary Gaussian series $X = \{X_n\}_{n\in\mathbb{Z}}$ with

$$\mathbb{E}X_n = 0, \quad \mathbb{E}X_n^2 = 1. \tag{5.8.1}$$

Let γ_X and ρ_X denote the autocovariance and autocorrelation functions of the series X. Because of (5.8.1), $\gamma_X(0) = 1$ and hence $\gamma_X = \rho_X$. We focus on the series $Y = \{Y_n\}_{n\in\mathbb{Z}}$ defined by

$$Y_n = G(X_n), \tag{5.8.2}$$

where $G : \mathbb{R} \mapsto \mathbb{R}$ is a deterministic function. Let γ_Y and ρ_Y denote the autocovariance and autocorrelation functions of the series Y. Let also F_Y be the marginal (cumulative) distribution function of Y.

The function G in (5.8.2) determines the marginal distribution F_Y and, as will be seen below, the autocovariance function γ_Y of the series Y. We will first discuss choices of G in terms of the marginal distribution, then study relationships between the autocovariance functions of X and Y, and finally consider the problem of generating series with a given marginal and autocovariance structure, using the construction (5.8.2).

5.8.1 Matching a Marginal Distribution

If a particular marginal distribution F_Y is sought, this may suggest a natural choice of the function G. For example, the choice

$$G(x) = x^2 \tag{5.8.3}$$

leads to the χ_1^2 (chi-square with one degree of freedom) distribution,

$$G(x) = e^{\sigma x+\mu}, \quad \sigma > 0, \mu \in \mathbb{R}, \tag{5.8.4}$$

leads to a log-normal distribution, and

$$G(x) = \begin{cases} 1, & \text{if } x \geq 0, \\ 0, & \text{if } x < 0, \end{cases} \tag{5.8.5}$$

leads to the Bernoulli distribution with parameter $1/2$. This is not the only way to obtain these marginal distributions.

A marginal distribution F_Y could also be obtained through a general construction. One such standard construction is

$$G_1(x) = F_Y^{-1}(\Phi(x)), \tag{5.8.6}$$

where $F_Y^{-1}(u) = \inf\{y : F_Y(y) \geq u\}$, $0 < u < 1$, is the generalized inverse (the quantile function) and Φ is the distribution function of a standard normal random variable. The basic idea behind (5.8.6) is that $\Phi(X)$ is a $U(0,1)$ (uniform on $(0,1)$) random variable for a standard normal variable X because, for $0 \leq y \leq 1$,

$$\mathbb{P}(\Phi(X) \leq y) = \mathbb{P}(X \leq \Phi^{-1}(y)) = \Phi(\Phi^{-1}(y)) = y,$$

and $F_Y^{-1}(U(0,1))$ has distribution F_Y. Another general construction is

$$G_2(x) = F_Y^{-1}(2(\Phi(|x|) - 1/2)), \tag{5.8.7}$$

by using the fact that $2(\Phi(|X|) - 1/2)$ is also distributed as $U(0,1)$ for a standard normal variable X. Indeed, since $\mathbb{P}(|X| \leq a) = 2\Phi(a) - 1$, then with $0 \leq y \leq 1$ and $a = (y+1)/2$, we have

$$\begin{aligned} \mathbb{P}(2(\Phi(|X|) - 1/2) \leq y) &= \mathbb{P}(\Phi(|X|) \leq a) = \mathbb{P}(|X| \leq \Phi^{-1}(a)) \\ &= 2\Phi(\Phi^{-1}(a)) - 1 = 2a - 1 = y. \end{aligned}$$

We gather these and other basic facts about the transformations (5.8.6) and (5.8.7) in the following proposition.

Proposition 5.8.1 *Let Y be a random variable with distribution function F_Y. Let X be a standard normal variable, and G_1 and G_2 be the transformations given in (5.8.6) and (5.8.7). Then, the distribution functions of the random variables $G_1(X)$ and $G_2(X)$ are both equal to F_Y. Moreover, G_1 is a non-decreasing function and, if $G_1 \in L^2(\mathbb{R}, \phi)$ with $\phi(dx)$ defined in (5.1.2), then the first coefficient in the Hermite expansion (5.1.6) of G_1 is*

$$g_1 = \mathbb{E}G_1(X)H_1(X) = \mathbb{E}G_1(X)X = \frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}} e^{-\frac{x^2}{2}} dG_1(x) > 0 \tag{5.8.8}$$

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and hence G_1 has Hermite rank 1. On the other hand, the function G_2 is even and, if $G_2 \in L^2(\mathbb{R}, \phi)$, then the first coefficient in the Hermite expansion of G_2 is $g_1 = 0$ and hence G_2 has Hermite rank greater than or equal to 2.

Proof The fact that $G_1(X)$ and $G_2(X)$ have distribution F_Y is argued before the proposition. G_1 is non-decreasing because Φ and F_Y^{-1} are non-decreasing. The formula (5.8.8) follows by integration by parts:

$$g_1 = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} G_1(x) x e^{-\frac{x^2}{2}} dx = -\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} G_1(x) d e^{-\frac{x^2}{2}} = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-\frac{x^2}{2}} dG_1(x) > 0,$$

since $\lim_{x\to\pm\infty} G_1(x) e^{-\frac{x^2}{2}} = 0$. The latter fact is a consequence of $\int_{\mathbb{R}} |G_1(x)| e^{-\frac{x^2}{2}} dx < \infty$, that is, $G_1 \in L^2(\mathbb{R}, \phi)$, and $G_1(x)$ being non-decreasing. The statements of the proposition about G_2 are obvious since G_2 is even. □

Proposition 5.8.1 shows that the functions G_1 and G_2 in (5.8.6) and (5.8.7) are very different.

Example 5.8.2 (*Transformations for various marginals*) The function G in (5.8.3) is G_2 for the χ_1^2 distribution. Indeed, note that in this case, for $y > 0$ and for a standard normal random variable X,

$$F_Y(y) = \mathbb{P}(X^2 \le y) = \mathbb{P}(-\sqrt{y} \le X \le \sqrt{y}) = \Phi(\sqrt{y}) - \Phi(-\sqrt{y}) = 2\Phi(\sqrt{y}) - 1$$

and hence

$$G_2(x) = F_Y^{-1}(2\Phi(|x|) - 1) = F_Y^{-1}(F_Y(x^2)) = x^2.$$

On the other hand, the functions G in (5.8.4) and (5.8.5) are G_1. In the case of (5.8.4), note that for $y > 0$,

$$F_Y(y) = \mathbb{P}(e^{\sigma X + \mu} \le y) = \Phi((\log y - \mu)/\sigma)$$

and hence

$$F_Y^{-1}(u) = e^{\sigma \Phi^{-1}(u) + \mu},$$

yielding $G_1(x) = F_Y^{-1}(\Phi(x)) = e^{\sigma x + \mu} = G(x)$. In the case of (5.8.5), Y is Bernoulli with parameter $1/2$. Then, the function G is $G_1(x) = F_Y^{-1}(\Phi(x))$, where

$$F_Y(y) = \begin{cases} 0, & \text{if } y < 0, \\ 1/2, & \text{if } 0 \le y < 1, \\ 1, & \text{if } y \ge 1, \end{cases} \qquad F_Y^{-1}(u) = \begin{cases} 0, & \text{if } 0 < u < 1/2, \\ 1, & \text{if } 1/2 \le u < 1, \end{cases}$$

yielding

$$F_Y^{-1}(\Phi(x)) = \begin{cases} 0, & \text{if } x < 0, \\ 1, & \text{if } x \ge 0 \end{cases} = G(x).$$

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5.8.2 Relationship Between Autocorrelations

We now turn to the relationships between the autocovariance functions of X and Y. By (5.2.1), we have

$$\gamma_Y(n) = \sum_{k=1}^{\infty} g_k^2 k!(\gamma_X(n))^k, \tag{5.8.9}$$

where g_k are the coefficients in the Hermite expansion (5.1.6) of G. The respective autocorrelation functions can then be related as

$$\rho_Y(n) = \sum_{k=1}^{\infty} \frac{g_k^2 k!}{\gamma_Y(0)} (\rho_X(n))^k, \tag{5.8.10}$$

since $\rho_X = \gamma_X$ by the assumption $\gamma_X(0) = 1$. We set

$$g(z) = \sum_{k=1}^{\infty} \frac{g_k^2 k!}{\gamma_Y(0)} z^k, \tag{5.8.11}$$

a function which will be used often below. We have then

$$\rho_Y(n) = g(\rho_X(n)). \tag{5.8.12}$$

Since an autocorrelation function takes values in $[-1, 1]$, we will be interested in the values of $g(z)$ for $z \in [-1, 1]$. Note that g is always well-defined for $z \in [-1, 1]$, $g(1) = \sum_{k=1}^{\infty} g_k^2 k!/\gamma_Y(0) = \mathrm{Var}(G(X_0))/\gamma_Y(0) = 1$, $g(z)$ is non-decreasing for $z \in (0, 1)$, and $|g(z)| \leq 1$ for $z \in (-1, 1)$.

Example 5.8.3 (*Functions g for various transformations*) (a) Consider the function $G(x) = x^2$ in (5.8.3) leading to the χ_1^2 distribution. Since

$$G(x) = x^2 = 1 + (x^2 - 1) = H_0(x) + H_2(x), \tag{5.8.13}$$

the only nonzero coefficients g_k in the Hermite expansion are $g_0 = 1$, $g_1 = 1$. Then, by (5.8.9), $\gamma_Y(0) = g_2^2 2! = 2$ and hence

$$g(z) = z^2, \tag{5.8.14}$$

so that

$$\rho_Y(n) = (\rho_X(n))^2. \tag{5.8.15}$$

(b) Consider the function $G(x) = e^{\sigma x + \mu}$, $\sigma > 0$, $\mu \in \mathbb{R}$, in (5.8.4) leading to a log-normal distribution. Suppose first that $\mu = 0$. Using the identity (4.1.2), we have

$$G(x) = e^{\sigma x} = \sum_{k=0}^{\infty} \frac{e^{\sigma^2/2}\sigma^k}{k!} H_k(x), \tag{5.8.16}$$

so that $g_k = e^{\sigma^2/2}\sigma^k/k!$. Since $\mathbb{E}H_0(X) = 1$ and $\mathbb{E}H_k(X) = 0$ for $k \geq 1$, we get $\mathbb{E}e^{\sigma X} = e^{\sigma^2/2}$ and $\mathbb{E}(e^{\sigma X})^2 = \mathbb{E}e^{2\sigma X} = e^{(4\sigma^2)/2} = e^{2\sigma^2}$, so that $\gamma_Y(0) = \mathrm{Var}(e^{\sigma X}) = \mathbb{E}e^{2\sigma X} -$

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$(\mathbb{E}e^{\sigma X})^2 = e^{2\sigma^2} - e^{\sigma^2} =: c_\sigma^2$. It follows that

$$g(z) = \sum_{k=1}^{\infty} \frac{g_k^2 k!}{\gamma_Y(0)} z^k = \sum_{k=1}^{\infty} \frac{e^{\sigma^2}\sigma^{2k}k!}{c_\sigma^2 (k!)^2} z^k = \frac{e^{\sigma^2}}{c_\sigma^2}\sum_{k=1}^{\infty}\frac{(\sigma^2 z)^k}{k!} = \frac{e^{\sigma^2}}{c_\sigma^2}(e^{\sigma^2 z} - 1) \qquad (5.8.17)$$

and hence

$$\rho_Y(n) = g(\rho_X(n)) = \frac{e^{\sigma^2}}{c_\sigma^2}(e^{\sigma^2\rho_X(n)} - 1). \qquad (5.8.18)$$

In the case of general μ, the multiplicative factor e^μ does not affect the correlation $\rho_Y(n)$ and the relation (5.8.18) continues to hold.

(c) Consider the function $G(x) = 1_{[0,\infty)}(x)$ in (5.8.5) leading to the Bernoulli distribution with parameter $1/2$. Instead of using the Hermite expansion of $G(x)$ as above (which is not that trivial for $G(x) = 1_{[0,\infty)}(x)$) to obtain the relationship between ρ_Y and ρ_X (and the corresponding function g), we shall specify the latter directly by using a well-known formula in Statistics. Observe that

$$\begin{aligned}\rho_Y(n) = \frac{\mathbb{E}Y_nY_0 - \mathbb{E}Y_n\mathbb{E}Y_0}{\mathrm{Var}(Y_0)} &= \frac{\mathbb{E}1_{[0,\infty)}(X_n)1_{[0,\infty)}(X_0) - \mathbb{E}1_{[0,\infty)}(X_n)\mathbb{E}1_{[0,\infty)}(X_0)}{\mathrm{Var}(1_{[0,\infty)}(X_0))} \\ &= 4\Big(\mathbb{P}(X_n \ge 0, X_0 \ge 0) - \frac{1}{4}\Big), \qquad (5.8.19)\end{aligned}$$

using the fact that since X_n is centered Gaussian and hence symmetric, we have $\mathbb{E}1_{[0,\infty)}(X_n) = 1/2$. The probability in (5.8.19) is that of a bivariate, mean zero normal vector belonging to the orthant $[0,\infty)^2$. More generally, the probability P_n of a n–variate, mean zero normal variable to belong to the orthant $[0,\infty)^n$, $n \ge 1$, has been studied extensively in Statistics. It is well known (e.g., Kendall and Stuart [552], Section 15.6; Kotz, Balakrishnan, and Johnson [577], Chapter 45, Section 5.2) that, in the bivariate case $n = 2$,

$$P_2 = \frac{1}{4} + \frac{1}{2\pi}\arcsin(\rho), \qquad (5.8.20)$$

where ρ is the correlation between the two components of the bivariate normal distribution. The relations (5.8.19) and (5.8.20) yield

$$\rho_Y(n) = \frac{2}{\pi}\arcsin(\rho_X(n)) =: g(\rho_X(n)). \qquad (5.8.21)$$

In fact, a commonly found proof of (5.8.21) is akin to using the Price theorem as we outline in Example 5.8.6 below.

Note from (5.8.11) and the examples above that through the coefficients g_k, the transformation G determines the function g in the relationship (5.8.12) between autocorrelations ρ_Y and ρ_X. Since the same marginal distribution can be achieved through various transformations G (e.g., (5.8.6) and (5.8.7)), the corresponding functions g and relationships between autocorrelation functions will be different. Consider, for example, the χ_1^2 distribution. It can be obtained through the transformation (5.8.3) but also through the transformation

$$G(x) = F_{\chi_1^2}^{-1}(\Phi(x)), \qquad (5.8.22)$$

where $F_{\chi_1^2}$ is the χ_1^2 distribution function. This is a special case of (5.8.6) which also leads to the χ_1^2 distribution. Its coefficients in the Hermite expansion can only be computed numerically. By Proposition 5.8.1, the first coefficient g_1 in the Hermite expansion of G in (5.8.22) is positive. As a consequence, the corresponding function g will take negative values (at least for negative z close to 0) and hence it will be very different from g in (5.8.14), which can only be nonnegative, associated with the transformation (5.8.13). As a further example, the log-normal distribution can be obtained through the transformation (5.8.4) but also through the transformation

$$G(x) = F_{LN}^{-1}(2(\Phi(|x|) - 1/2)), \tag{5.8.23}$$

where F_{LN} is the log-normal distribution function. The (nonnegative) function g corresponding to (5.8.23) will be different from the g in (5.8.17). For further discussion, see Section 5.8.4.

5.8.3 Price Theorem

The function G relates X_n to Y_n in the relation $Y_n = G(X_n)$ and the function g relates the autocorrelation of $\{X_n\}$ to the autocorrelation of $\{Y_n\}$. The Price theorem provides a differential equation for g, which can be solved to obtain g. In order to obtain this differential equation, we need to examine the derivatives of g. In this regard, the following auxiliary result will be useful. Recall the definition of the measure $\phi(dx)$ in (5.1.2).

Proposition 5.8.4 *Let $G \in L^2(\mathbb{R}, \phi)$ be n–times differentiable with $G^{(\ell)} = d^\ell G/dx^\ell \in L^2(\mathbb{R}, \phi)$, $\ell = 1, \dots, n$. Suppose $\lim_{x\to\pm\infty} e^{-x^2/2}x^k G^{(\ell)}(x) = 0$ for $l = 1, \dots, n-1$, and all $k \geq 1$. Then, for $\ell = 1, \dots, n-1$, and $k \geq 0$,*

$$g_k^{(n)} = (k+1)(k+2)\dots(k+n-\ell)g_{k+n-\ell}^{(\ell)} = (k+1)(k+2)\dots(k+n)g_{k+n}, \tag{5.8.24}$$

where g_k and $g_k^{(\ell)}$ are the Hermite coefficients of G and $G^{(\ell)}$, respectively.

Proof Observe that by (5.1.7), (4.1.1) and integration by parts,

$$\begin{aligned}
g_k^{(n)} &= \frac{1}{k!}\mathbb{E}G^{(n)}(X)H_k(X) = \frac{1}{k!}\int_{\mathbb{R}} G^{(n)}(x)H_k(x)e^{-x^2/2}\frac{dx}{\sqrt{2\pi}} \\
&= \frac{1}{k!}\int_{\mathbb{R}} G^{(n)}(x)(-1)^k\frac{d^k}{dx^k}(e^{-x^2/2})\frac{dx}{\sqrt{2\pi}} = -(-1)^k\frac{1}{k!}\int_{\mathbb{R}} G^{(n-1)}(x)\frac{d^{k+1}}{dx^{k+1}}(e^{-x^2/2})\frac{dx}{\sqrt{2\pi}} \\
&= (k+1)\frac{1}{(k+1)!}\int_{\mathbb{R}} G^{(n-1)}(x)(-1)^{k+1}\frac{d^{k+1}}{dx^{k+1}}(e^{-x^2/2})\frac{dx}{\sqrt{2\pi}} = (k+1)g_{k+1}^{(n-1)}.
\end{aligned}$$

The relations in (5.8.24) follow by iterating the argument above. □

Proposition 5.8.4 yields the following result, known as the Price theorem, after Price [832].

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Theorem 5.8.5 (*Price theorem.*) *Let g be the function in (5.8.11). Under the assumptions and notation of Proposition 5.8.4, we have for $z \in (-1, 1)$,*

$$\frac{d^n g}{dz^n}(z) = \frac{1}{\gamma_Y(0)} \sum_{k=0}^{\infty} (g_k^{(n)})^2 k! z^k = \frac{1}{\gamma_Y(0)} \mathbb{E} G^{(n)}(X_0) G^{(n)}(X_1) \Big|_{\gamma_X(1)=z}. \tag{5.8.25}$$

Proof The result follows from

$$\begin{aligned}
\frac{d^n g}{dz^n}(z) &= \sum_{k=n}^{\infty} \frac{g_k^2 k!}{\gamma_Y(0)} k(k-1)\dots(k-n+1) z^{k-n} \\
&= \frac{1}{\gamma_Y(0)} \sum_{\ell=0}^{\infty} g_{\ell+n}^2 (\ell+n)!(\ell+n)(\ell+n-1)\dots(\ell+1) z^{\ell} \\
&= \frac{1}{\gamma_Y(0)} \sum_{\ell=0}^{\infty} \Big(g_{\ell+n}(\ell+1)\dots(\ell+n) \Big)^2 \ell! z^{\ell} \\
&= \frac{1}{\gamma_Y(0)} \sum_{\ell=0}^{\infty} (g_{\ell}^{(n)})^2 \ell! z^{\ell} = \frac{1}{\gamma_Y(0)} \mathbb{E} G^{(n)}(X_0) G^{(n)}(X_1) \Big|_{\gamma_X(1)=z},
\end{aligned}$$

where we have used (5.8.24) and, in the last step, (5.2.1). The differentiation in the first step above is valid since $\sum_{\ell=0}^{\infty} (g_{\ell}^{(n)})^2 \ell! = \sum_{k=n}^{\infty} g_k^2 k! k(k-1)\dots(k-n+1) < \infty$ by the assumption $G^{(n)} \in L^2(\mathbb{R}, \phi)$. □

We illustrate next how Theorem 5.8.5 can be used. On one hand, as the following example shows, it can be used to derive an expression for g.

Example 5.8.6 (*Functions g through Price theorem*) (*a*) Consider the transformation $G(x) = x^2$ in (5.8.13). If X has the $\mathcal{N}(0, 1)$ distribution, then $Y = X^2$ satisfies $\gamma_Y(0) = \mathrm{Var}(Y) = \mathrm{Var}(X^2) = 2$. Applying (5.8.25) with $n = 1$ and observing that $G^{(1)}(x) = 2x$ yields

$$\frac{dg}{dz}(z) = \frac{4}{2} \mathbb{E} X_0 X_1 \Big|_{\gamma_X(1)=z} = 2z.$$

Solving this differential equation for $g(z)$ yields $g(z) = z^2 + C$. The constant $C = 0$ since $g(0) = 0$. This yields $g(z) = z^2$, which agrees with (5.8.14).

(*b*) In the case $G(x) = e^{\sigma x}$ in (5.8.16), we get that

$$\frac{dg}{dz}(z) = \frac{\sigma^2}{c_\sigma^2} \mathbb{E} e^{\sigma X_0} e^{\sigma X_1} \Big|_{\gamma_X(1)=z}.$$

Note that, by (5.8.12) and the fact that X_n is standardized (so that $\rho_X(n) = \gamma_X(n)$), we have $\gamma_Y(n) = \mathbb{E} e^{\sigma X_n} e^{\sigma X_0} - \mathbb{E} e^{\sigma X_n} \mathbb{E} e^{\sigma X_0} = \gamma_Y(0) g(\gamma_X(n))$. Hence, after setting $n = 1$ and $\gamma_X(1) = z$,

$$\mathbb{E} e^{\sigma X_0} e^{\sigma X_1} \Big|_{\gamma_X(1)=z} = \gamma_Y(0) g(z) + (\mathbb{E} e^{\sigma X})^2.$$

As noted in Example 5.8.3, (b), $\gamma_Y(0) = e^{2\sigma^2} - e^{\sigma^2} =: c_\sigma^2$ and $\mathbb{E}e^{\sigma X} = e^{\sigma^2/2}$. Hence, the differential equation for g becomes

$$\frac{dg}{dz}(z) = \sigma^2 g(z) + \frac{\sigma^2 e^{\sigma^2}}{c_\sigma^2}.$$

The solution to this differential equation is $g(z) = Ce^{\sigma^2 z} - e^{\sigma^2}/c_\sigma^2$. The initial condition $g(0) = 0$ yields $C = e^{\sigma^2}/c_\sigma^2$ and hence

$$g(z) = \frac{\sigma^2 e^{\sigma^2}}{c_\sigma^2}(e^{\sigma^2 z} - 1),$$

which agrees with (5.8.18).

(c) Strictly speaking, the case $G(x) = 1_{[0,\infty)}(x)$ in (5.8.5) is not covered by the Price theorem since the function $G(x)$ is not differentiable at $x = 0$. But approximating G by, for example,

$$G_\epsilon(x) = \int_{\mathbb{R}} G(y)\frac{1}{\sqrt{2\pi}\epsilon}e^{-\frac{(x-y)^2}{2\epsilon^2}}dy = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}\epsilon}e^{-\frac{z^2}{2\epsilon^2}}dz,$$

as $\epsilon \to 0$, we have for its derivative $G_\epsilon^{(1)}(x) = \frac{1}{\sqrt{2\pi}\epsilon}e^{-\frac{x^2}{2\epsilon^2}}$ and the relation (5.8.25) becomes

$$\frac{dg_\epsilon}{dz}(z) = \frac{1}{\mathrm{Var}(G_\epsilon(X_0))}\mathbb{E}G_\epsilon^{(1)}(X_0)G_\epsilon^{(1)}(X_1)\Big|_{\gamma_X(1)=z}.$$

As $\epsilon \to 0$, $\mathrm{Var}(G_\epsilon(X_0)) \to \mathrm{Var}(G(X_0)) = 1/4$ and

$$\begin{aligned}\mathbb{E}G_\epsilon^{(1)}(X_0)G_\epsilon^{(1)}(X_1)\Big|_{\gamma_X(1)=z} &= \int_{\mathbb{R}}\int_{\mathbb{R}} \frac{e^{-\frac{x_0^2}{2\epsilon^2}}}{\sqrt{2\pi}\epsilon}\frac{e^{-\frac{x_1^2}{2\epsilon^2}}}{\sqrt{2\pi}\epsilon}\frac{1}{2\pi(1-z^2)^{1/2}}e^{-\frac{1}{2(1-z^2)}(x_0^2+x_1^2-2zx_0x_1)}dx_0dx_1\\ &\to \frac{1}{2\pi(1-z^2)^{1/2}},\end{aligned}$$

since $\frac{1}{\sqrt{2\pi}\epsilon}e^{-\frac{x_j^2}{2\epsilon^2}}$, $j = 0, 1$, act like kernel (or Dirac) functions localizing the integrand at $x_0 = x_1 = 0$. This suggests that, in the limit $\epsilon \to 0$,

$$\frac{dg}{dz}(z) = \frac{2}{\pi(1-z^2)^{1/2}}.$$

A general solution to this equation is

$$g(z) = \frac{2}{\pi}\arcsin(z) + C$$

and the constant $C = 0$ since $g(0) = 0$. The resulting function g is then the same as given in (5.8.21).

Here is another useful application of Theorem 5.8.5.

Corollary 5.8.7 *If $G \in L^2(\mathbb{R}, \phi)$ is an increasing function with $G^{(1)} \in L^2(\mathbb{R}, \phi)$, then $g(z)$ is an increasing function on $z \in (-1, 1)$.*

Proof By (5.8.25),

$$\frac{dg}{dz}(z) = \frac{1}{\gamma(0)} \mathbb{E} G^{(1)}(X_0) G^{(1)}(X_1) \Big|_{\gamma_X(1)=z}.$$

(There is no need to assume $\lim_{x\to\pm\infty} e^{-x^2/2} x^k G(x) = 0$ as this is satisfied arguing as in the proof of Proposition 5.8.1.) Since G is increasing, $G^{(1)}(x) > 0$ and hence $dg/dz(z) > 0$. □

Corollary 5.8.7 applies to the functions G in (5.8.6) when F_Y is continuous. Note also that the corollary is not obvious for $z \in (-1, 0)$ only. For $z \in (0, 1)$, the function g is always increasing by construction.

5.8.4 Matching a Targeted Autocovariance for Series with Prescribed Marginal

Consider now the problem of generating the stationary series $\{Y_n\}$ with a prescribed autocorrelation function ρ_Y and marginal distribution function F_Y. We focus on the constructions of the type $Y_n = G(X_n)$. The marginal distribution Y_n can then be obtained by using a suitable transformation G (Section 5.8.1). Such a transformation is not unique and its choice will affect the procedure for matching the autocovariance, and other order properties of the series, as well.

Suppose a transformation G was chosen to match a desired marginal F_Y. Is there an autocorrelation function ρ_X of the underlying Gaussian series $\{X_n\}$ so that $\{Y_n\}$ has the desired autocorrelation function ρ_Y? The relation (5.8.12) suggests to set

$$\rho_X(n) = g^{-1}(\rho_Y(n)), \tag{5.8.26}$$

where g^{-1} is the inverse of the function g.

The choice (5.8.26), however, may neither be possible nor yield a valid autocorrelation function. More precisely, on one hand, the choice of G and hence that of g may restrict the domain of g^{-1} for which $g^{-1}(\rho_Y(n))$ is even defined. Denote this domain by

$$\mathcal{R}_g = \{g(z) : z \in [-1, 1]\}, \tag{5.8.27}$$

which is also the range of g on $[-1, 1]$. For example, for $G(x) = x^2$ in (5.8.13), $g(z) = z^2$ so that $\mathcal{R}_g = [0, 1]$ and hence $g^{-1}(\rho_Y(n))$ is defined only for $\rho_Y(n) \geq 0$. In other words, with the transformation (5.8.13), only nonnegative autocorrelation functions ρ_Y could possibly be reached. The same conclusion holds for the general even transformation G in (5.8.7). In the case of G in (5.8.6), when F is continuous, g is increasing by Corollary 5.8.7 and hence $\mathcal{R}_g = [g(-1), g(1)]$. We have $g(1) = \sum_{k=1}^{\infty} g_k^2 k!/\gamma_Y(0) = \mathrm{Var}(G(X_0))/\gamma_Y(0) = 1$ and, similarly, $g(-1) = \mathrm{Cov}(G(X_0), G(-X_0))/\gamma_Y(0)$. Thus, in this case,

$$\mathcal{R}_g = [-g^*, 1] \quad \text{with} \quad g^* = -\frac{\mathrm{Cov}(G(X_0), G(-X_0))}{\mathrm{Var}(G(X_0))}. \tag{5.8.28}$$

There are examples where $(-g^*) > -1$ (see Example 5.8.9 below). See also Exercise 5.10.

Moreover, even if $\rho_Y(n) \in \mathcal{R}_g$ so that $\rho_X(n) = g^{-1}(\rho_Y(n))$ are well-defined, the resulting series $\rho_X(n)$ may not be a valid autocovariance function. Here is an example.

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Example 5.8.8 (*(Non-)Valid autocovariances*) Consider $G(x) = x^2$ in (5.8.13), the corresponding $g(z) = z^2$ in (5.8.14) and the relationship $\rho_Y(n) = (\rho_X(n))^2$ between autocorrelations in (5.8.15). Then, (5.8.26) becomes $\rho_X(n) = \pm\sqrt{\rho_Y(n)}$ or

$$\gamma_X(n) = \pm\sqrt{\frac{\gamma_Y(n)}{2}}, \tag{5.8.29}$$

since $\mathrm{Var}(X) = \mathbb{E}X^2 - (\mathbb{E}X)^2 = 3 - 1 = 2$. Even if $\gamma_Y(n) \geq 0$ so that the right-hand side of (5.8.29) is well-defined, the left-hand side of (5.8.29) may not define a valid autocovariance structure. Indeed, consider an MA(1) autocovariance structure

$$\frac{\gamma_Y(n)}{2} = \begin{cases} 1, & \text{if } n = 0, \\ b, & \text{if } n = \pm 1, \\ 0, & \text{otherwise.} \end{cases} \tag{5.8.30}$$

It is known (Brockwell and Davis [186], Example 1.5.1) that (5.8.30) is a valid autocovariance structure if and only if $|b| \leq 1/2$. Substituting (5.8.30) into (5.8.29) leads to

$$\gamma_X(n) = \begin{cases} 1, & \text{if } n = 0, \\ \pm\sqrt{b}, & \text{if } n = \pm 1, \\ 0, & \text{otherwise.} \end{cases} \tag{5.8.31}$$

This is now a valid autocovariance structure if and only if $\sqrt{b} \leq 1/2$ or $0 \leq b \leq 1/4$. Thus, if $1/4 < b \leq 1/2$, $\gamma_Y(n)$ in (5.8.30) is a valid autocovariance structure but $\gamma_X(n)$ in (5.8.31) is not.

The argument above could also be repeated for moving average series of finite length $n = 0, \dots, N-1$. It is known (Grenander and Szegö [430], Section 5.3, (a)) that $\gamma_Y(n)$, $n = 0, \dots, N-1$, is a valid autocovariance structure if and only if $|b| \leq 1/(2\cos(\pi/(N+1)))$.

We now present a method to choose a *valid* underlying autocovariance $\rho_X = \gamma_X$, and to generate the underlying Gaussian series $\{X_n\}$ in (5.8.2). The method is a natural modification of the circulant matrix embedding method in Section 2.11.2. The series $\{X_n\}$ will be generated for $n = 0, \dots, N-1$; that is, of length N. We suppose that a function G in (5.8.2) has been selected, and that the targeted autocorrelation $\rho_Y(n) \in \mathcal{R}_g$, where $\mathcal{R}_g$ is defined in (5.8.27).

Algorithm for generating an approximate underlying Gaussian series:

- *Step 1:* Compute
$$\widetilde{\gamma}(n) = g^{-1}(\rho_Y(n)), \quad n = 0, \dots, N-1. \tag{5.8.32}$$
- *Step 2:* Form a circulant matrix $\widetilde{\Sigma}$ in (2.11.6) of size $2M = 2(N-1)$ using $\widetilde{\gamma}$ instead of γ. Compute the eigenvalues $\widetilde{\lambda}_m$, $m = 0, \dots, 2M-1$, of $\widetilde{\Sigma}$ as in (2.11.8) using $\widetilde{\gamma}$ instead of γ.[8]
- *Step 3:* Set
$$\lambda_m = \begin{cases} \widetilde{\lambda}_m, & \text{if } \widetilde{\lambda}_m \geq 0, \\ 0, & \text{if } \widetilde{\lambda}_m < 0, \end{cases} \qquad m = 0, \dots, 2M-1.$$

[8] The eigenvalues of $\widetilde{\Sigma}$ are denoted $\widetilde{\lambda}_m$ instead of g_m to avoid confusion with the Hermite coefficients.

Let $\Lambda = \mathrm{diag}\{\lambda_0, \dots, \lambda_{2M-1}\}$ and set $\widetilde{\Sigma}_0 = F\Lambda F^*$ in analogy to (2.11.7).

- *Step 4:* Generate a Gaussian series in (2.11.9) using λ_j instead of $\widetilde{\lambda}_j$. Denoting it $\widetilde{X}^0$ instead of $\widetilde{Y}$ and using Proposition 2.11.2, $\Re(\widetilde{X}^0)$ and $\Im(\widetilde{X}^0)$ have the autocovariance structure $\widetilde{\Sigma}_0$.
- *Step 5:* Take the first N values of $\Re(\widetilde{X}^0)/(\mathbb{E}\Re(\widetilde{X}^0)^2)^{1/2}$ or $\Im(\widetilde{X}^0)/(\mathbb{E}\Im(\widetilde{X}^0)^2)^{1/2}$ for the time series $X_n, n = 0, \dots, N-1$.

Here are a few comments on the steps of the algorithm. In Step 1, $\widetilde{\gamma}(n)$ are computed either by using an explicit form of g^{-1} when it is available (as for g's in Example 5.8.3), or numerically (as discussed in Example 5.8.9 below). Since the circulant matrix $\widetilde{\Sigma}$ is symmetric, the eigenvalues $\widetilde{\lambda}_m$ in Step 2 are real-valued in general, and if some of them are negative, they are set to zero in Step 3. The rest of the argument is the same as in the circulant matrix embedding in Section 2.11.2 except that the nonnegative eigenvalues λ_m replace the eigenvalues $\widetilde{\lambda}_m$. The rescaling by $(\mathbb{E}\Re(\widetilde{X}^0)^2)^{1/2}$ or $(\mathbb{E}\Im(\widetilde{X}^0)^2)^{1/2}$ in Step 5 is there to ensure that the resulting series has variance 1.

We next shed light on the algorithm by discussing several important points. Parts of the discussion below are heuristic.

First, note that if $\widetilde{\lambda}_m \geq 0$ for all $m = 0, \dots, 2M-1$, then $\widetilde{\gamma}(n), n = 0, \dots, N-1$, is a valid autocovariance structure and the algorithm generates a Gaussian series having $\widetilde{\gamma}(n)$ as autocovariance function. On the other hand, note again from Step 2 that

$$\widetilde{\lambda}_m = \widetilde{\gamma}(0) + \widetilde{\gamma}(M)(-1)^m + 2\sum_{j=1}^{M-1} \widetilde{\gamma}(j)\cos\left(\frac{\pi jm}{M}\right).$$

For large N, one expects that

$$\widetilde{\lambda}_m \approx 2\pi \widetilde{f}\Big(\frac{\pi m}{M}\Big) \tag{5.8.33}$$

(see the proof of Proposition 2.11.4), where

$$\widetilde{f}(\lambda) = \frac{1}{2\pi}\Big(\widetilde{\gamma}(0) + 2\sum_{j=1}^{\infty} \widetilde{\gamma}(j)\cos(j\lambda)\Big).$$

One has $\widetilde{f}(\lambda) \geq 0$ if and only if $\widetilde{\gamma}(n), n \in \mathbb{Z}$, is a valid autocovariance structure. Hence, for large N, the fact that $\widetilde{\lambda}_m < 0$ for some m strongly suggests that $\widetilde{\gamma}(n), n = 0, \dots, N-1$, is not a valid autocovariance structure. The algorithm thus provides, in particular, a very fast and almost exact method to determine whether $\widetilde{\gamma}(n), n = 0, \dots, N-1$, is a valid autocovariance structure.

Second, one can argue that the algorithm is optimal in a certain sense. Consider

$$I := \min \sum_{n=-\infty}^{\infty} |\rho_Y(n) - \rho_{app,Y}(n)|^2\, w_n,$$

where the minimum is over $\rho_{app,Y}$, which is the autocorrelation function of a series $Y_{app,n} = g(X_{app,n})$ for a Gaussian series $X_{app,n}$ having autocovariance $\gamma_{app,X}$, $w_n \geq 0$ are certain weights to be chosen below and "app" stands for approximation. Supposing that $\tilde{\gamma}(n)$ and $\gamma_{app,X}(n)$ are close and using the Taylor approximation, we get that

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$$I = \min \sum_{n=-\infty}^{\infty} |g(\widetilde{\gamma}(n)) - g(\gamma_{app,X}(n))|^2 \, w_n$$
$$\approx \min \sum_{n=-\infty}^{\infty} |\widetilde{\gamma}(n) - \gamma_{app,X}(n)|^2 |g'(\widetilde{\gamma}(n))|^2 \, w_n = \min \sum_{n=-\infty}^{\infty} |\widetilde{\gamma}(n) - \gamma_{app,X}(n)|^2,$$

if $w_n = |g'(\widetilde{\gamma}(n))|^{-2}$. Since $\widetilde{\gamma}(n) - \gamma_{app,X}(n)$ are the Fourier coefficients of the function $\widetilde{f}(\lambda) - f_{app,X}(\lambda)$, we can further write the above as

$$I \approx \min \frac{1}{2\pi} \int_{-\pi}^{\pi} |\widetilde{f}(\lambda) - f_{app,X}(\lambda)|^2 d\lambda.$$

Since the minimum is over $f_{app,X}(\lambda) \geq 0$, the optimal choice is

$$f_{app,X}(\lambda) = \begin{cases} \widetilde{f}(\lambda), & \text{if } \widetilde{f}(\lambda) \geq 0, \\ 0, & \text{if } \widetilde{f}(\lambda) < 0. \end{cases}$$

In view of (5.8.33), this corresponds to choosing λ_m as in Step 3.

Finally, we conclude by illustrating the algorithm and some ideas above with a concrete numerical example (see also Helgason, Pipiras, and Abry [464]).

Example 5.8.9 (*Generating series with χ_1^2 marginal and AR(1) autocovariance*) Suppose a series $\{Y_n\}$ with χ_1^2 marginal and autocovariance function $\gamma_Y(n) = 2\phi^{|n|}$, $|\phi| < 1$, of AR(1) series is targeted. Suppose also that negative correlations are sought; that is, $-1 < \phi < 0$. Consider the transformation G in (5.8.22) for which negative correlations are possible (see (5.8.28)). After computing the coefficients in its Hermite expansion numerically and using (5.8.11), the plot of the corresponding function g is given in Figure 5.2, the

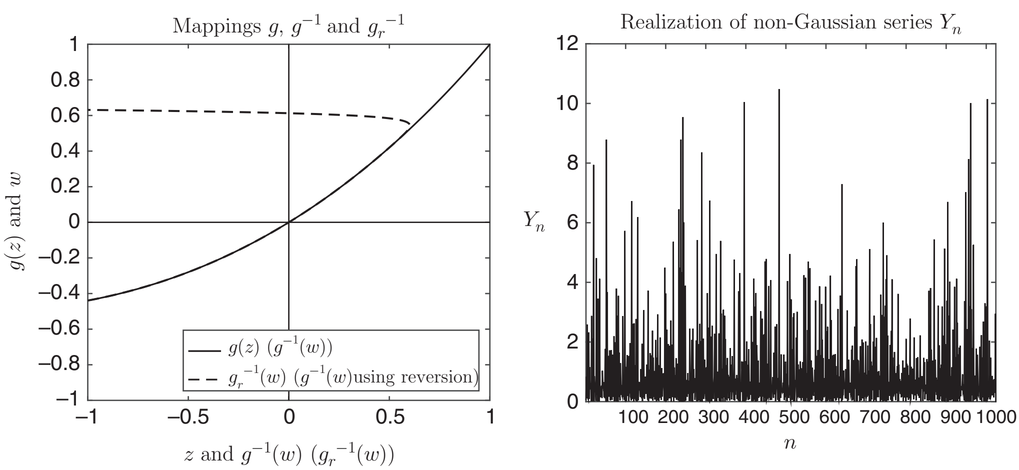

Figure 5.2 Left: The solid line is a graph $w = g(z)$ and by viewing the vertical axis as the w axis, it is a graph of $z = g^{-1}(w)$, where $w \in [-0.44, 1]$. The dashed line is the reversion approximation $g_r^{-1}(w)$ of $g^{-1}(w)$. The two coincide for $w \in [-0.44, 0.47]$ (the dashed line is obscured by the solid line), but diverge for $w > 0.47$. Right: Realization of the non-Gaussian series $\{Y_n\}$ in Example 5.8.9.

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left plot. The function g is increasing with $\mathcal{R}_g = [-g^*, 1]$ in (5.8.27) and (5.8.28), where $(-g^*)$ is about -0.44. Take $\phi = -0.35$ so that $\rho_Y(n) = \phi^{|n|}$ fall in the domain $\mathcal{R}_g$. Take the sample size $N = 1024$.

Computing $\widetilde{\gamma}(n)$ in (5.8.32) requires g^{-1}. This is the function $g^{-1}(w)$, $w \in [-g^*, 1]$, seen (in solid line) in the same Figure 5.2, the left plot, but viewing the vertical (horizontal, resp.) axis as being horizontal (vertical, resp.). An explicit form of g^{-1} is not available in this example. One possibility is to solve (5.8.12) numerically for each $\rho_Y(n)$ in terms of $\rho_X(n)$, which is facilitated by the fact that g is increasing. A convenient alternative is to use the so-called *reversion* operation. The function g has a Taylor expansion in (5.8.11), with numerically computed coefficients. Denote the expansion by $g(z) = \sum_{k=1}^{\infty} b_k z^k$. It is expected that g^{-1} will also have a Taylor expansion $g^{-1}(w) = \sum_{k=1}^{\infty} d_k w^k$ (see, e.g., Henrici [467]). A substitution of the Taylor series into the equation $g(g^{-1}(w)) = w$, that is, $\sum_{j=1}^{\infty} b_j (\sum_{k=1}^{\infty} d_k w^k)^j = w$, and an identification of the coefficients yields $b_1 d_1 = 1$, $b_1 d_2 + b_2 d_1 = 0$, $b_1 d_3 + 2b_2 d_1 d_2 + b_3 d_1^3 = 0, \dots$ Hence, $d_1 = b_1^{-1}$, $d_2 = -b_1^{-3} b_2$, $d_3 = b_1^{-5}(2b_2^2 - b_1 b_3)$, and so on. The latter operation on series (sequences) is known as *reversion* (see, e.g., Henrici [467]). Denote by $g_r^{-1}(w)$ the inverse function of g defined through the Taylor series and the reversion operation. A disadvantage of $g_r^{-1}(w)$ is that it does not need to coincide with $g^{-1}(w)$ over the whole range where the latter is defined (see, e.g., Henrici [467]). Indeed, Figure 5.2, the left plot, also depicts $g_r^{-1}(w)$ (dashed line): it coincides with $g^{-1}(w)$ for $w \in [-g^*, g_1]$ with $g_1 \approx 0.47$, where the dashed line is obscured by the solid line corresponding to the inverse $g^{-1}(w)$, but then diverges quickly for $w > g_1 \approx 0.47$ (the dashed line would continue below $z = -1$ if the z axis were extended to include negative values smaller than -1). We note, however, that $g_r^{-1}(w)$ can still be used to evaluate $g^{-1}(\gamma_Y(n))$ in (5.8.32) since most of the autocovariances $\gamma_Y(n)$ are expected to be small (and, in particular, belong to $[-g^*, g_1]$ where g_r^{-1} and g^{-1} coincide). In fact, in the example here, $\rho_Y(n) = 0.35^{|n|}$ belong to $[-g^*, g_1] \approx [-0.44, 0.47]$ for all $n \geq 1$.

In Figure 5.2, the right plot, depicts a realization of the non-Gaussian series $\{Y_n\}$. In Figure 5.3, for the first few lags n, we also plot the targeted autocovariance $\gamma_Y(n)$ and its approximation $\gamma_{app,Y}(n)$ obtained from the algorithm described above, as well as the difference between them.

Remark 5.8.10 The AR(1) covariance structure considered in Example 5.8.9 corresponds to SRD. Instead, we could have considered, for example, the FARIMA$(0, d, 0)$ covariance structure associated with LRD. Alternatively, if the χ_1^2 marginal is desired and just the LRD property of the resulting series $\{Y_n\}$ is sought (that is, without a specific resulting LRD covariance structure), one can consider the series $Y_n = X_n^2$ where $\{X_n\}$ is LRD. Indeed, supposing X_n is standardized, the series $\{Y_n\}$ will indeed have the χ_1^2 marginal and be LRD with the parameter $2d - 1/2 \in (0, 1/2)$ by Proposition 5.2.4, where $d \in (1/4, 1/2)$ is the LRD parameter of the series $\{X_n\}$.

Remark 5.8.11 In this section, we have focused on generation of non-Gaussian time series with prescribed marginals and targeted autocovariance structures. For estimation issues in related models, see Livsey, Lund, Kechagias, and Pipiras [640].

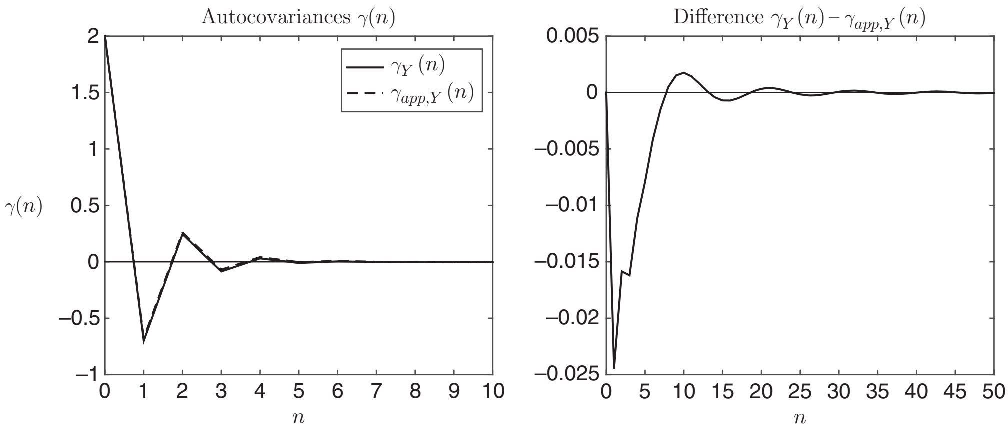

Figure 5.3 Left: The targeted autocovariance $\gamma_Y(n)$ and its approximation $\gamma_{app,Y}(n)$ in Example 5.8.9. Right: The difference $\gamma_Y(n) - \gamma_{app,Y}(n)$.

5.9 Exercises

The symbols * and ** next to some exercises are explained in Section 2.12.

Exercise 5.1 Let g_n, $n \geq 0$, be the coefficients (5.1.7) in the Hermite expansion (5.1.6) of a function $G \in L^2(\mathbb{R}, \phi)$. Show that the definition (4.1.1) of Hermite polynomials and integration by parts yield

$$g_n = -\frac{1}{n!}\mathbb{E}G'(X)H_{n-1}(X)$$

under suitable to-be-specified conditions. Conclude under these suitable conditions that $(n!)^{1/2}|g_n| \leq C/n^{1/2}$ for a constant $C > 0$. Generalize this result to obtain a bound of the kind $C/n^{r/2}$ with integer $r \geq 1$. *Hint:* See Proposition 5.8.4.

Exercise 5.2 If $G \in L^2(\mathbb{R}, \phi)$ is an even function – that is, $G(x) = G(-x)$, $x \in \mathbb{R}$ – what can its Hermite rank be? What about the Hermite rank of an odd function G satisfying $G(x) = -G(-x)$, $x \in \mathbb{R}$?

Exercise 5.3 Do functions $G(x)$, $x \in \mathbb{R}$, and $G(ax + b)$, $x \in \mathbb{R}$, for constants $a \neq 0, b$ have the same Hermite rank?

Exercise 5.4 (a) Prove the convergence (5.3.19) and that the limit can be expressed as (5.3.20). (b) Prove the more general convergence (5.3.14), and that the limit is continuous.

Exercise 5.5 Prove Lemma 5.4.4. *Hint:* If S_N denotes the left-hand side of (5.4.17), write $|S_N| \leq \int_0^a f_N(u)du$, where $f_N(u) = \sum_{n_1=1}^{[Na]} c_{N,n_1} 1_{(\frac{n_1-1}{N}, \frac{n_1}{N}]}(u)$ and $c_{N,n_1} = \sum_{n_2=1}^{[Nb]-[Na]} |c([Na] + n_2 - n_1)|$, and use the dominated convergence theorem.

https://doi.org/10.1017/CBO9781139600347.007 Published online by Cambridge University Press

Exercise 5.6** The non-central limit theorem (Theorem 5.3.1) assumes that $d \in ((1-1/k)/2, 1/2)$ and the central limit theorem (Theorem 5.4.1) that $d \in (0, (1-1/k)/2)$. What happens in the boundary case $d = (1-1/k)/2$?

Exercise 5.7 Under the assumptions of Theorem 5.6.3, let $G(z) = z^2$ and $1/4 < d < 1/2$. (i) Show that in this case

$$\begin{aligned} G(X_n) &= \sum_{0 \le p_1 \neq p_2 < \infty} \psi_{p_1}\psi_{p_2}\epsilon_{n-p_1}\epsilon_{n-p_2} + \sum_{0 \le p < \infty} \psi_p^2(\epsilon_{n-p}^2 - \mu_2) + \mathbb{E}G(X_n) \\ &=: A_n + B_n + \mathbb{E}G(X_n), \end{aligned}$$

where $\mu_2 = \mathbb{E}\epsilon_1^2$. (ii) Show that $\mathbb{E}(\sum_{n=1}^N A_n)^2 \sim L_1(N)^4 N^{4d}$ as $N \to \infty$. (iii) Show that the autocovariance of B_n is summable so that $\mathbb{E}(\sum_{n=1}^N B_n)^2 = O(N)$. (iv) Prove Theorem 5.6.3 in this case using Proposition 5.6.4.

Exercise 5.8 Prove that the bound in (5.7.39) is finite when $d < \frac{1}{2}(1-\frac{1}{k})$. *Hint:* One way is to use a similar bound with d replaced by d^* such that $\max(0, d) < d^* < \frac{1}{2}(1-\frac{1}{k})$, and use Proposition 2.2.9.

Exercise 5.9** The partial sum processes of the type (5.0.1) were considered throughout Sections 5.3–5.7 for $t \ge 0$. Formulate the results of these sections for partial sum processes for $t \in \mathbb{R}$.

Exercise 5.10** Prove or disprove the following conjectures. (a) As in (5.8.27), let $\mathcal{R}_g$ be the range of the function g, associated with a function G matching a marginal distribution F_Y. Then, $\mathcal{R}_g \subset \mathcal{R}_{g_1}$, where the function g_1 is associated with the function G_1 in (5.8.6). (b) Similarly and stronger, let $\mathcal{C}_g = \{\{g(\gamma_X(n))\}_{n\in\mathbb{Z}} :$ Gaussian stationary $X = \{X_n\}$ with $\mathbb{E}X_n = 0, \mathbb{E}X_n^2 = 1\}$ be the collection of possible autocovariances of the series $\{G(X_n)\}$ with a function G matching a marginal distribution F_Y and giving rise to the function g. Then, $\mathcal{C}_g \subset \mathcal{C}_{g_1}$, where the function g_1 is associated with the function G_1 in (5.8.6).

5.10 Bibliographical Notes

Section 5.1: The results in the key Proposition 5.1.1 and its consequences (e.g., Proposition 5.1.4) are sometimes referred to as *Mehler's formula*. Mehler's formula (Mehler [714]) reads, under the assumptions of Proposition 5.1.1,

$$q(x, y) = \sum_{k=0}^{\infty} \frac{(\mathbb{E}XY)^k}{k!} H_k(x)H_k(y)q(x)q(y),$$

where $q(x, y)$ is the joint density of (X, Y) and $q(x), q(y)$ are the marginal densities of X, Y. Assuming that $\mathbb{E}H_m(X)H_n(X) = \int_{\mathbb{R}} H_m(x)H_n(x)q(x)dx = n!$ if $m = n$ and $= 0$ if $m \neq n$ (which is stated as Corollary 5.1.2 but can also be proved simpler by using basic properties of Hermite polynomials), note that the Mehler's formula above implies (5.1.1) since

$$\mathbb{E}H_n(X)H_m(Y) = \int_{\mathbb{R}}\int_{\mathbb{R}} H_n(x)H_m(y)q(x,y)dxdy$$
$$= \sum_{k=0}^{\infty} \frac{(\mathbb{E}XY)^k}{k!} \int_{\mathbb{R}} H_n(x)H_k(x)q(x)dx \int_{\mathbb{R}} H_m(y)H_k(y)q(y)dy.$$

Section 5.2: The notion of Hermite rank was introduced by Taqqu [943]. A similar notion, that of "Laguerre rank" when the Gaussian distribution is replaced by a gamma distribution can be found in Leonenko, Ruiz-Medina, and Taqqu [615], together with a corresponding non-central limit theorem. Testing for Hermite rank in Gaussian subordinated processes is considered in Beran, Möhrle, and Ghosh [128].

Section 5.3: Non-central limit theorem for Gaussian stationary series was obtained by Dobrushin and Major [321], Taqqu [943, 945]. See also Rosenblatt [860, 861], Ibragimov [514], Berman [132], Maejima [661, 662]. A random field version can be found in Leonenko, Ruiz-Medina, and Taqqu [616]. The proof presented in Section 5.3 follows the approach of Dobrushin and Major [321]. The rate of convergence to the Rosenblatt-type distribution in the non-central limit theorem is studied in Anh, Leonenko, and Olenko [37].

Sly and Heyde [911] have an interesting result involving a sequence $G(X_n)$ where X_n is a linear Gaussian sequence but where $G(X_n)$ has heavy tails with exponent $0 < \alpha < 2$. Although $G(X_n)$ does not have finite variance, Hermite polynomials are still involved. They show that the limit can be an Hermite process or an α-stable Lévy motion depending on the values of the parameters. In the boundary case, the limit is a sum of independent copies of these two processes.

Section 5.4: The proof of the central limit theorem for Gaussian stationary series presented in Section 5.4 follows the approach of Breuer and Major [184]. Some earlier related work can be found in Sun [931, 932], Cuzick [272], and some further work in Maruyama [693, 694], Giraitis and Surgailis [396], Ho and Sun [478], Chambers and Slud [209, 210]. The convergence of densities in the central limit theorem for functions of Gaussian stationary series is studied in Hu, Nualart, Tindel, and Xu [496].

Section 5.5: Theorem 5.5.1 is proved in Nualart and Peccati [771] by means of stochastic calculus techniques. For books dealing with the fourth moment condition, see Peccati and Taqqu [797], Nourdin and Peccati [763], Nourdin [760] and Tudor [963]. Peccati and Zheng [799] consider functionals of a Poisson random measure.

Section 5.6: The material of Section 5.6.1 is based on Surgailis [935], Lévy-Leduc and Taqqu [618]. Other work and a related approach based on Appell polynomials, for both central and non-central limit theorems, can be found in Surgailis [934, 936, 937], Giraitis [392, 393, 394], Giraitis and Surgailis [397, 398], Avram and Taqqu [59], Surgailis and Vaičiulis [938].

Nourdin, Peccati, and Reinert [767] established the following universality result: if a sequence of off-diagonal homogeneous polynomial forms in i.i.d. standard normal random variables converges in distribution to a normal, then the convergence also holds if one replaces these i.i.d. standard normal random variables in the polynomial form by any independent standardized random variables with uniformly bounded third absolute moment. The result, which was stated for polynomial forms with a finite number of terms, was extended by Bai and Taqqu [79] to allow an infinite number of terms in the polynomial forms. Based on a contraction criterion derived from this extended universality result, they obtain a central limit theorem for long-range dependent nonlinear processes, whose memory parameter lies at the boundary between short and long memory.

The material of Section 5.6.2 follows Ho and Hsing [477], Lévy-Leduc and Taqqu [618]. Related work, for both central and non-central limit theorems, includes Ho and Hsing [476], Giraitis and Surgailis [400], Hsing [490], Wu [1012].

Section 5.7: The multivariate results involving functions of Gaussian random variable are based on Bai and Taqqu [74]. In the SRD case, the multivariate convergence to the multivariate Gaussian distribution also follows from the univariate convergence and the convergence of the covariances, as shown in Peccati and Tudor [798]. In Bai and Taqqu [74], mixed convergence was established up to Hermite order two in order to ensure that the moment-determinancy of the limit distributions. This restriction has now been eliminated due to the recent results of Nourdin et al. [768] and Nourdin and Rosiński [764], as stated in Theorem 5.7.10 and Corollary 5.7.11.

Proofs for the corresponding multivariate results for multilinear processes presented in Section 5.7.4 can be found in Bai and Taqqu [75].

Section 5.8: The material of this section is based on Helgason et al. [464]. But probabilistic modeling based on transformations of Gaussian stationary time series is quite popular in engineering mechanics (e.g., Grigoriu [431], Bocchinia and Deodatis [164], Nichols, Olson, Michalowicz, and Bucholtz [745]), signal processing (e.g., Liu and Munson [639], Scherrer, Larrieu, Owezarski, Borgnat, and Abry [887]), econometrics (e.g., Dittmann and Granger [318]) and other areas.

Other notes: Central and non-central limit theorems for Gaussian stationary vector time series are considered in Berman [133], Ho and Sun [478], Arcones [42, 43], Sánchez de Naranjo [884, 885]. Several of the references given above also cover the case of central and non-central limit theorems for Gaussian stationary random fields, including Dobrushin and Major [321], Breuer and Major [184]. For other related work, see Maejima [663, 665], Leonenko and Parkhomenko [614], Leonenko and Olenko [613], Lahiri and Robinson [588] and the textbooks by Ivanov and Leonenko [525], Leonenko [612]. Limit theorems for tempered Hermite processes can be found in Sabzikar [875].

The following papers of Bai and Taqqu contain recent results. The paper [77] introduces Generalized Hermite processes, the paper [78] deals with the convergence of long-memory discrete kth-order Volterra processes, the paper [82] studies the impact of diagonals of polynomial forms on limit theorems with long memory and the paper [76] analyzes the structure of the third moment of the generalized Rosenblatt distribution.

Recent results on limit theorems for multiple Wiener integrals have exploited the Malliavin calculus. See the monographs on the subject by Nourdin and Peccati [763], Nourdin [760]. Several notable papers are Nualart and Peccati [771], Nourdin and Peccati [762]. The basics of the Malliavin calculus are given in Appendix C, and used in Chapter 7.